GEK-65280E

DRIVE SYSTEM INSTRUCTIONS

MARK II SPEEDTRONIC® TURBINE CONTROL PANEL

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL@) ELECTRIC

(GEK-65280E)

MK II SPEEDTRONIC®
Turbine Control Panel

The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Department, USA.

'INTRODUCTION

The MKII turbine control panel (See Figure 1-1), along with sensors and output servo valves, controls the startup, protection and operation of the General Electric gas turbines. Figure 1-2 shows a typical Input Filter Nodule and Figure l-3 shows a typical Relay Module. The primary functions provided by the MKII SPEEDTRONIC ® Control are shown in Figure 1-4.

Control System

The fuel control system consists of three major loops. These are startup, load turbine speed and temperature. These loops are illustrated in Figure l-5. These loops are tied together into a minimum value: gate such that the loop calling for the least amount of fuel will be in command. Two shaft turbines have, in addition to the fuel control loops, a nozzle control loop. This loop positions the angle of the nozzle to control high pressure (HP speed). The control system is designed to provide maximum running reliability. Redundant sensors' are frequently used to supply feedback signals. Reliability is further enhanced by *redundance by association'. For example, if the system is operating on speed control, the temperature control will back it up,

Sequential Circuitry

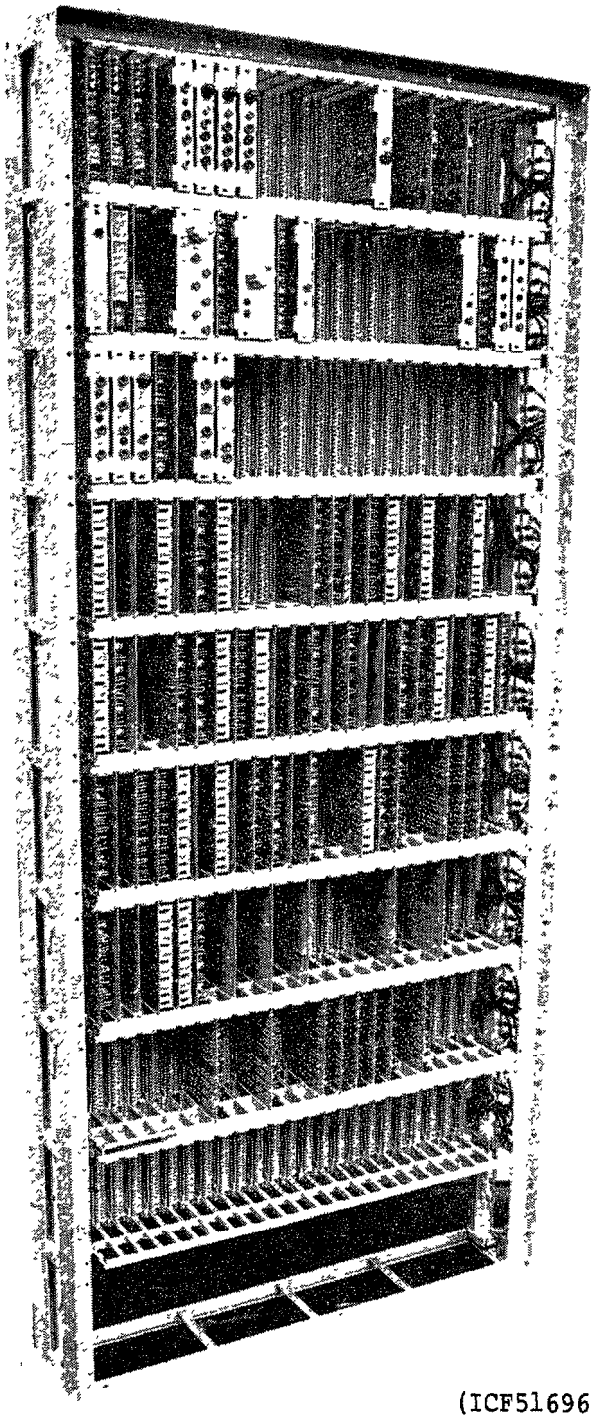
Sequential circuitry is provided to monitor the turbine process, starting operation and auxiliaries to determine if all systems are normal. The return signals from these devices serve as permissives to operate the turbine. Abnormal conditions are annunciated via local and remote (optional1 annunciators. The sequential circuitry also generates logic signals to facilitate starting and stopping the turbine in a prescribed manner. These logic signals include speed signals, digital setpoint control, starting means control, timers and other operations.

Protection Circuitry

Protection circuitry is provided to trip the turbine when critical operating parameters *are* exceeded or when critical equipment fails. Alarm annunciation is provided to indicate when critical levels are being approached and to indicate abnormal equipment operation.

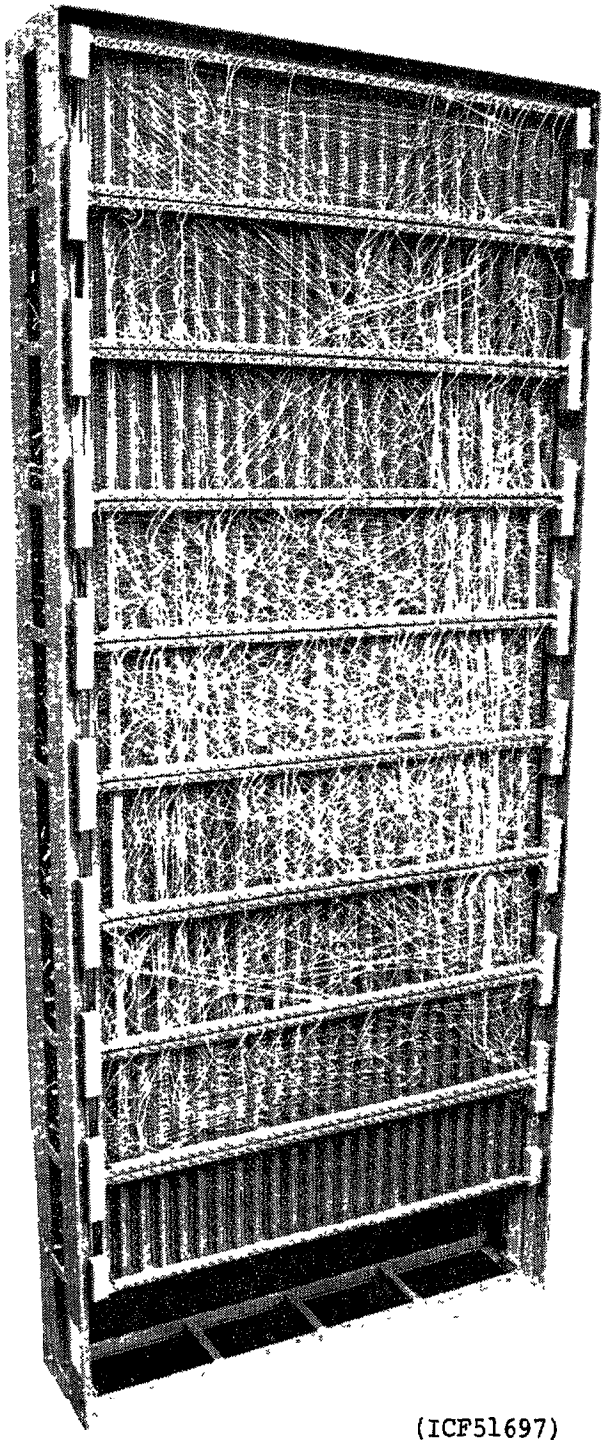
Power Supply

The control panel contains its own highly reliable solid state power supply. The power supply consists of a DC-to-DC converter and final regulators for P28, P12, P5.3 and N12 volts. The converter will operate satisfactorily with an input voltage within the range of 95 to 140 volts DC. A source selector which permits operation from a 95 to 140 volt DC battery, commercial power or inductor alternator is optional.



(ICF51696)

Front View



(ICF51697)

Backplane Wiring

Figure 1-1
SPEEDTRONIC® Governor Page (Typical)

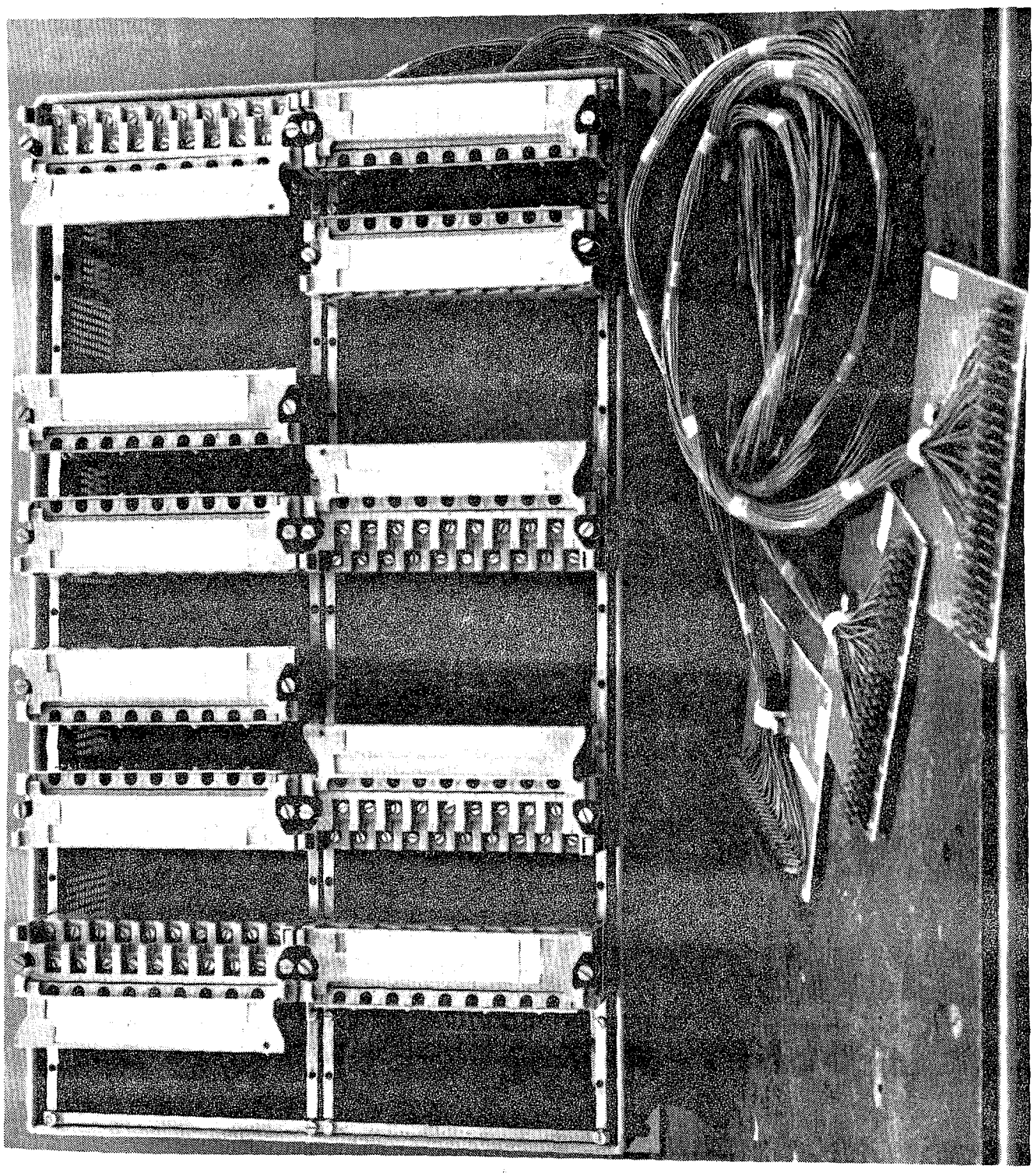
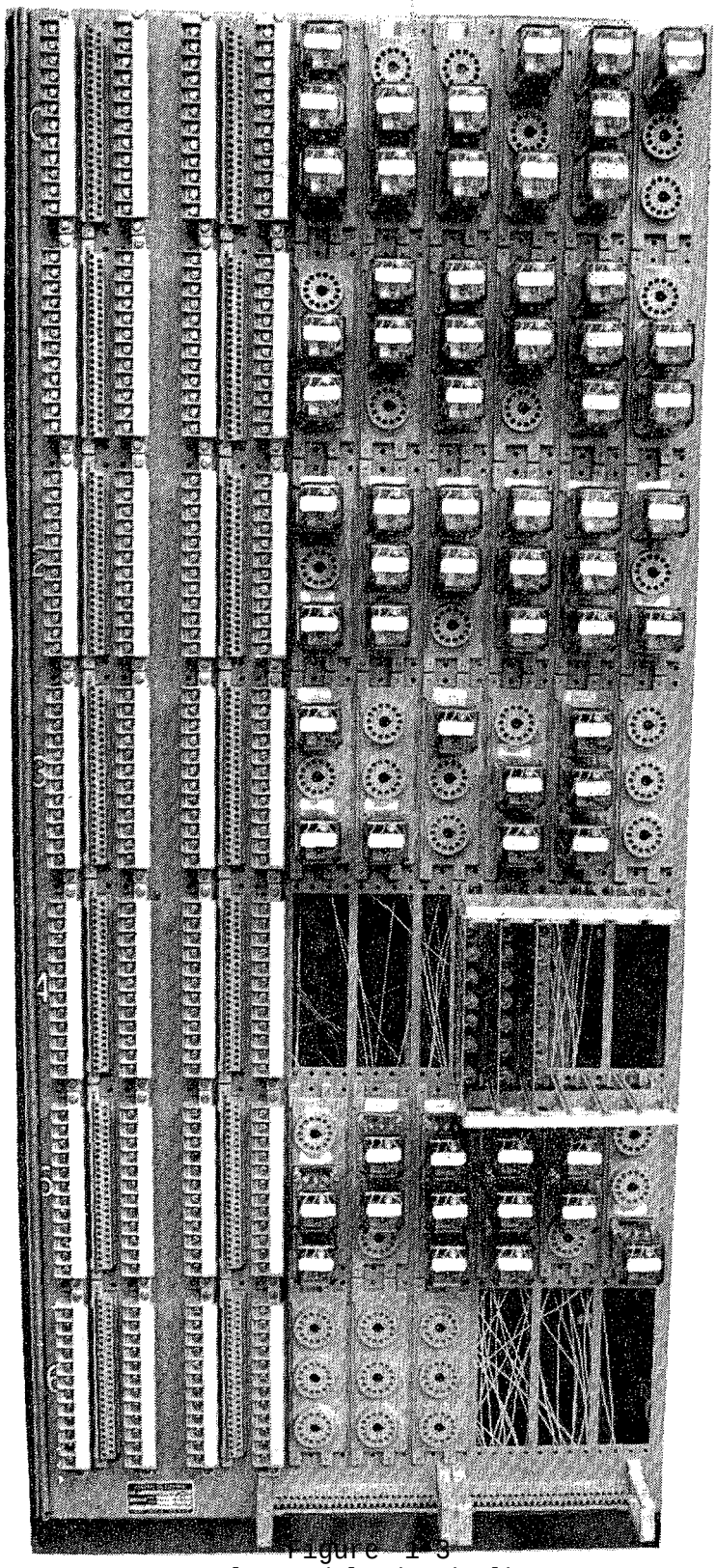


Figure 1-2 Input Filter Module (Typical) (ICF51698)



(ICF51699)

Figure 1-3
Relay Module (Typical)

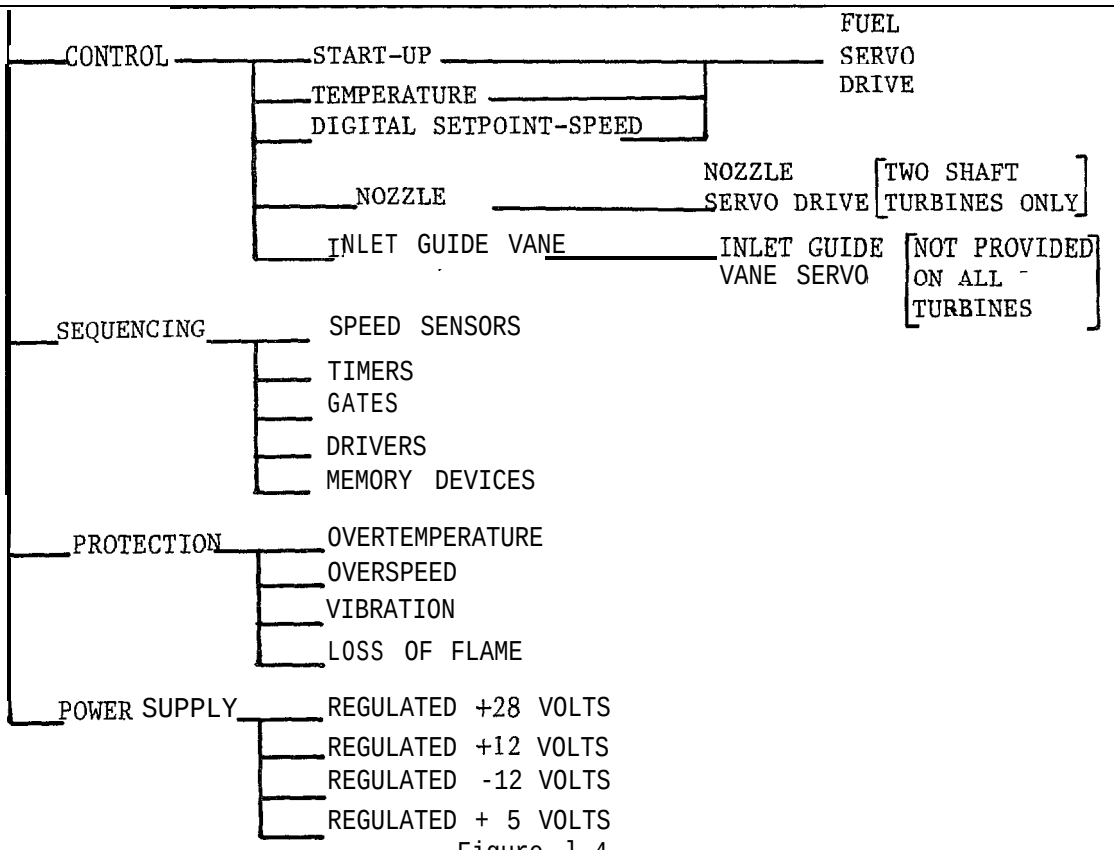


Figure l-4
SPEEDTROMIC* MKII Control Functions

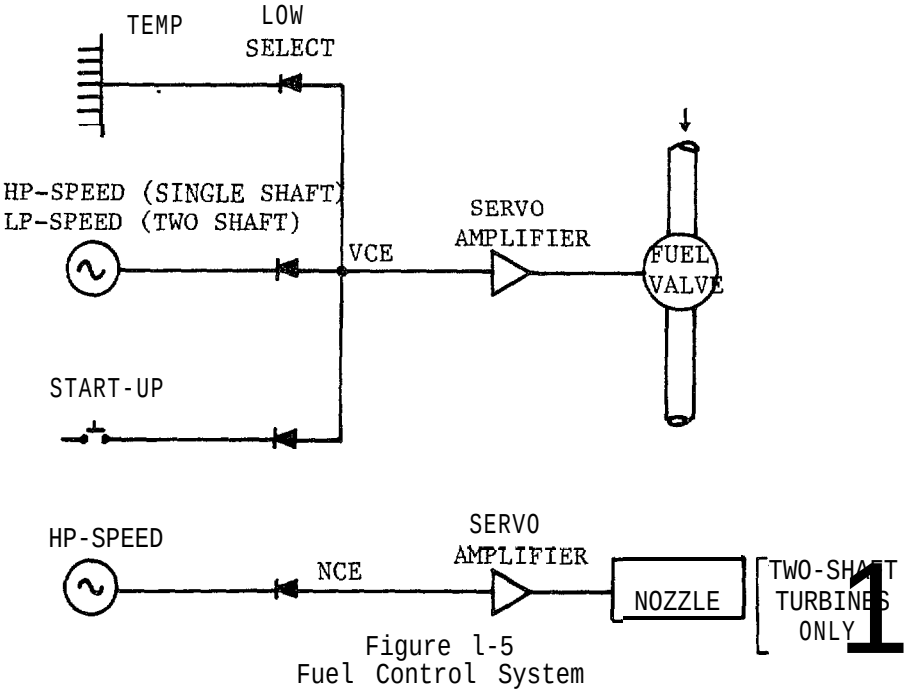


Figure l-5
Fuel Control System

*Trademark of General Electric Company, U.S.A.

Publications and Repair Notes

The intent of this instruction book is for use in conjunction with the (1) Turbine Control Specification, (2) Control Panel Elementary Diagram, (3) Schematic Piping Diagram, (4) Operating Instructions, and (5) Turbine Control Elementary Reading Aid. This instruction book is typical and does not purport to cover all details or variations that may arise on a particular application.

When troubleshooting the Turbine Control Panel, the test points provided on the calibrator (if available) and card fronts should be used whenever possible to avoid damage to the cards that often results when connecting test probes to the pins on back of the SPEEDTRONIC page. If it is absolutely necessary to troubleshoot on the back of the page, insulated test probes should be used. Recommended procedures concerning the removal or replacement of cards with power applied is given in Technical Bulletin 060 TB l-01-001.

Failed cards and modules should be returned to the General Electric Company for repairs. No attempt should be made to repair the card in the field. The procedure for returning cards and modules can be obtained from your nearest General Electric representative.

****
* *
* CAUTION *
* *
****

THIS EQUIPMENT IS DESIGNED TO OPERATE IN A NORMAL INDUSTRIAL ATMOSPHERE. OPERATING RADIO TRANSMITTERS NEARBY MAY CAUSE THE CONTROL EQUIPMENT TO MISOPERATE. IF IT IS NECESSARY TO OPERATE.RADIO TRANSMITTERS WITHIN THE VICINITY OF THE CONTROL PANELS, MAKE SURE THE PANEL DOORS ARE CLOSED, THE TRANSMITTER IS NO CLOSER THAN ONE METER TO THE EQUIPMENT, AND THAT THE TRANSMITTER OUTPUT POWER IS NO GREATER THAN FIVE WATTS.

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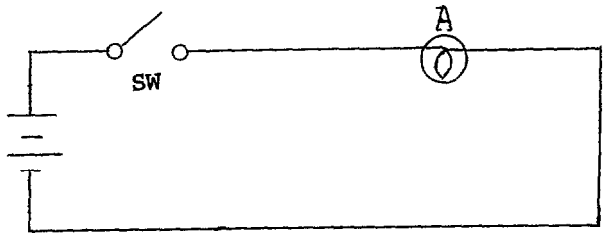
SEQUENTIAL CONTROL

Logic Interpretation

A brief discussion of logic signals is presented to aid in understanding the sequential circuitry. A comparison is made between relay logic and solid state logic to help those familiar with relay logic, but not familiar with solid state logic, to read solid state logic.

A logic signal has only two possible states. These states have various names in different literature. In this instruction book, the two states are defined as "one" (1) and "zero" (0). A logic 0 indicates that some event has not occurred or is not occurring. In the MKII system, a logic 1 state is usually represented by +5 volts ahd a logic 0 state is usually represented by zero volts.

If the state of some event is designated by using the variable A, then when A equals 1, the event is occurring or has occurred. When A equals 0, the event is not occurring or has not occurred. For example, in the circuit below, the event is defined as the illumination of the light, and the variable A is used to represent the state of the event. When the switch is closed, A is equal to 1 (lamp illuminated) and when the switch is open, A is equal to 0 (lamp is extinguished).



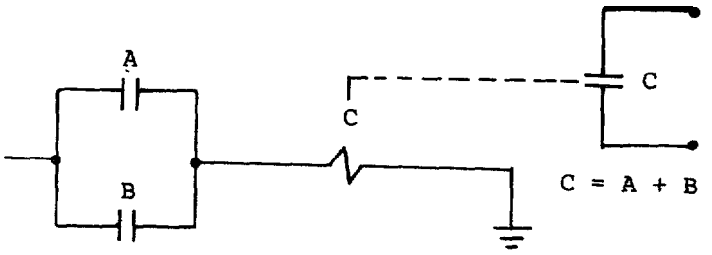
The mathematics of logic signals is referred to as Boolean Algebra. A very brief discussion of Boolean Algebra is given subsequently. For each Boolean operation, both relay logic and DIRECTO-MATIC ® II logic, to perform the operation, are given.

There are three basic operations associated with Boolean Algebra. These are:

(1) LOGIC ADDITION ("OR" FUNCTION): Logic addition is designated by the symbol "+". The addition operation is defined as:

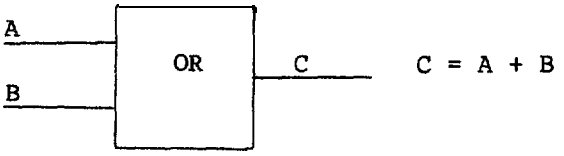
1 + 1 = 1
1 + 0 = 1
0 + 1 = 1
0 + 0 = 0

(a) Relay logic for addition (OR).



Relay C will be picked up if either contact A or B is closed.

(b) DIRECTO-MATIC II solid state logic for addition (OR).

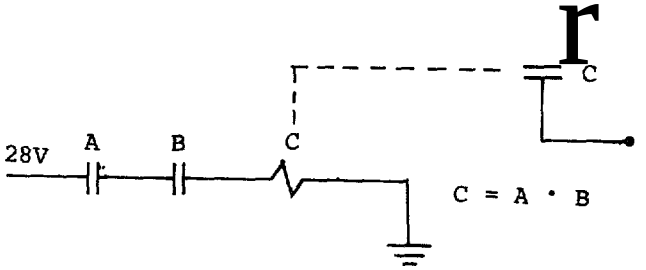


Output C will be 1 (+5 volts) if either output A or B is a 1 (+5 volts).

(2) LOGIC MULTIPLICATION ("AND" FUNCTION): Logic multiplication is designated by the symbol ".". The multiplication operation is defined as :

1 · 1 = 1
1 · □ = □
□ · 1 = □
0 · 0 = 0

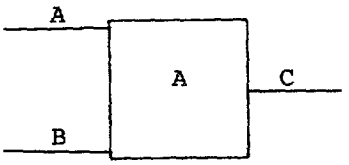
(a) Relay implementation:



If A and B are made, relay C will be picked up.

NOTE: The symbol "·" is often omitted, i.e. A·B = AB

(b) DIRECTO-MATIC II logic implementation:



$C = A \cdot B$

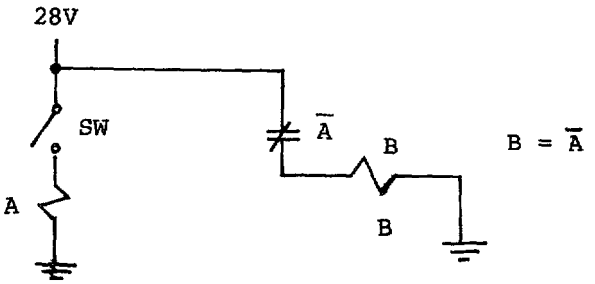
If A = 1 and B = 1 then C will be 1. For any other combination of A and B, C is 0.

(3) LOGIC NEGATION ("NOT" FUNCTION): Logic negation (also called inverse or complement) is designated by placing a bar above the variable. (Example: If A is the variable, then $\bar{A}$ is the inverse of A). The inverse of A is defined as follows:

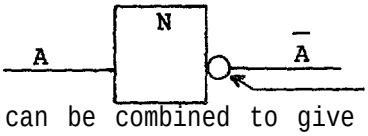
When A = 1, $\bar{A} = 0$

When A = 0, $\bar{A} = 1$

(a) Relay implementation:



(b) DIRECTO-MATIC II implementation:



Logic operations can be combined to give logic equations such as:

$$A = (C \bullet B) + D$$

NOTE: The priority of the operations are ·, +, NOT. Therefore, if no parentheses are included, all the "·" operations are completed first; then the "+" and the negation is last. Example: A = C + B .D is understood to be A = C + (B .D).

Logic equations containing terms such as $A = \overline{B \cdot D}$ can be simplified using the following technique:

To simplify an expression such as $B \cdot D$, substitute "+" for "•" and complement each individual variable. For example, the expression $B \cdot D$ simplifies to $\overline{B+D}$. Likewise, $\overline{B+D}$ simplifies to $\overline{B}'\overline{D}$. This procedure can be extended to more complex expressions. The following example illustrates the technique used:

Example:

$$E = \overline{\overline{A \cdot B} \cdot (C + D)}$$

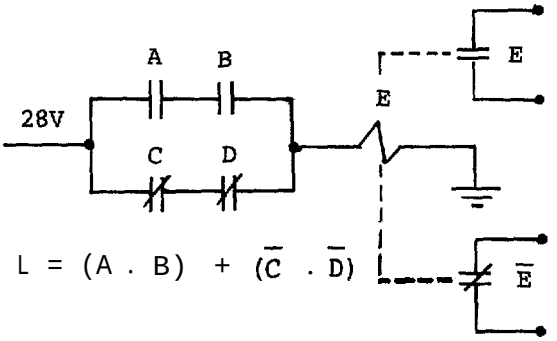
$$(1)\quad E = \overline{\overline{A \cdot B}} + \overline{(C + D)}$$

$$(2)\quad E = A \cdot B + \overline{(C + D)}$$

$$(3)\quad E = A \cdot B + (\overline{C} \cdot \overline{D})$$

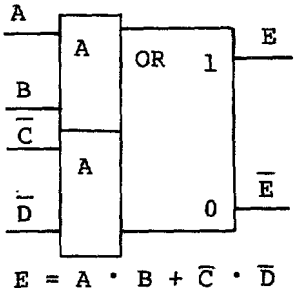
The implementation of logic equations using both relay and DIRECTO-MATIC II logic is illustrated below:

(a) Relay:



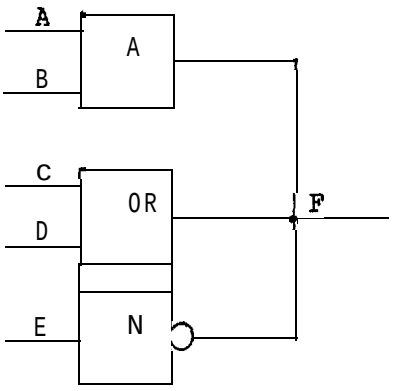
$L = (A \cdot B) + (\overline{C} \cdot \overline{D})$

(b) DIRECTO-MATIC II



$E = A \cdot B + \overline{C} \cdot \overline{D}$

The 1 and 0 located inside the symbol adjacent to outputs E and $\overline{E}$ respectively are often used to designate the normal and inverse outputs. The circuit is said to be in the true state when the output adjacent to 1 is a logic one. It is said to be in the false state when the output adjacent to 0 is a logic one. It should be noted here that in the case where the outputs of more than one logic element are tied together, a 0 will dominate over a 1. The following circuit illustrates this:



F is equal to 0 if any of the following conditions are true:

(1) A + B = 0

(2) C . D = 0

(3) E = 1

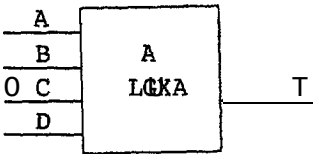
PRECAUTION: A 1 can be forced to a 0 by jumpering the signal to common; however, trying to force a 0 to a 1 by jumpering the signal to i-5.3 volts could result in damage to the circuit.

DIRECTO-MATIC II Logic Cards and Symbols +

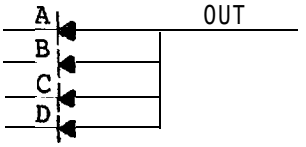
t This section is merely to familiarize the reader with DIRECTO-MATIC II logic cards. For specific information, consult the **DIRECTO-MATIC II** Card Application Data Book.


The presence if the "•" at the input terminal of an element denotes the ability to accept either normal logic outputs or the output of a LGXA.

IC3600LGXA - "AND" Gate Expander.



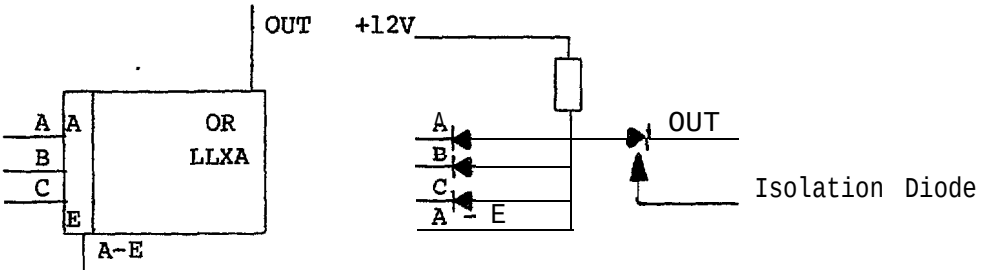
9 Circuits per Card



Circuit

The LGXA provides a cluster of diodes for expanding the AND inputs of many DIRECTO-MATIC II logic cards. An illustration of how the LGXA is used is given in the LCDA discussion. When the OUT terminal of the LGXA is connected to a A-B input of another logic element, a 0 on any input will pull OUT to a 0. A logic 1 on all inputs will allow OUT to go to a **logic** 1.

IC3600LLXA - OR Logic Expander:

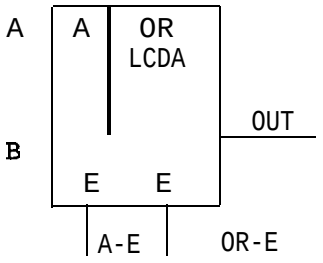


9 CIRCUITS PER CARD

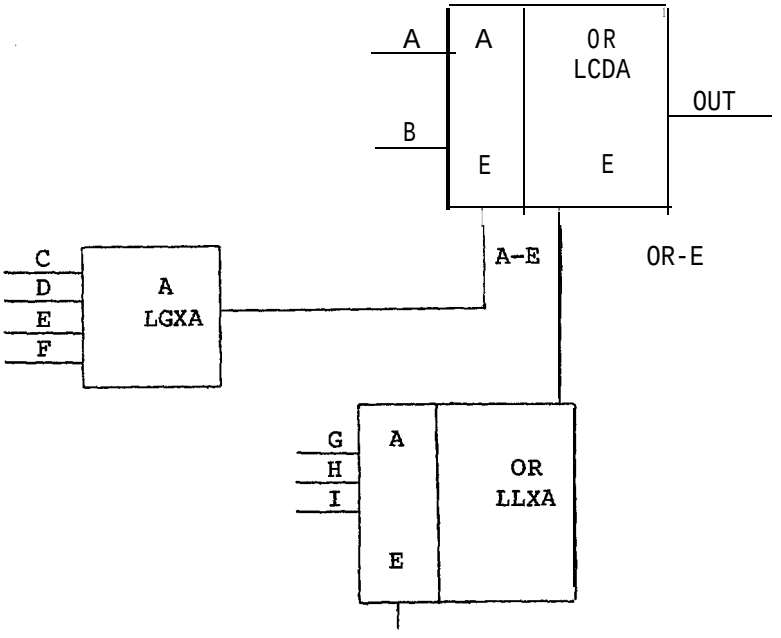
The LLXA provides circuitry for input expansion of logic elements having an OR-E input. If all inputs are 1, the isolation diode will be forward biased and pull OR-E input of connected logic element to a 1 which makes the normal output of that element a 1. If any input is a 0, the isolation diode will be reverse biased and will allow the output of the connected element to be controlled and will allow the output of the connected element to be controlled by its normal inputs. An A-E input is provided for expanding the AND inputs using a LGXA. The use of the LLXA will be illustrated in the LCDA discussion,

NOTE: For most DIRECTO-MATIC II cards, an unused input left open is recognized as a 1.

IC3600LCDA - Clock Driver



The LCDA clock driver contains a power amplifier which will drive 32 "L" logic loads. In addition to the two normal AND inputs, A-E and OR-E inputs are provided. If no connection is made to A-E or OR-E, the circuit performs as a normal two-input AND gate with power amplifier. An illustration of how the A-E and OR-E inputs could be used is illustrated and followed by its truth table.



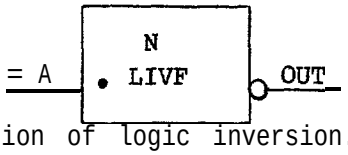
Out = (A • B • C • D • E • F) + (G • H • I)

TRUTH TABLE*

A	B	C	D	E	F	G	H	I	OUT
0	0	0	1	1	1	0	1	1	0
0	0	0	1	1	1	1	1	1	1
0	0	1	1	1	1	0	1	1	0
1	0	1	1	1	1	1	1	1	1
1	0	0	1	1	1	0	1	1	0
1	0	1	1	1	1	0	1	1	0
1	1	0	1	1	1	0	1	1	0
1	1	1	1	1	1	0	1	1	1

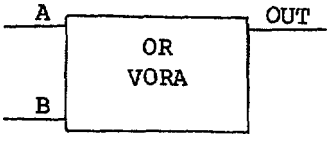
* In order to simplify table, assume inputs D, E, F, H and I are always a 1. For example, only inputs A, B, C and G can change states.
0 equals "don't care" if 1 or 0.

IC3600LIVF - Inverter

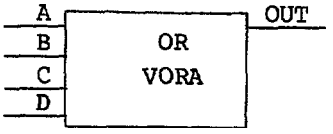


The LIVF provides the function of logic inversion. A 1 in will give a 0 out and a 0 in will give a 1 out.

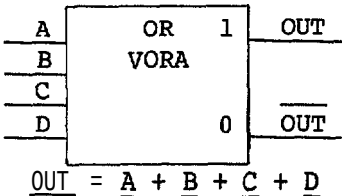
IC3600VORA - Logic OR



OUT = A + B



OUT = A + B t C + D

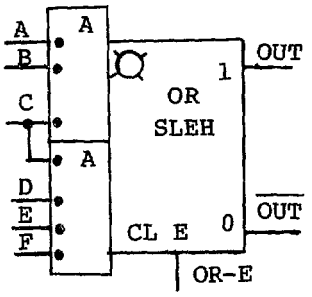


$\underline{OUT} = \underline{A} + \underline{B} + \underline{C} + \underline{D}$
$OUT = \underline{A} \cdot \underline{B} \cdot \underline{C} \cdot \underline{D}$

2 CIRCUITS PER CARD

This card provides a true OR function. A 1 on any input will give a 1 on OUT. Two circuits provide an inverse output OUT. For this circuit, a 1 on any input will give a 1 on OUT and a 0 on $\overline{OUT}$. If all inputs are 0, OUT is 0 and $\overline{OUT}$ is 1.

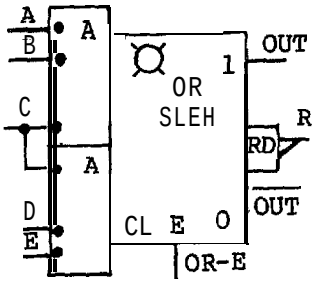
IC3600SLEH - Logic Element



2 CIRCUITS PER CARD

$OUT = A \cdot B \cdot C + D \cdot E \cdot F \cdot C$

$\overline{OUT} = (\overline{A}+\overline{B}+\overline{C}) \cdot (\overline{D}+\overline{E}+\overline{F}+\overline{C})$



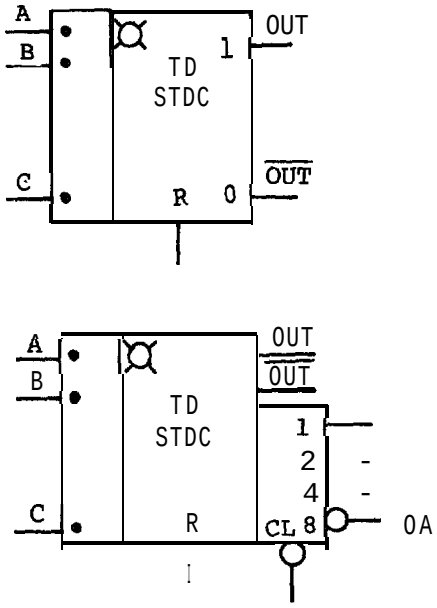
3 CIRCUITS PER CARD

$OUT = A \cdot B \cdot C + D \cdot E \cdot C$

$\overline{OUT} = (\overline{A}+\overline{B}+\overline{C}) \cdot (\overline{D}+\overline{E}+\overline{C})$

The basic SLEH card consists of an OR logic element gated by two AND gates. If either AND gate has all "1" inputs, out will be 1. An OR-E expander input is provided for use with the LLKA. The relay driver provides a closure to common when OUT is a 1.

IC3600STDC - Time Delay



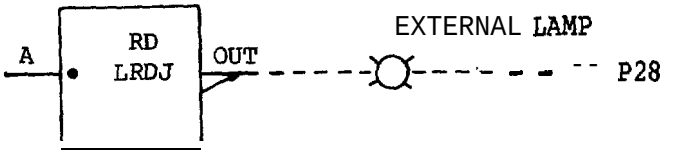
1 CIRCUIT PER CARD

This card provides a time delay function for relatively long periods of one second to 5 minutes. The timing is initiated by all inputs being a 1. If all inputs remain 1, OUT will become a 1 and $\overline{OUT}$ a 0 after the time delay set by adjustment potentiometer. OUT will go from a 1 to a 0 immediately if any input goes to a 0. The LED will be illuminated from the instant timing is initiated until OUT becomes a 1.

The time delay provided is divided into two ranges to give more adjustment resolution. The short range (pin R open) is adjustable from 1 to 25 seconds.

The long range (pin R to N12) is adjustable from approximately 20 seconds to 5 minutes. There are four circuits per card. One circuit provides an eight-count counter for additional time delay. By connecting $\overline{OUT}$ of this circuit to one of the inputs, and OA to another, the timer will cycle eight times before stopping with OA equal 0. This will provide a delay up to 40 minutes.

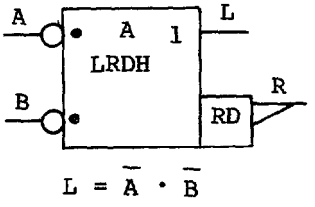
IC3600LRDJ - Relay Driver



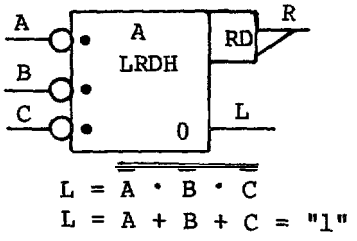
16 CIRCUITS PER CARD

This circuit provides a closure to common in response to a 1 input, an open circuit in response to a 0 input. The circuit is intended to drive a load such as a relay connected between OUT and positive supply voltage not greater than P28.

IC3600LRDH - Relay Driver



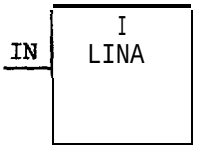
5 CIRCUITS PER CARD



2 CIRCUITS PER CARD

These circuits provide a closure to common for a load between P28 and the R output in response to proper input logic. In addition, they also provide a logic output signal at L in response to the input logic. Circuits 1 thru 5 require both inputs to be a logic 0 to provide closure to common and to generate a logic 1 at the output. Circuits 6 and 7 are similar to 1 thru 5 except that a J-input AND gate is provided instead of a 2-input gate and the logic output is a 0 when closure to common is provided.

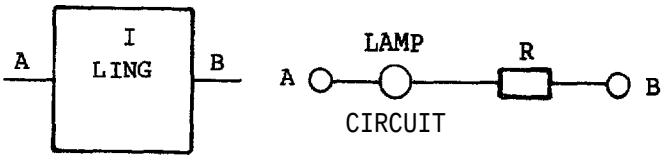
IC3600LINA - Light Indicator



16 CIRCUITS PER CARD

The LINA includes a light amplifier and indicating light. The light will be illuminated when the input is a 1. Inputs for unused circuits must be tied to common to prevent the light from remaining illuminated.

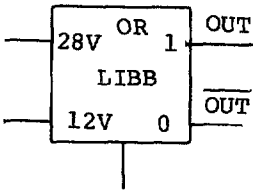
IC3600LING - Indicator



16 CIRCUITS PER CARD

The LING provides indicating lamps to be driven by display drivers, relay drivers, or contact closing. Each circuit is completely independent. This card cannot be operated by a normal logic signal.

IC3600LIBB - Input Buffer

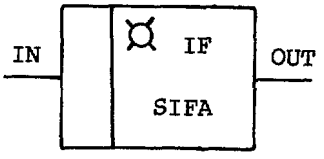


9 CIRCUITS PER CARD

The LIBB converts relay contact inputs to logic levels. Inputs are provided for contact closure to either 28 volts or 12 volts. When contact to supply is closed, OUT is a 1 and $\overline{OUT}$ is a 0. The 28-volt input should not be used off - page. When 28 volts is connected to an off-page contact, the signal must be conditioned by an input filter circuit.

NOTE: The LIBA card functions the same as the LIBB card but has a different rating and timing. DO NOT interchange cards.

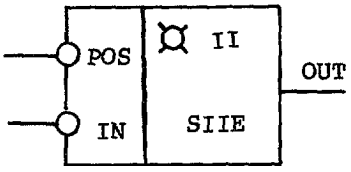
IC3600SIFA Non-isolated Input Filter



9 CIRCUITS PER CARD

The SIFA card is designed to receive a 28 volt signal, filter the signal, and provide drive to the 12 volt input of the LIBB CARD. An LED is on when input is at 28 volts.

IC3606SIIE - Isolated nput Filter

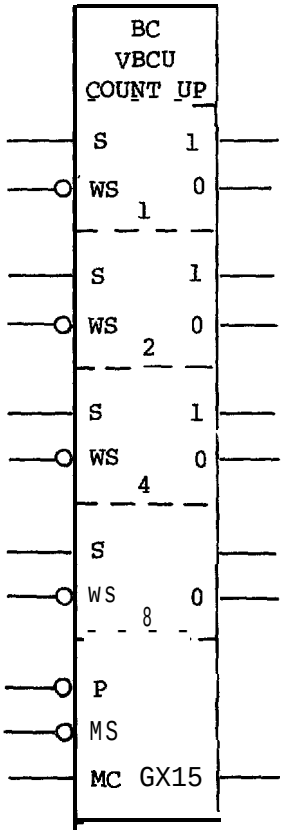


9 CIRCUITS PER CARD

The SIIE provides isolation and filtering of high level (nominal 125 volts DC) signals. The output is designed to drive the 12-volt input of the LIBB card.

IC3600VBCU - Binary Counter

2 CIRCUITS PER CARD



The VBCU card contains two 4-bit counters. The counter will count up one count each time the signal on input P changes from a 0 to a 1. Each bit has a normal and inverse output. A logic 1 on MC will clear all bits. A logic 0 on MS will set all bits whose S input is a 1. Input WS is provided for wired OR expansion of set operation. S must be a 0, or MS must be a 1 for this function. Output GX15 will be a 1 if all four bits are set. The two 4-bit counters may be operated as an 8-bit counter by connecting the 0 output of the lower order bit 8 count to the P input of the second counter.

Annunciator Alarm

The annunciator on the DIRECTO-MATIC II turbine control panel provides the circuitry to annunciate up to 60 alarm conditions. The alarms are latching and will remain, even if the alarm condition is momentary, until cleared by a pushbutton on the panel. An alarm may be annunciated by the sounding of a bell and the flashing of the alarm's associated lights.

The alarm indicators are referred to as "drops". Special features, discussed subsequently, are provided to handle multiple alarms. A block diagram is given in Figure 2-2. It consists of one annunciator master card (VANB) and up to 15 annunciator cards (VANA). Each VANA card contains four circuits; therefore, up to 60 drops are available. The standard system normally consists of 40 drops.

There are three operator pushbuttons associated with the annunciator. (1) The LAMP TEST pushbutton provides a means of checking for burned-out bulbs and faulty lamp drivers. (2) The ALARM RESET pushbutton will silence the alarm bell and cause the flashing light to become steady on. It also enables an additional alarm to come on flashing. (3) The LAMP RESET pushbutton will clear the light if the alarm condition has been removed. An automatic lamp reset is available to automatically reset the lamp when the alarm condition is removed.

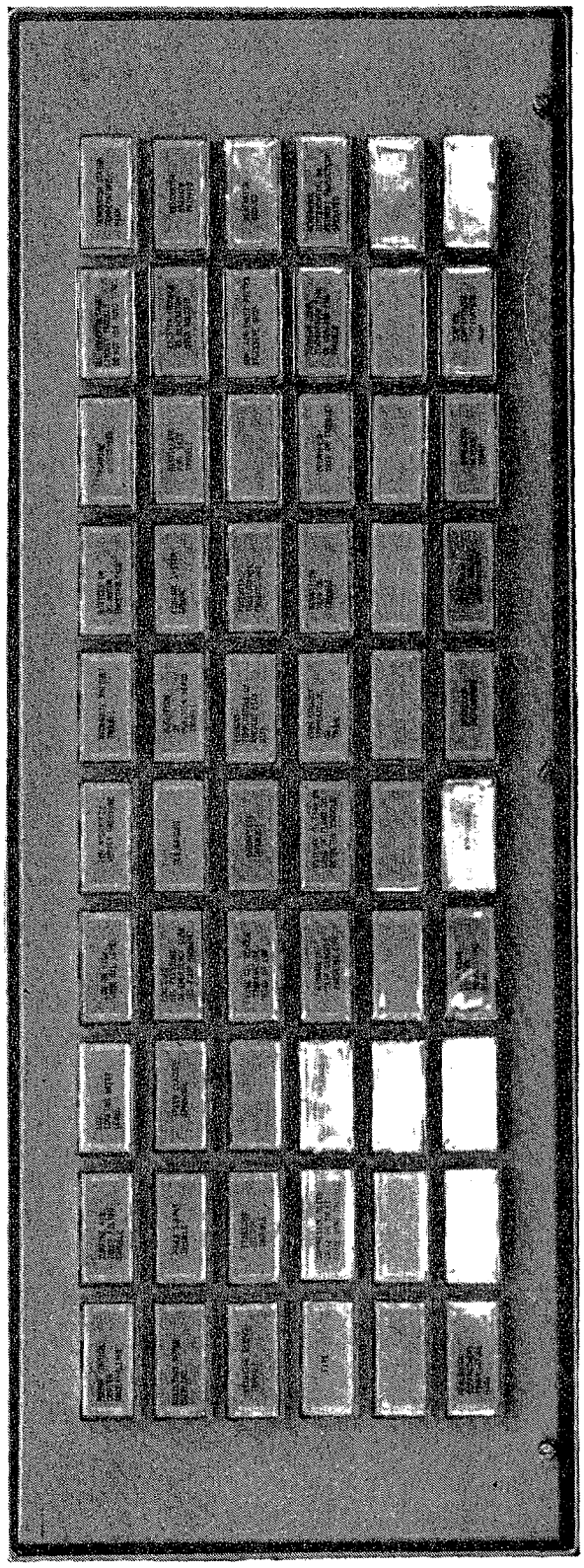
A flow chart (See Figure 2-4) is included to illustrate some typical sequences that may occur in the case of multiple alarm conditions. Alarms must occur at least 36 milliseconds apart for the annunciator to determine which occurred first. If there is less than 36 milliseconds between alarms, both will come on flashing.

The VANB card provides the control signals to the VANA card. (See Figure 2-2). It also contains the driver for relay 30X whose contacts are used to drive the alarm bell. VANB output FL provides the flashing signal to all VANA cards. It is present.at all times.

The signal flow for a typical condition will be given to aid in explaining the annunciator.

Assuming that no alarm condition is present, output AHD of VANB will be a 1. This is a flashing permissive to all VANA cards. Now assume that a fault occurs (a 1 on L). On the VANA card, the flashing signal appears on LT and the corresponding indicator light flashes, and output $\overline{AH}$ becomes a 0. This 0 goes to VANB input $\overline{AH}$ and performs two functions. It places a 0 on HRD and picks up relay 30X which causes the bell to sound, and output $\overline{AHD}$ becomes a 0 which removes the flashing permissive signal.

The first fault remains flashing but any more faults occurring will come on steady. Suppose that the ALARM RESET pushbutton is now depressed, the VANB output ACK will become a 1 which also performs two functions in the VANA card. It removes the flashing signal from LT and replaces it with a steady 0 which makes the light remain steady on. It also causes output AH to become a 1. This 1 on $\overline{AH}$ goes to VANB and silences the bell. It also allows VANB , output $\overline{AHD}$ to become a 1.



(ICF-52018)

Figure 2-1
Annunciator, Front View

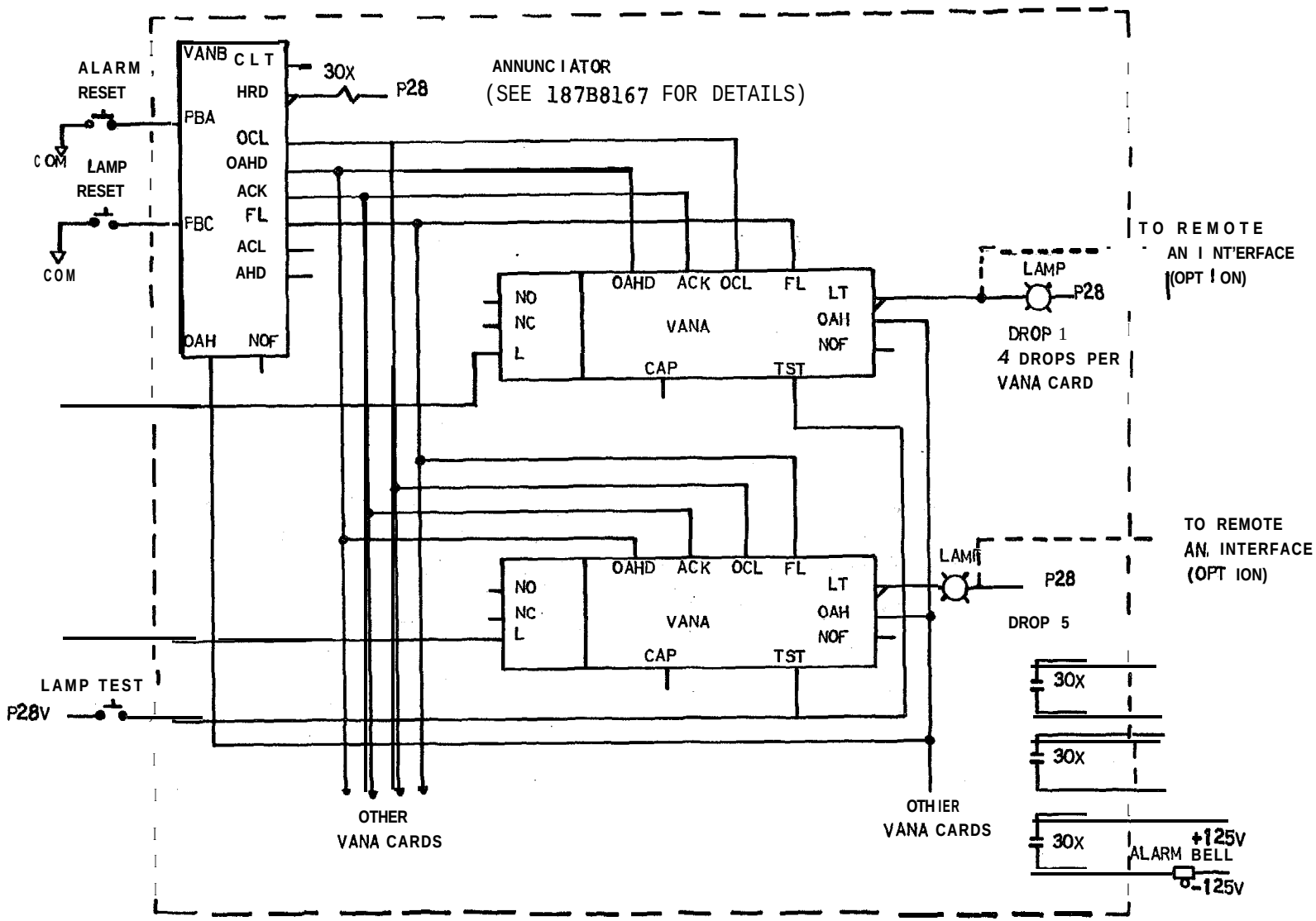
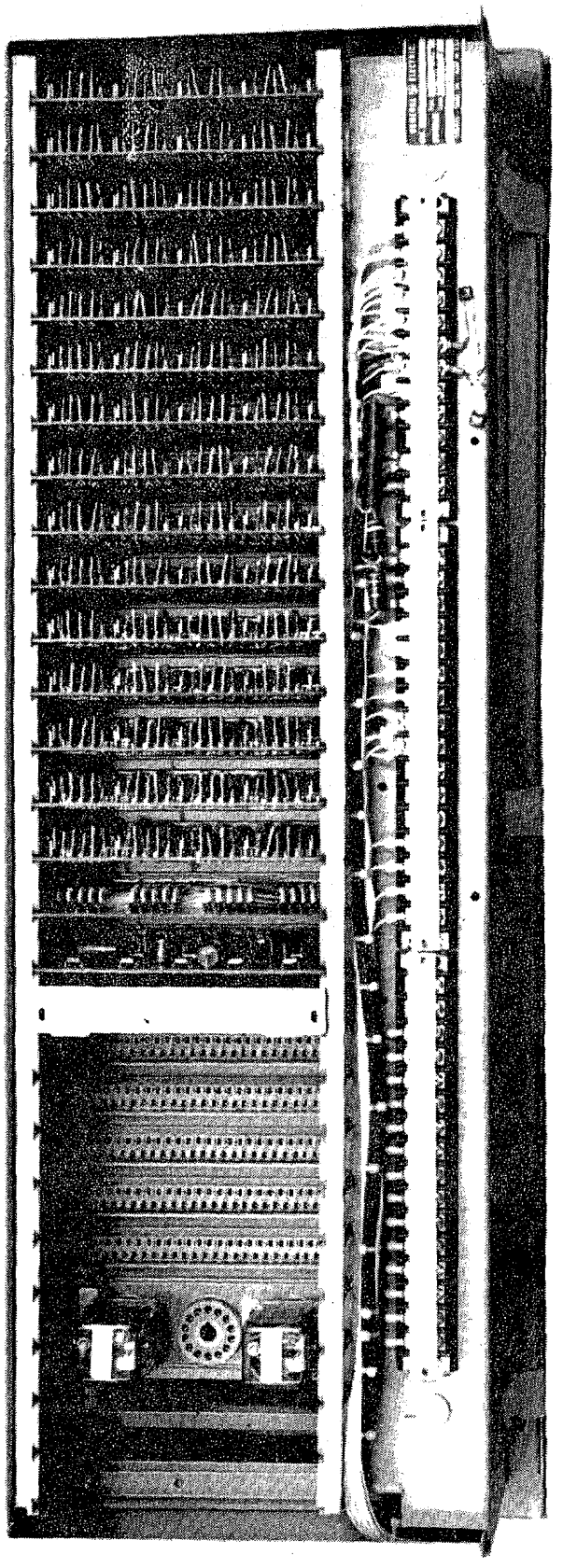


Figure 2-2
Annunciator Block Diagram (Typical Circuitry)

This 1 on AHD is a permissive to all other VANA cards to flash if an additional alarm occurs.

Now, assume that no other alarm condition occurs, and the present fault condition is cleared. Depressing the LAMP RESET pushbutton places a 0 on the CL bus and the LT signal on VANA becomes a 1 which turns the lamp off.



(ICF-50875)

Figure 2-3
Annunciator - Rear View

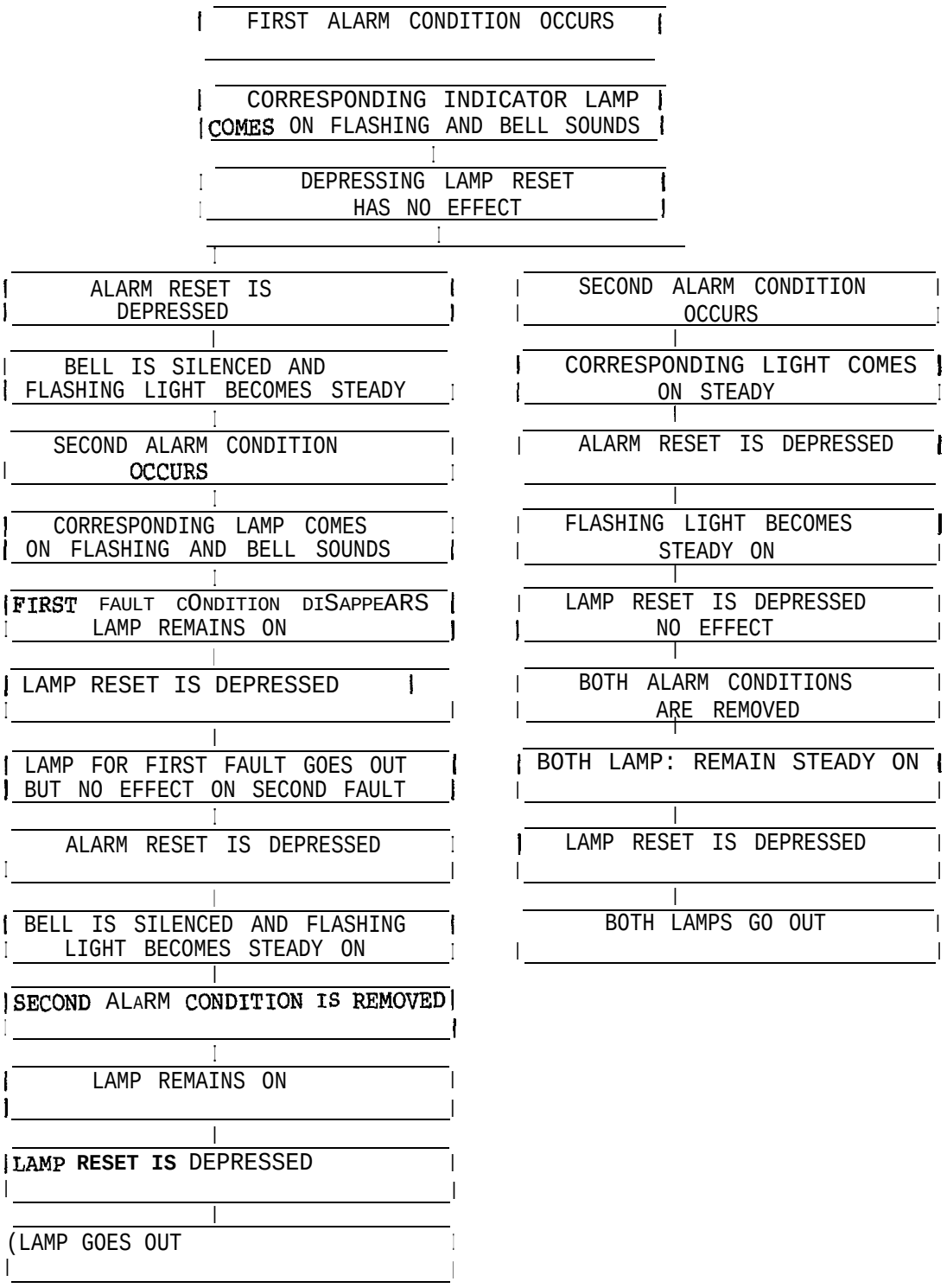
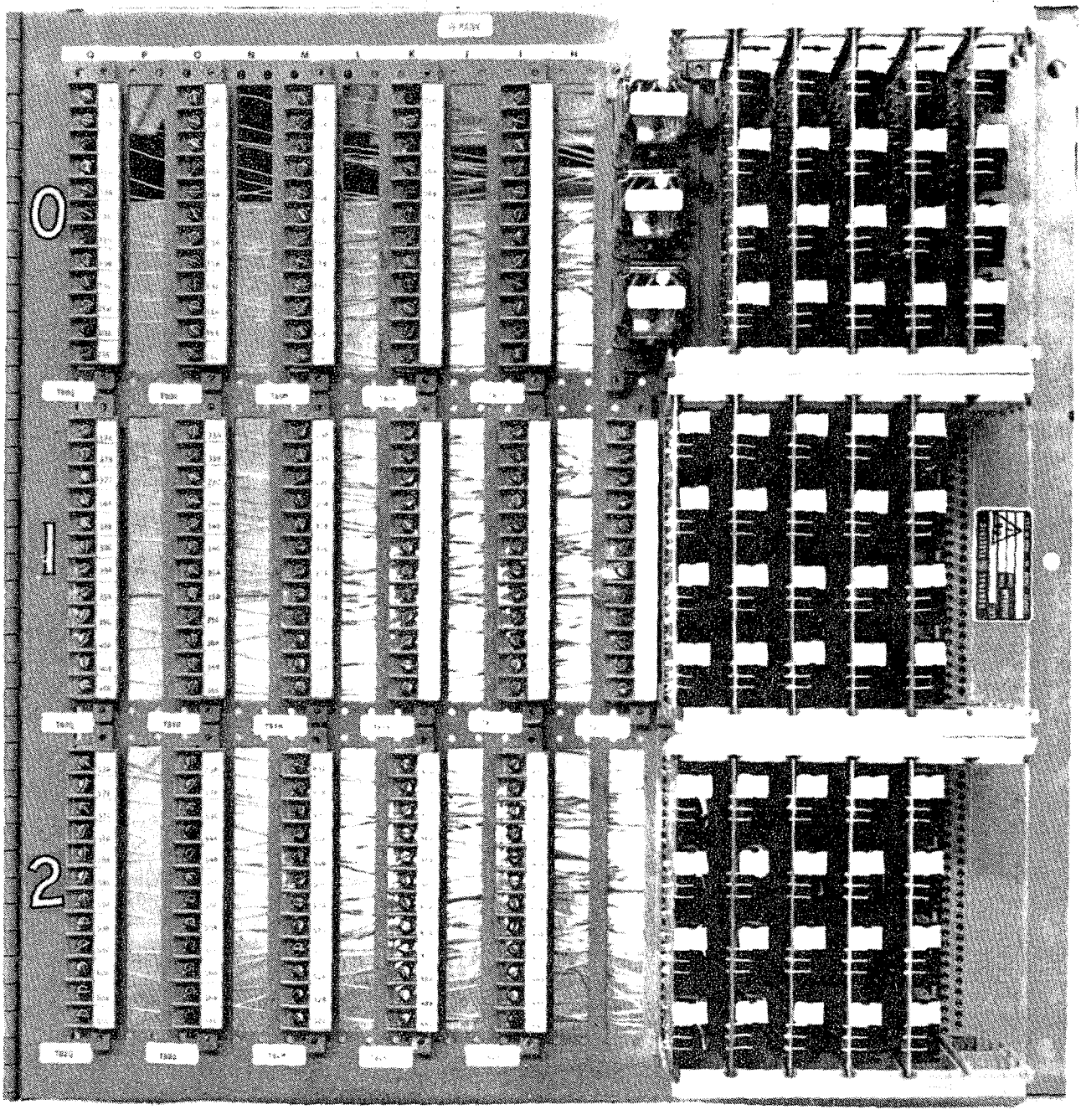


Figure 2-4
Alarm Condition Flow Chart

Remote Annunciator Interface Module (Option)

A remote annunciator interface module is available to provide alarm indications at a remote location (See Figure 2-5). The remote terminal is completely isolated from the turbine control panel. The remote terminal provides its own power supply voltage and alarm indication is via relay contact closure in the remote annunciator interface module. Refer to turbine control panel elementary for connection between remote interface module and local annunciator. Provision is made for three operator pushbuttons to be located at the remote terminal to perform the same function as the three local pushbuttons. The remote pushbuttons pick up relays in the interface module. The contacts of these relays are either paralleled or connected in series (whichever is appropriate) with the local pushbuttons.

When an alarm occurs, the local annunciator operates as discussed previously. The output of the VANA card that drives the alarm lamp is also routed to a corresponding relay in the interface module. Therefore, the relay in the remote annunciator interface module is picked up when the lamp in the local annunciator is on. The relay is normally connected such that it is latching and does not follow the flashing light. However, a flashing remote alarm is available. In the case where the relay is latching when the LAMP RESET pushbutton is depressed the latching path is removed and the relay will drop out if the alarm condition has been removed. A Form "C" interlock is provided for "open to alarm" or "closed to alarm" indications.



(ICF50880)

Figure 2-5
Remote Annunciator Interface Module

Speed Sensors

The speed sensor cards receive analog NHP and NLP signals from the PR/A converter on the SSZB cards, and frequency inputs from magnetic pickups. These sensor cards provide logic signals to indicate that the turbine shaft, or other rotation device has reached a certain speed.

SVSE Card

This card (See Figures 2-6 and 2-7) contains five speed sensors with given (fixed or adjustable) pickup and dropout characteristics. These are:

14HM - minimum speed
14HA - accelerate speed
14HS - operating speed
14HR - zero speed
NMR - speed monitor

The first three sensors operate from a common input signal NHP. The zero-speed sensor operates directly from an electromagnetic pickup. The speed monitor receives a 0 to 10 volt signal and gives a logic output at MR and $\overline{MR}$ for a preset input level.

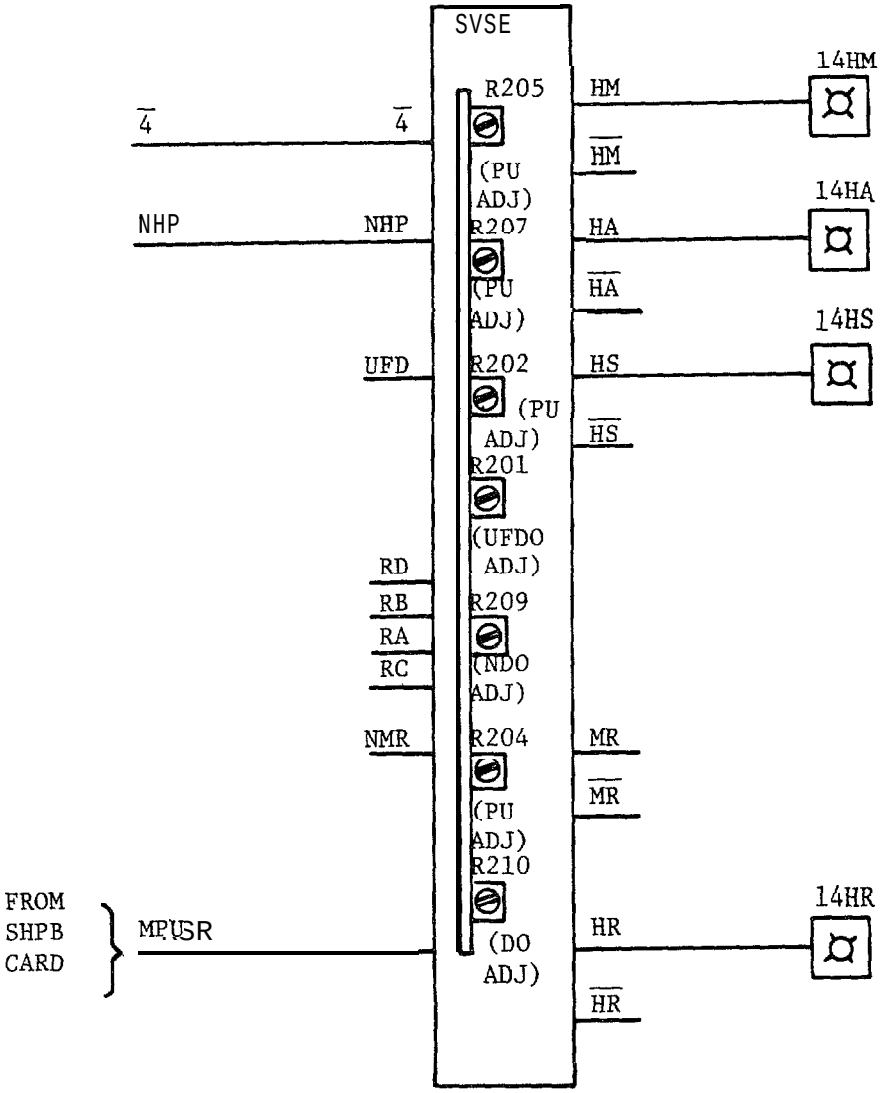


Figure 2-6. HP Speed Sensor Block Diagram

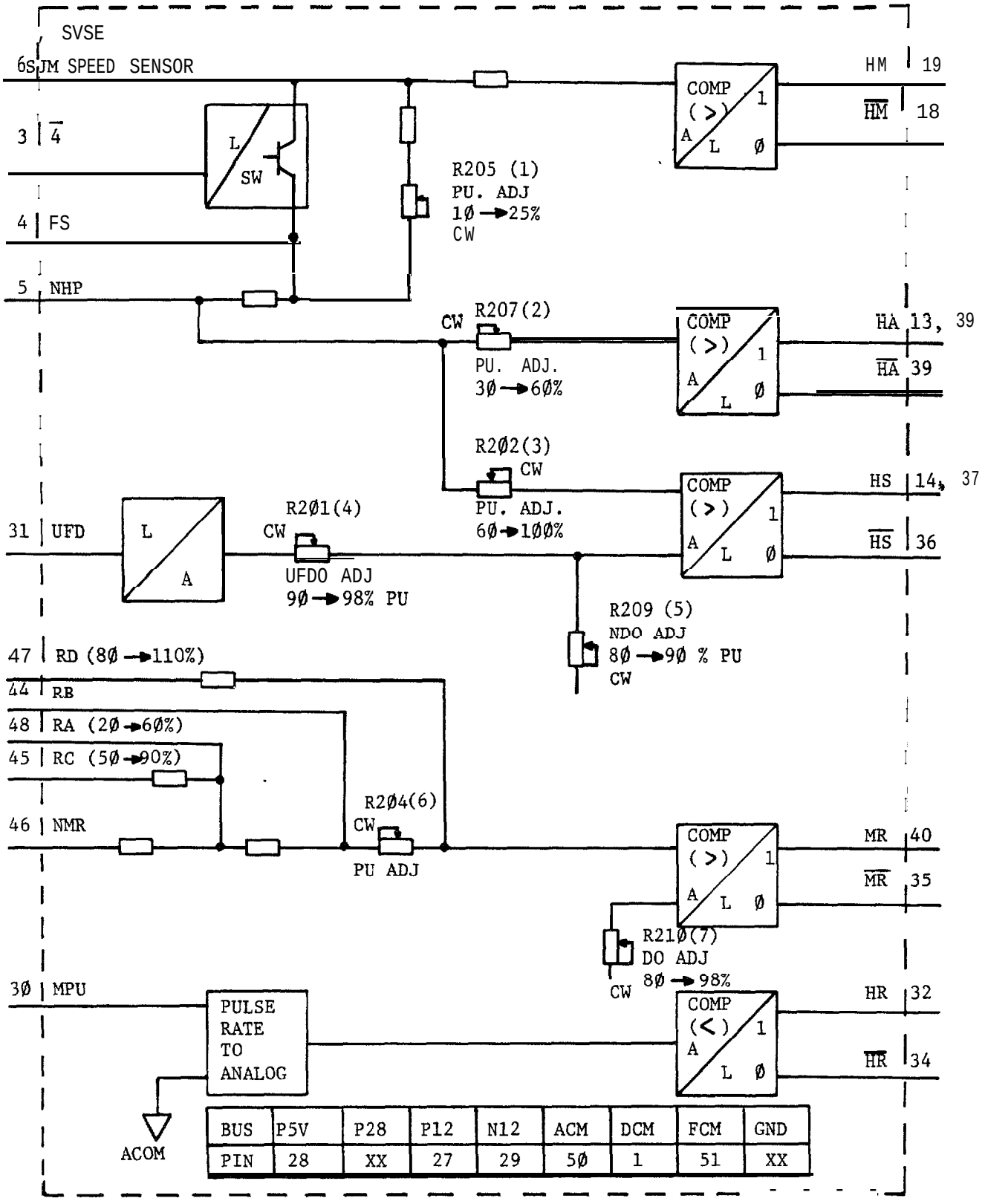


Figure 2-7. HP Speed Sensor Card.

The 14HM pickup and dropout points are programmable by logic inputs on 4, FS and SJM. A 1 on 4 reprograms the pickup point to 2% and dropout to 1.8%. Resistors can be switched between FS and SJM to reprogram the pickup and dropout. The dropout of 14HM is fixed at 90% of pickup. No separate adjustment is provided. The 14HM sensor outputs are on pins HM and $\overline{HM}$.

The 14HA sensor looks at NHP and provides logic outputs on HA and $\overline{HA}$ for a preset level of NHP. The dropout point is dependent on the pickup setting and does not have a separate adjustment. The dropout is 75% of pickup value.

The 14HS sensor looks at NHP and provides logic signals at HS and $\overline{HS}$. The pickup and dropout level is adjustable. The dropout is also programmable with a logic signal on input UFD. For UFD equal 0, dropout is 80 to 98% of pickup. For UFD equal 1, dropout is 90 to 98% of pickup. Potentiometer R201 (UFDO) is used to set dropout level when UFD is a 1. After R201 is set, R209 (NDO) can be set for normal dropout. (See control specification for adjustments.)

The speed monitor sensor has adjustable pickup ranges that are jumper selectable. The pickup ranges and jumpers are:

RANGE (% Speed)	JUMPER
20-60	RA to RB
50-90	RC to RB
80-110	RD to RB

The 14HR sensor receives a frequency on MPU proportional to RPM and generates an analog signal. This analog signal is compared to a fixed reference and gives a 1 on HR if analog signal is less than reference. Unlike the other sensors, 14HR is picked up at zero speed and drops out as speed is increased.

ADJUSTMENT	POTENTIOMETER	RANGE (% OF SPEED)
14HM PU	R205	10-25
14HA PU	R207	30-60
14HS PU	R202	60-100
14HS DO	R209	90-98
14HS UFDO	R201	80-98

SLPA Card

The SLPA card (See Figures 2-8 and 2-9) contains three speed sensors with given (fixed or adjustable) pickup and dropout characteristics. These are:

14LS - Operating Speed
14SR - Zero Speed
14ST - Speed Monitor

The 14LS sensor operates from the input signal, NLP. The 14LS pickup is adjustable by jumper selected ranges. These ranges are:

RANGE (% SPEED)	JUMPER
20-60	RA to RB
50-90	RC to RB
80-110	RD to RB

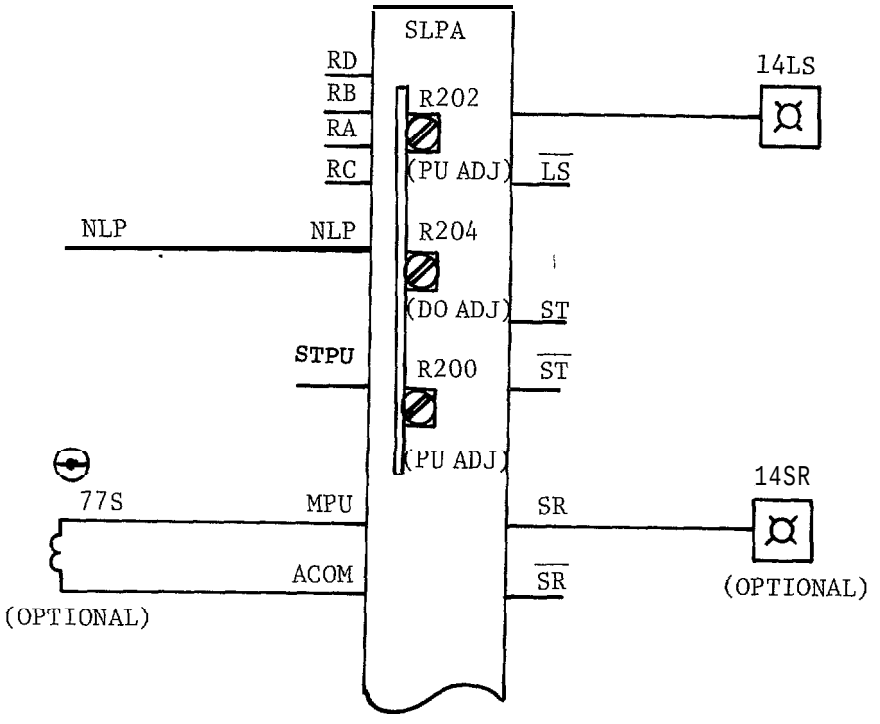


Figure 2-8
L.P. Speed Sensor Block Diagram

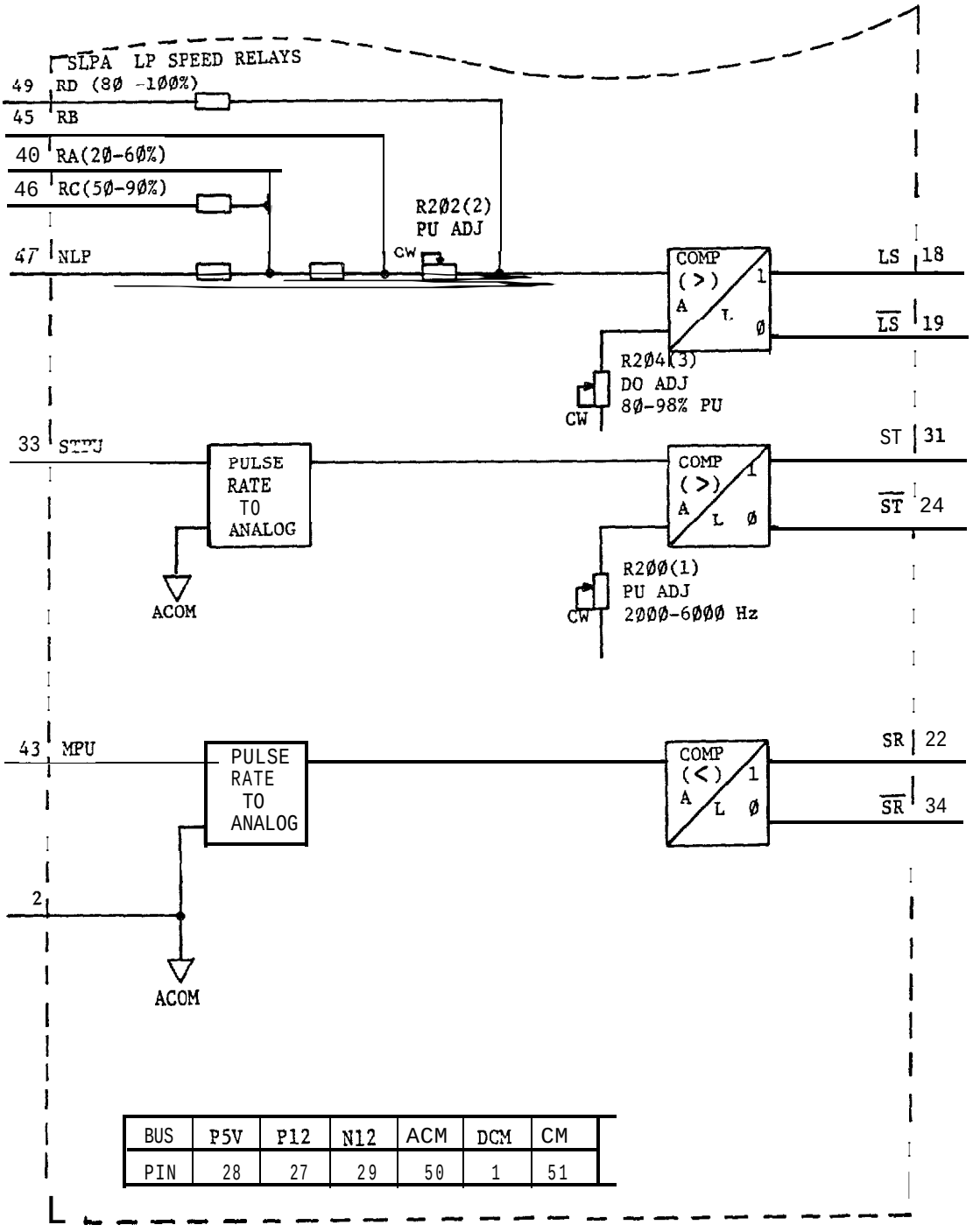


Figure 2-9
LP Speed Sensor Card

Potentiometer R202 is used to set the exact pickup value within the selected range. The dropout value is adjustable from 80 to 98% of the pickup value, using potentiometer R204. This sensor provides logic signals at points LS and LS for use in sequencing turbine operation.

The 14SR sensor inputs a frequency proportional to RPM, on terminal MPU, and generates a corresponding analog signal. This signal is then compared to a fixed reference. Terminal SR goes to a 1 if the signal is less than the reference. This sensor differs from the others in that it is picked up at zero speed and drops out as speed increases.

The 14ST sensor also inputs a frequency proportional to RPM, on terminal STPU, and generates an analog signal. This signal is then compared to an adjustable reference. Terminal ST goes to a 1 if the signal level exceeds the reference. The reference is adjustable from 2000 to 6000 Hz, using potentiometer R200. The dropout value is fixed at less than 90% of the pickup value.

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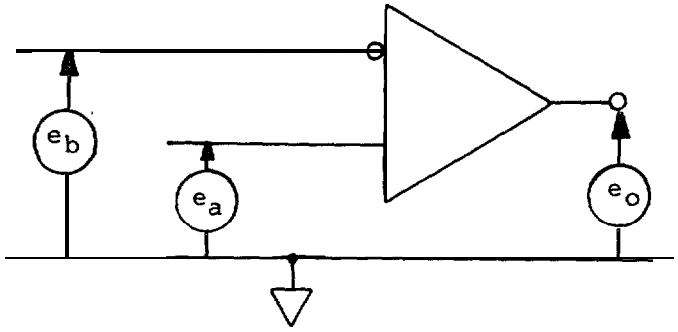
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CONTROL CIRCUITS

Operational Amplifiers

The DIRECTO-MATIC II SPEEDTRONIC control uses the operational amplifier (normally referred to as an op amp) extensively, therefore, to understand the control panel, one must have a basic understanding of the op amp. A brief discussion of the op amp enhanced with several examples is given subsequently. The intent of this text is not to present a rigorous analytical description of the op amp, but an introduction to the op amp in simplified terms. It is hoped that from this ground work, the user can follow the turbine control elementary diagram and gain an understanding of the SPEEDTRONIC control.

The operational amplifier is generally considered a high gain, high input impedance, low output impedance amplifier capable of amplifying DC as well as AC signals. The standard symbol for the op amp is as follows:



The input with the circle is the inverting input and the other input is non-inverting. The output is:

$$e_o = K(e_a - e_b)$$

where K is the gain of the amplifier. As mentioned previously, the op amp has a very high gain (typically 10^5 in DIRECTO-MATIC II), therefore, a very small difference between e_a and eb Will give a large e_o.

However, e_o is limited by the supply voltages of the op amp. When e_o reaches approximately the supply voltage, it is said to be in saturation. Typical supply voltages in DIRECTO-MATIC II are +12 volts.

Using a +12 volt supply and a gain of 10^5, the value of e_a - eb for saturation is:

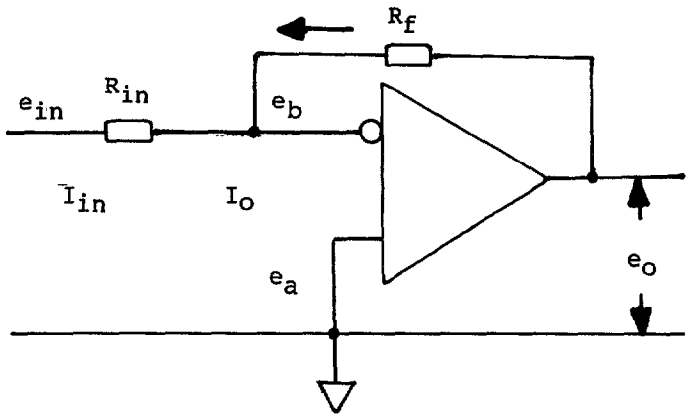
$$12 = 10^5 \text{ (ea - eb)}$$

$$(e_a - e_b) = \frac{12}{10^5} = 12 \times 10^5$$

$$(e_a - e_b) = 120 \text{ Microvolts}$$

Because of this higb gain, the op amp is practically useless when used alone; however, with negative feedback, its uses are limited *only* by the designer's imagination.

The most common circuit of the op amp is as follows:



In this case, the non-inverting input e_a is tied to common. The useful, operating region of the op amp is when it is not in saturation; therefore, e_b is approximately equal to e_a. In this case, e_a is tied to common; therefore, e_b is approximately zero volts. Operating in this mode, eb is often referred to as virtual common. To determine e_o as a function of e_{in}, the currents at eb are summed together. This junction is often referred to as the summing junction.

$$I_{in} + If - Io = 0$$

The op amp has a very high input impedance and I, can be considered zero. Then:

$$I_{in} + If = 0$$

The op amp has a very low output impedance and will supply the necessary current to make the above equation true.

I_{in} and If can be found as follows:

$$I_{in} = \frac{E_{in}}{R_{in}} \qquad I_f = \frac{e_o}{R_f}$$

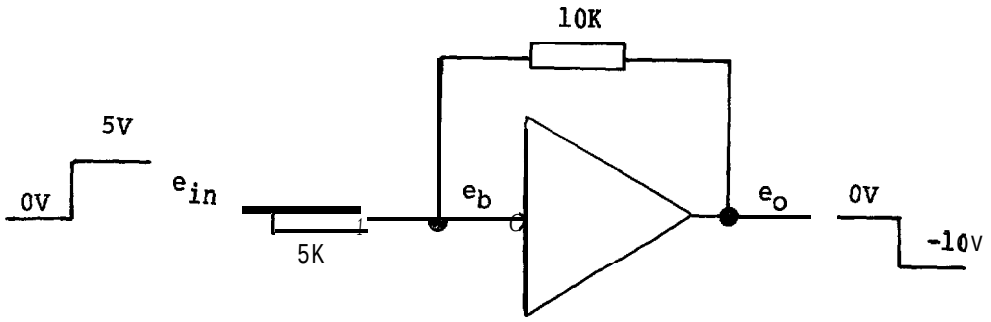
Substituting gives:

$$\frac{e_{in}}{R_{in}} + \frac{e_o}{R_f} = 0$$

$$e_o = - \frac{e_{in}}{R_{in}} \quad R_f$$

$$e_o = - e_{in} \frac{R_f}{R_{in}}$$

This equation applies as long as the op amp is not in saturation. If it goes into saturation, e_b will no longer be at virtual common and the equation loses validity. To further illustrate the inverter, the circuit is repeated below with values assigned. The connecting of e_a to common on the symbol is often omitted with the understanding that it exists.

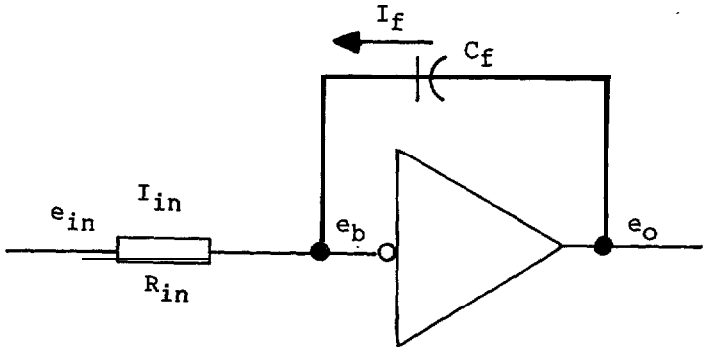


$$e_o = - e_{in} \frac{10K}{5K} = -2e_{in}$$

If e_{in} is a +5V signal, then:

$$e_o = -2(5)V = -10V$$

The second op amp circuit discussed is the integrator. The following is an integrator circuit:



Again, summing the currents at e_b gives:

$$I_{in} + I_f = 0 \quad \text{Where}$$

$$I_{in} = \frac{e_{in}}{R_{in}} \quad \text{And}$$

$$I_f = C_f \frac{d(e_o)}{dt}$$

$$\frac{e_{in}}{R_{in}} + \frac{d(e_o)}{dt} C_f = 0$$

$$C_f \frac{d(e_o)}{dt} = -\frac{e_{in}}{R_{in}}$$

$$d(e_o) = -\frac{e_{in}}{R_{in}C_f} dt$$

$$d(e_o) = -\frac{e_{in}}{R_{in}C_f} dt$$

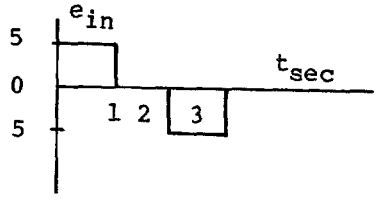
$$e_o = \frac{1}{R_{in}C_f} - e_{in}\, dt$$

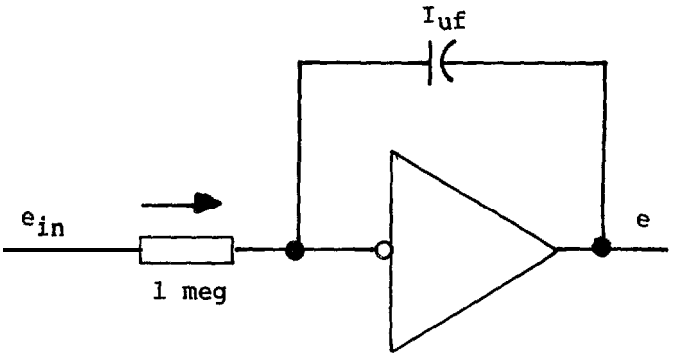
If e_{in} is a constant or step input, then:

$$e_o = -\frac{e_{in}}{R_{in}C_f} t \quad \text{OR}$$

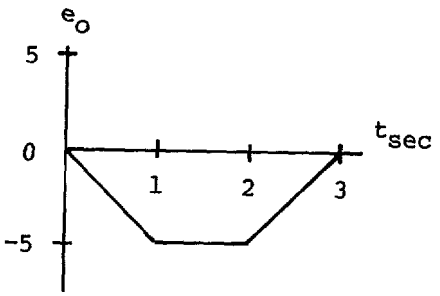
$$e_o = -\frac{I_{in}}{C_f} t \quad t = \text{Seconds}$$

The integrator circuit is redrawn below with values assigned:



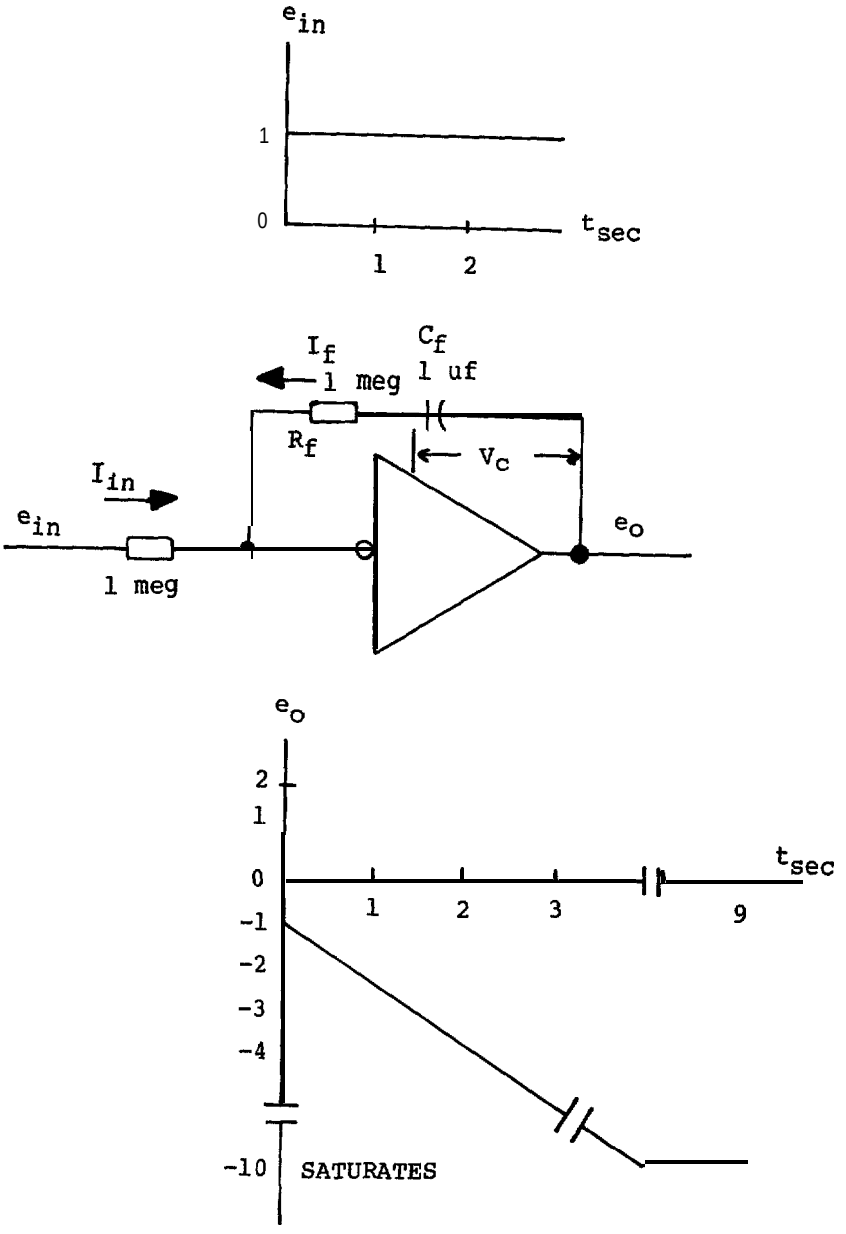


$$e_o = - \frac{5}{(10^6)\ (10^{-6})}\ t = -5t$$



As can be seen from the curves adjacent to the circuit, in response to a step input, the output increases on a ramp. If the input returns to zero, the capacitor remains charged and e_c remains at the level prior to input going to zero. If a step input of opposite polarity is applied, the capacitor will begin discharging.

For the next circuit, the input-output relationship will not be developed mathematically; instead an intuitive approach will be presented. The circuit is often referred to as a proportional plus integral.



To begin, assume that e_{in} and e_o are zero (Vc = 01. Now assume a step input of 1 volt is applied at e_{in}:

$$I_{in} = \frac{e_{in}}{R_{in}} = \frac{IV}{IM} = 1 \text{ ua}$$

$$If = - lin = - 1 \text{ ua}$$

Now e_o is equal to the voltage across Rf and Cf. The voltage across R_f is:

$$V_t = I_f R_f = - IinRf$$

$$V_t = 1 \text{ volt}$$

The voltage across a capacitor subjected to a constant current is:

$$V_c = \frac{I}{C} t$$

Therefore, if e_{in} is a step input

$$e_o = - IinRf - \frac{*in}{C} t$$

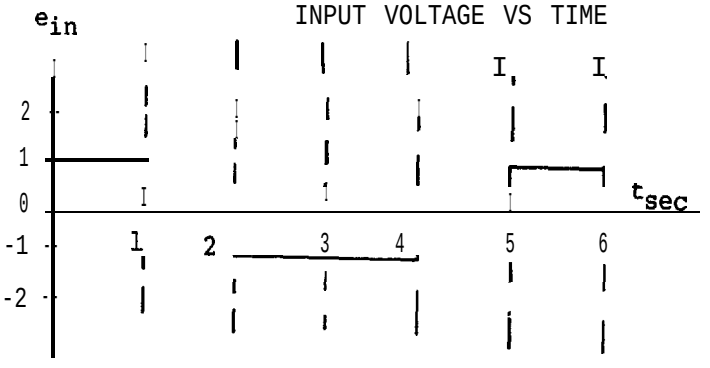
at the instant that the step is applied (t=0)

$$e_o = - IinRf$$

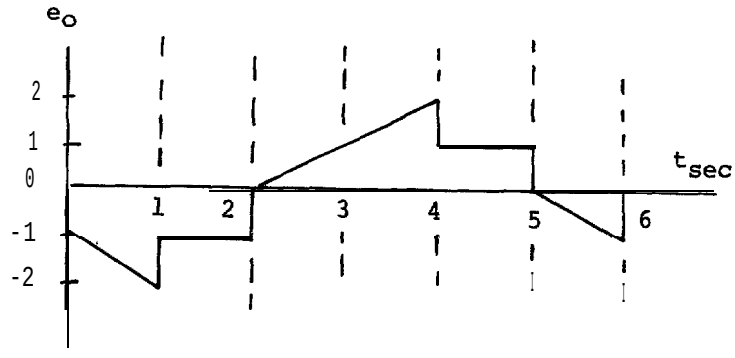
$$e_o = -(lua)(1 \text{ meg})$$

$$e_o = -1 \text{ volt}$$

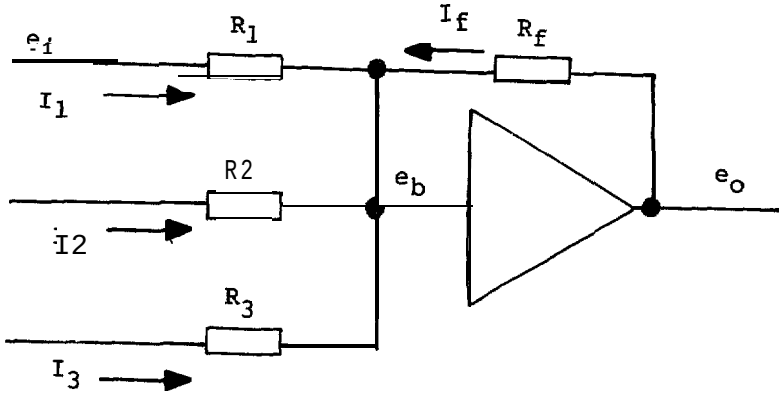
As time increases, the capacitor charges up as shown in the graph adjacent to the circuit. Following are curves showing output response to a varying input voltage.



OUTPUT VOLTAGE VS TIME



The next op amp circuit discussed is the summer. The circuit is as follows:



Summing the currents 'at eb yields:

$$I_1 + I2 + I3 + If = 0$$

$$\frac{e_1}{R_1} + \frac{e_2}{R2} + \frac{e_3}{R3} + \frac{e_o}{R_f} = 0$$

$$\frac{e_o}{R_f} = -\left(\frac{e_1}{R_1} + \frac{e_2}{R_2} + \frac{e3}{R_3}\right)$$

$$e_o = -\left(\frac{R_f e_1}{R_1} + \frac{R_f e_2}{R2} + \frac{R_f e_3}{R3}\right)$$

As implied by its name, the circuit sums the inputs with each input weighted by the ratio of its input resistor to the feedback resistor, R_f.

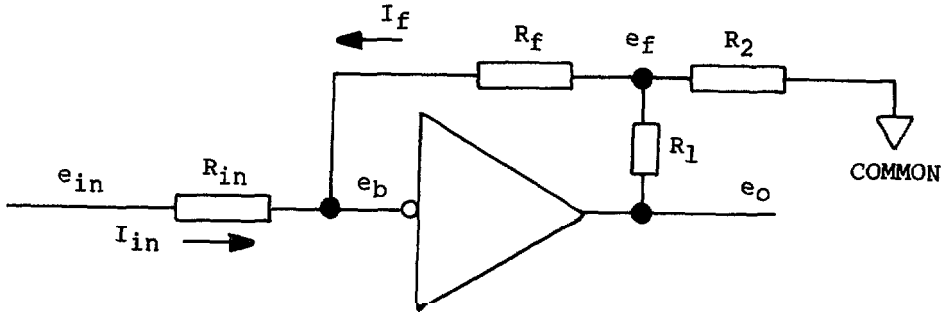
The multiple input configuration can also be applied to the integrator.

Referring back to the integrator equation of a step input:

$$e_o = -\frac{I_{in}}{Cf}\ t$$

For multiple inputs, the algebraic sum of the input currents is substituted for I_{in}.

Occasionally the feedback voltage is not derived directly from e_o. Following is an example of such a circuit:



Again, summing the currents at eb gives:

$I_{in} + If = 0$ (Summing junction equation) where:

$$I_{in} = \frac{e_{in}}{R_{in}}$$

$$If = \frac{e_f}{Rf}$$ (If Equation) But:

$$e_f = e_o\left(\frac{\frac{R_f\ R2}{Rf + R_2}}{R_1 + \frac{RfR2}{Rf + R_2}}\right)$$ Let: $$Req = \frac{Rf\ R2}{R_f + R2}$$ Then:

$$e_f = e_o\left(\frac{R_{eq}}{R_1 + R_{eq}}\right)$$

Substituting this in If equation gives:

$$If = \frac{e_o}{R_f}\left(\frac{R_{eq}}{R_1 + R_{eq}}\right)$$

The summing junction equation now becomes:

$$\frac{e_o}{R_f}\left(\frac{R_{eq}}{R_1 + R_{eq}}\right) = -\frac{e_{in}}{R_{in}}$$

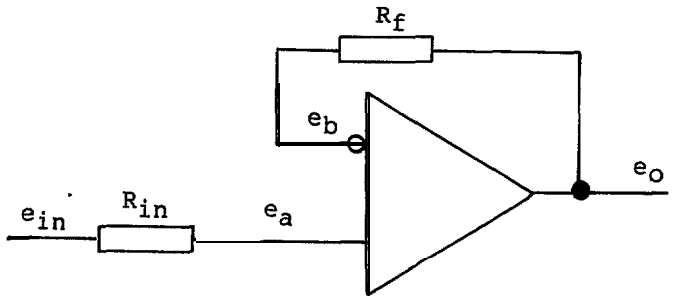
$$e_o = -e_{in}\frac{Rf}{R_{in}}\left(\frac{R_1 + R_{eq}}{R_{eq}}\right)$$

Up to this point, we have only discussed the inverting mode of the op amp. Before going into the non-inverting mode, two important features of the op amp discussed previously will be repeated.

1. The op amp has a very high input impedance, therefore, it is safe to assume that it draws zero input current.

2. A very small difference between e_a and eb causes the op amp to go into saturation, therefore, in the active region, it is assumed that e_a equals e_b.

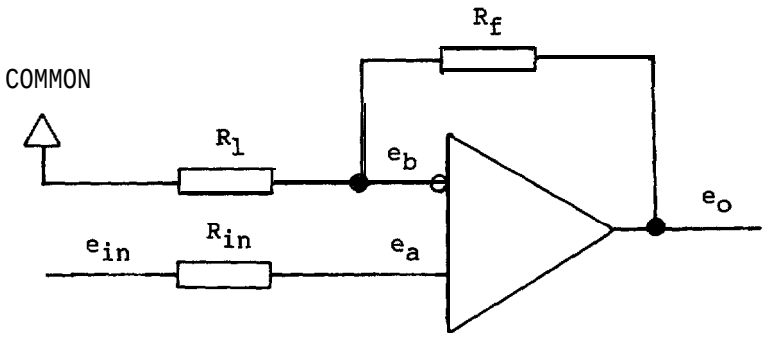
The first non-inverting op amp circuit to be discussed is the voltage follower. The circuit appears as follows:



Since the op amp input draws no current, e_a is equal to e_{in}. Per feature number 2, e_b equals e_a and no current is drawn by the op amp; therefore, e_o equals eb which, in turn, is equal to e_{in}. The input-output equation is:

$$e_o = e_{in}$$

To obtain a gain greater than unity, the following circuit is used:



As in the previous circuit, eb equals ea which equals e_{in} because no current flows into the op amp. eb is also equal to:

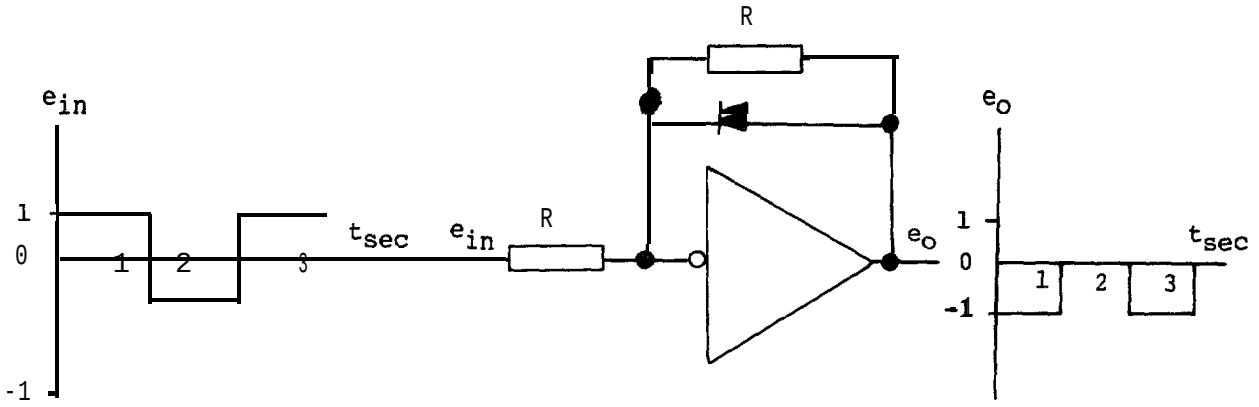
$$e_b = e_o \frac{R_1}{R_1 + Rf} \quad \text{OR}$$

$$e_o = \left(\frac{R_1 + R_f}{R_1}\right) e_{in}$$

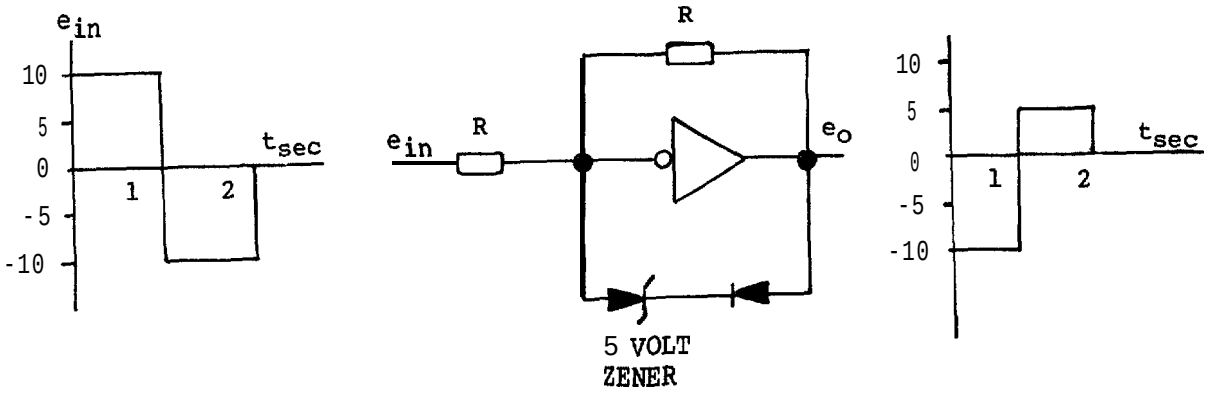
$$e_o = \left(\frac{R_1 + R_f}{R_1}\right) e_{in} \qquad e_b = e_{in}$$

$$e_o = \left(1 + \frac{R_f}{R_1}\right) e_{in}$$

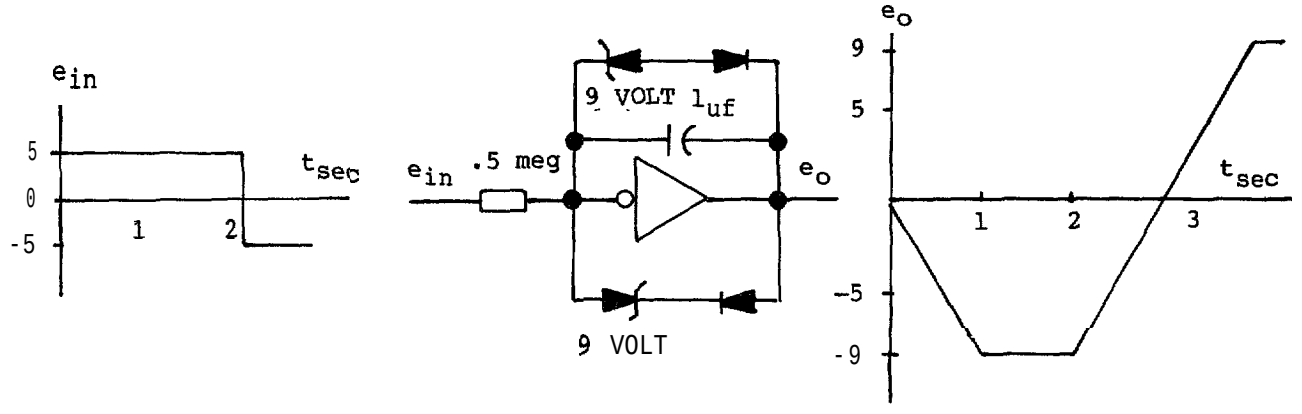
As seen from this equation, the gain of this circuit is always greater than unity. It is often desired to clamp the op amp output at some value. The following circuits illustrate the use of diodes and zener diodes to clamp the output.



The above circuit limits the positive swing to approximately zero.



Above circuit limits positive swing to approximately 5 volts.



The above circuit limits positive and negative swings to 9 volts. This circuit is often used to prevent an op amp from saturating and summing junctions from rising above common.

Start-Up

The turbine control panel contains start-up, load turbine speed and temperature control loops which modulate variable control electrons (VCE). VCE serves as a reference to the fuel control loops which regulate fuel flow to the turbine. On two shaft turbines, a nozzle control loop modulates nozzle control electrons (NCE) to control H.P. speed. A typical control system elementary diagram for a single shaft turbine is shown in Figure 3-1, and for a two shaft turbine in Figure 3-2.

The start-up control is an open loop system that operates in conjunction with the sequential control circuitry to protect preset fuel levels to the turbine during start-up. In addition to the above, it also sets the minimum and maximum limits on VCE after the start-up period is over.

A typical VCE start-up curve is shown in Figure 3-3. A typical control system is shown in Figure 3-4.

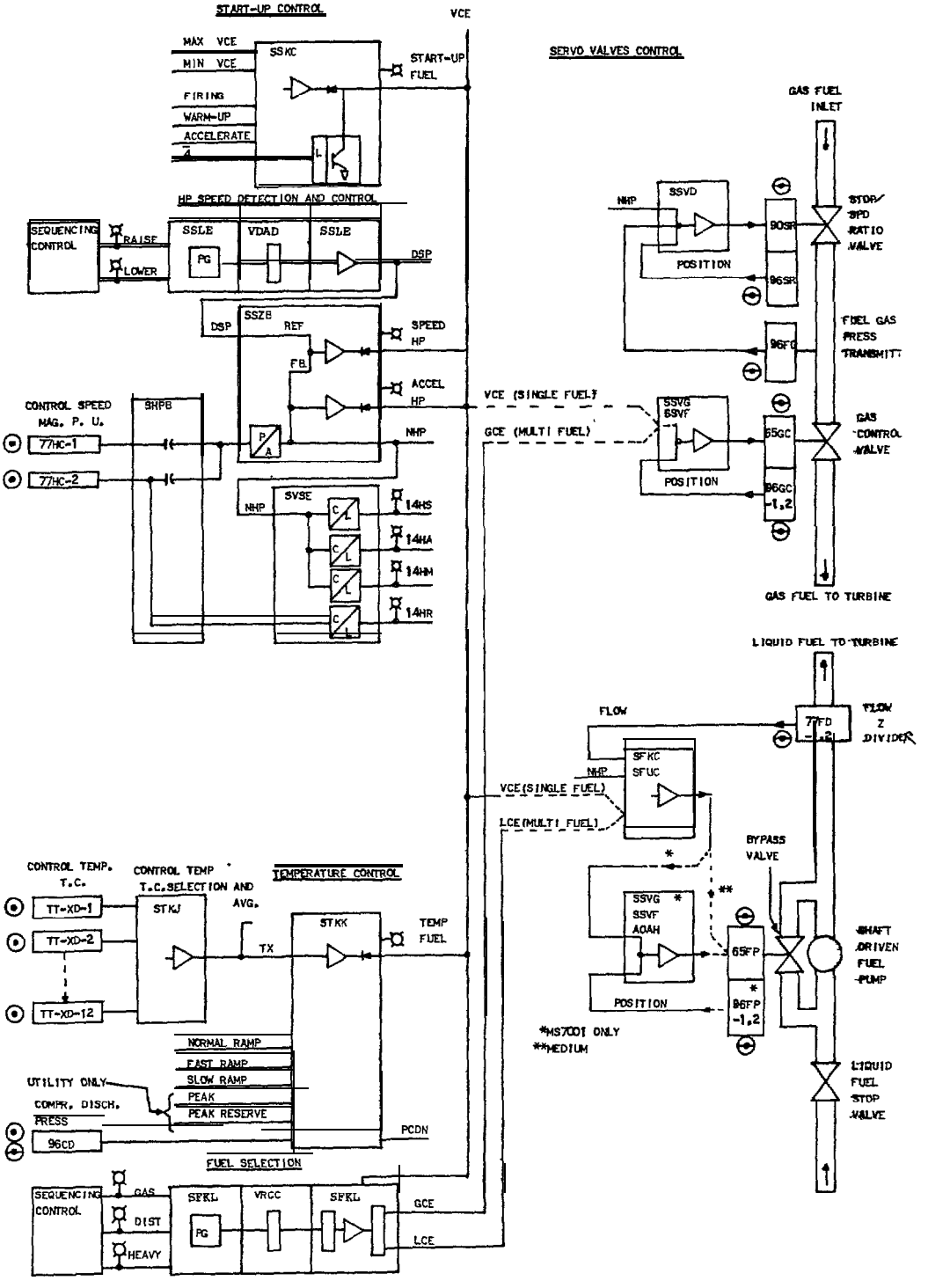


Figure 3-1
Typical Control System-Single Shaft Turbine

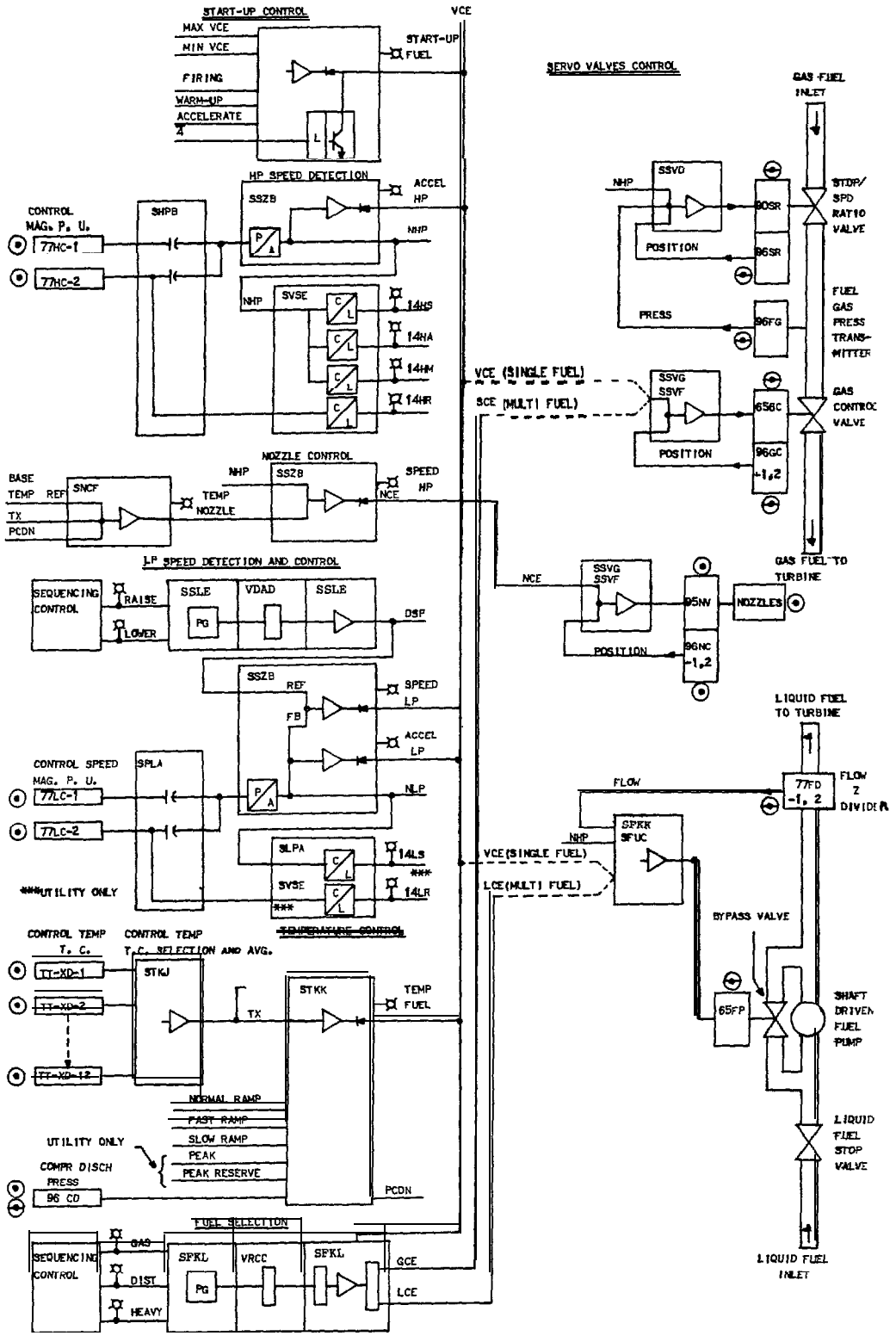


Figure 3-2
Typical Control System P-Shaft Turbine

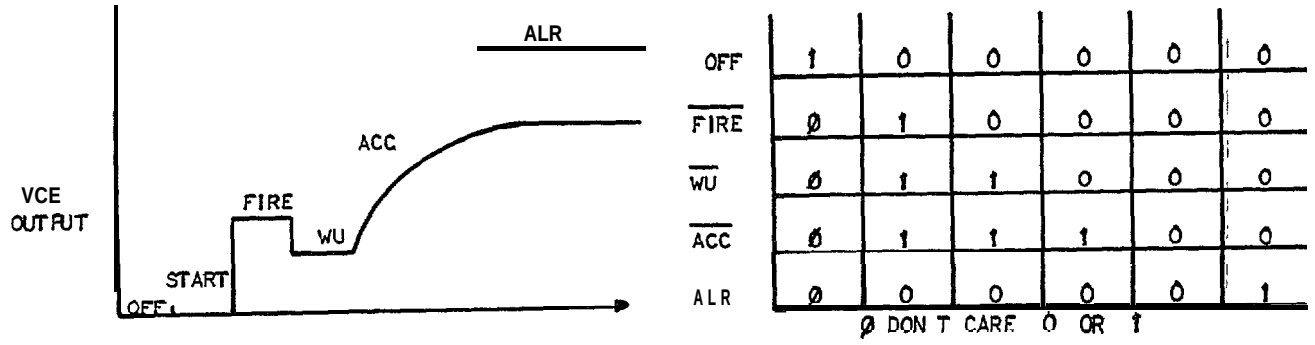


Figure 3-3
Open Loop Start-Up Profile

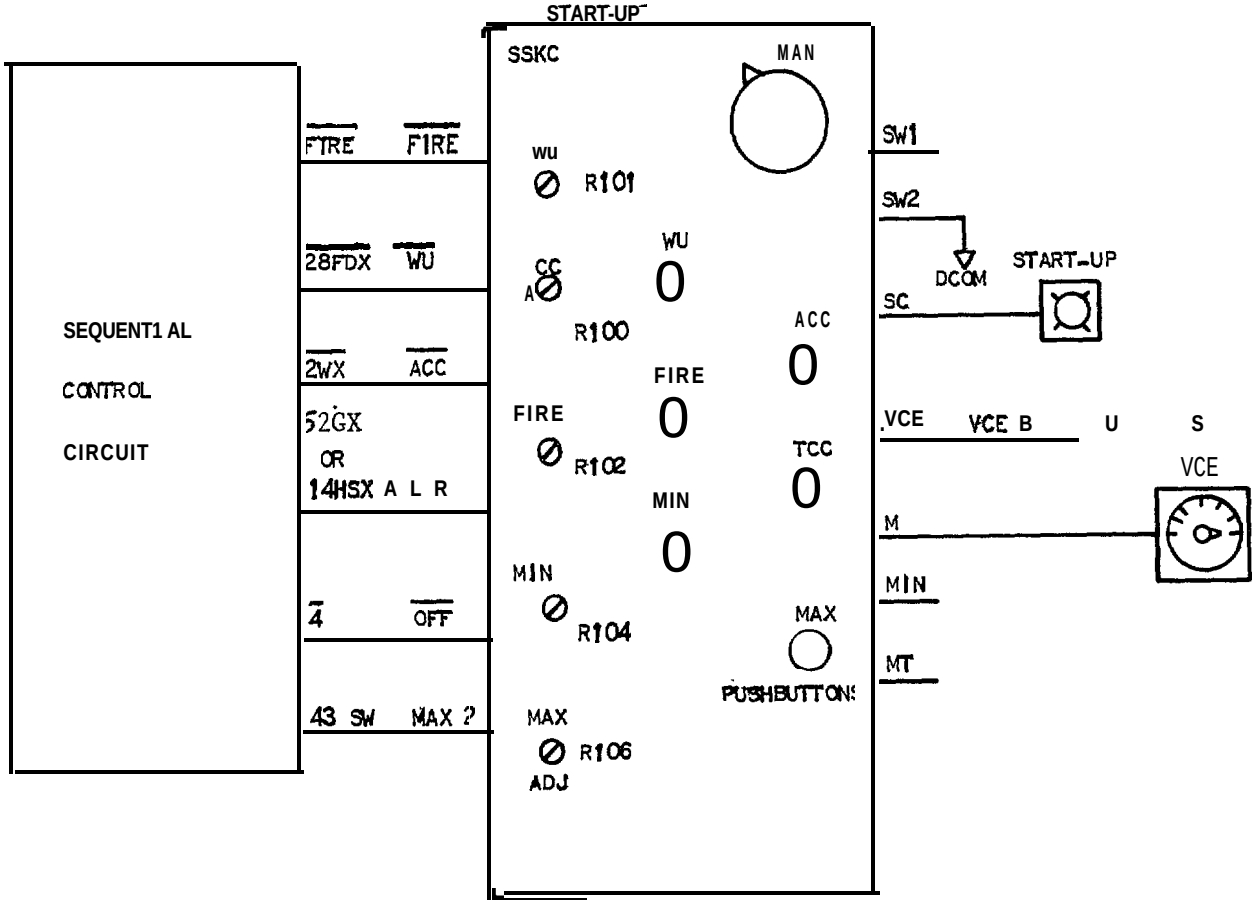


Figure 3-4
Typical Start-Up Control System

SSKC Card

(See Figure 3-5) A logic 1 on input OFF or input FIRING will Clamp output VCE to zero volts. If OFF is a 0, a 0 occurring on FIRING will remove the clamp on VCE. To facilitate a description of the card function, assume that OFF is a 0, ALR is a 0, and FIRING, WU and ACC are 1. Under these conditions, VCE is clamped to zero volts. Now assume that a 0 occurs on input FIRING. VCE will jump to FIRE level as set by R102. A 0 occurring now on WU will cause VCE to drop from FIRE to WARM-UP level which is set by R101. VCE will remain at this level until ACC becomes 0 or VCE *is* pulled down by some other control loop. Upon receiving a 0 on ACC, VCE will be released to increase on an exponential curve until a limit, set by R100, is reached. This limit on VCE will remain until input ALR becomes a 1. A 1 on ALR will remove the limit and allow VCE to increase to a maximum limit set by R106.

A minimum VCE circuit monitors VCE and provides a negative bias at output MIN when VCE decreases below a MIN LEVEL set by R104.

Typically, the bias out of MIN is connected to the speed and acceleration controls and prevents them from driving VCE below the MIN level. A test circuit is provided that supplies a positive bias out of MT when MIN pushbutton is depressed. This can be used as an input to the speed card to simulate speed feedback which tells the speed card to decrease VCE. The MIN circuit should then prevent VCE from going below the preset level.

Pushbuttons (MAX, MIN, FIRE, WU, ACC) are provided on the card for testing and adjusting their corresponding preset levels. Depressing either the MIN, FIRE or WU pushbuttons will set VCE to its corresponding voltage level (See ranges below). Depressing the MAX pushbutton will set VCE at its maximum value, only if the "43" switch is in the OFF or CRANK position (See range below). Depressing the ACC pushbutton will allow VCE to rise exponentially to its accelerate limit. Depressing the TCC and ACC pushbuttons simultaneously will reduce the exponential time constant by about a factor of 10. With any of the previous ranges selected, the actual value of VCE can be adjusted by the potentiometers listed below:

Potentiometer RH1 is provided for manual control of VCE. VCE can be adjusted from MAX VCE to zero using RH1 if some other control loop is not pulling VCE down. A limit switch on the potentiometer ties pins SW1 and SW2 together when RH1 is not fully counter-clockwise.

ADJUSTMENT	POTENTIOMETER	RANGE (Volts)
MAX	R106	8.00 to 10.50
MIN	R104	2.75 to 4.00
FIRE	R102	3.25 to 5.00
WU	R101	2.50 to 4.00
ACC	R100	3.50 to 6.50

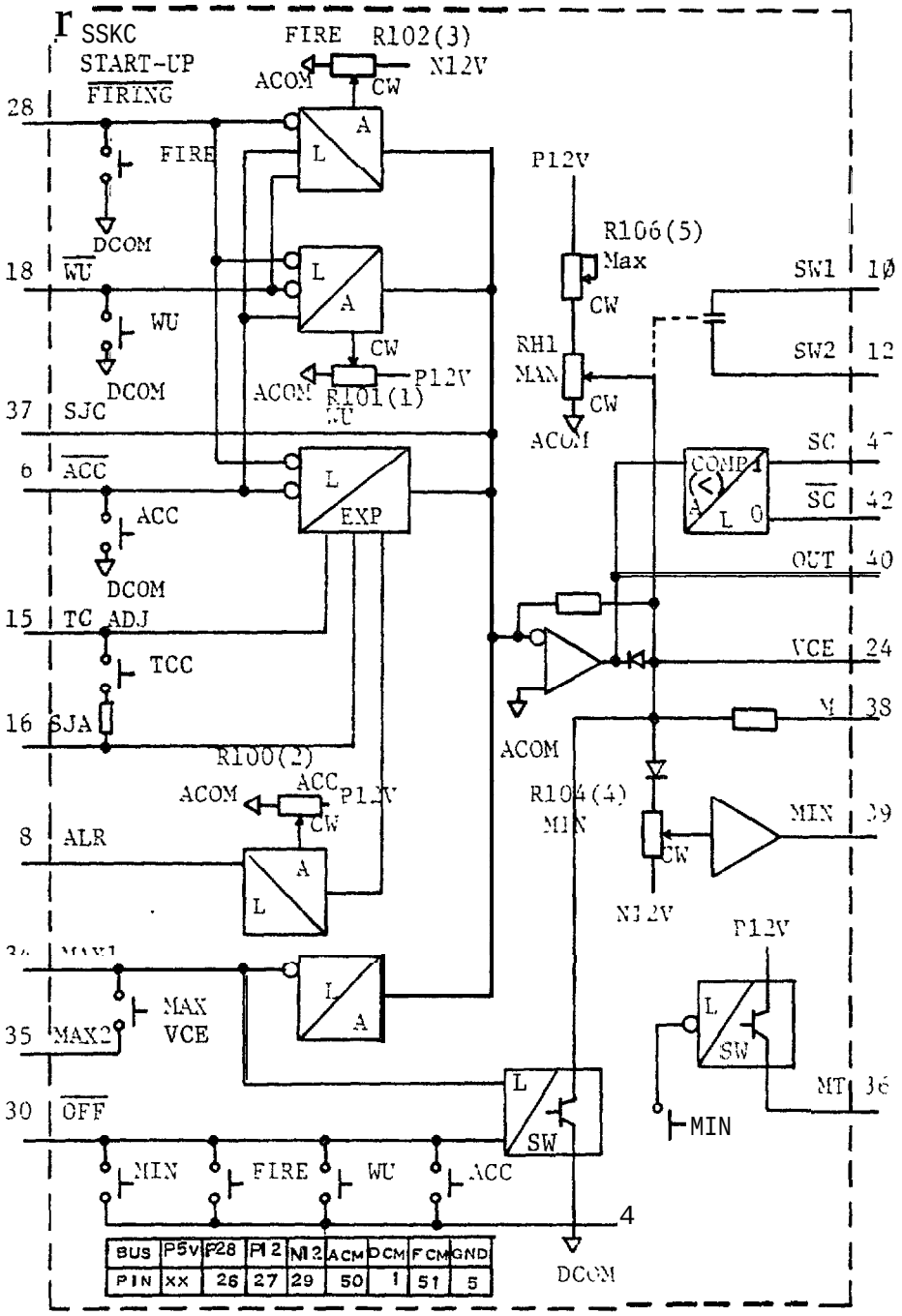


Figure 3-5
Startup Control Card

Digital Setpoint

The digital setpoint produces a programmable analog signal (DSP) used as a speed or load signal to the speed control card. The system consists of cards VDAC, SSLD and associated control signals. (See Figure 3-6.)

VDAC Card

(See Figure 3-7) The VDAC card contains a 12-bit reversible counter, a digital-to-analog (D/A) converter and twelve light emitting diodes (LEDs) to indicate the output of the counter. A logic 0 on the ENAB input permits the counter to count down or up as a function of the logic state of the DN/UP input. With ENAB at logic 0 and DN/UP at logic 1, the counter will count down. With ENAB at logic 0 and DN/UP at logic 0, the counter counts up. The CCP input is a pulse rate signal which sets the rate at which the counter counts. CCP is a pulse train signal which causes the counter to increment/decrement in response to the steering logic. When the counter is empty and a down count is present, or when the counter is full and an up-count is present, output signals MM1, MM2 and MM3 will all be a logic 1.

SAOUT is the amplifier output from the D/A converter. The following full scale voltage output ranges are as follows:

1. Nominal 0 to 10V; no jumper between SJA and SJB.
2. Nominal -5V to +5V; jumper SJA to SJB.

Binary inputs 1MSB and 1B2 thru 1B12, along with signal load, provide the VDAC card with a "preset" capability. To set and clear desired bits, the following three actions are taken simultaneously: (1) apply a logic 1 to the bits to be set, (2) a logic zero to the bits to be cleared, and (3) apply a single shot signal (⅂⌐) to the LOAD line.

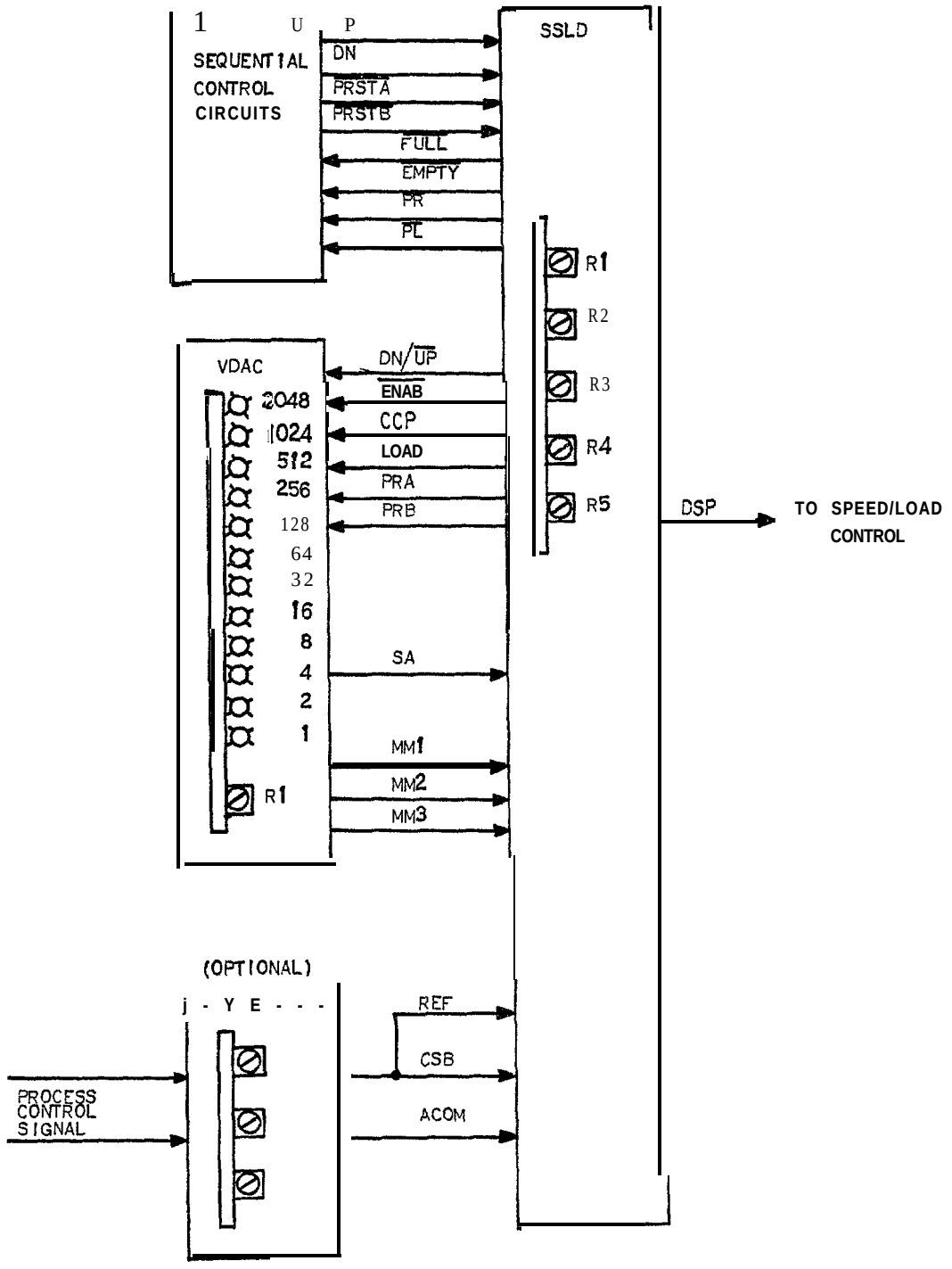


Figure 3-6
Digital Setpoint Block Diagram

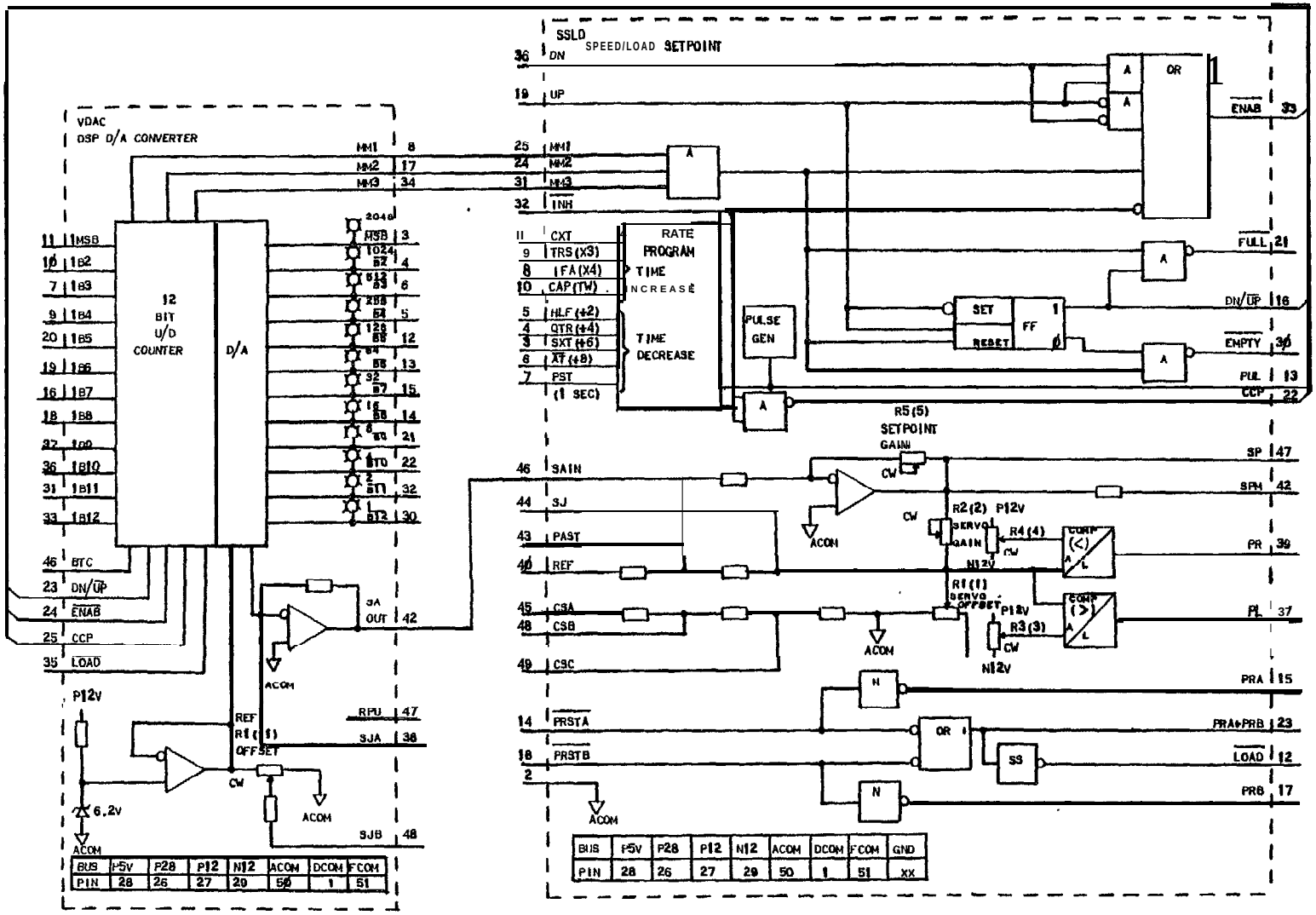


Figure 3-7
Digital Setpoint cards

SSLD Card

(See Figure 3-7) The SSLD card is used in conjunction with the previously described VDAC card to provide the digital setpoint signal for speed/load control. The D/A output signal from the VDAC card is fed into the SSLD card and amplified to produce the digital setpoint signal. The op amp on the SSLD card provides signal inversion and potentiometer R5 provides for signal gain adjustment. Signal offset is provided by R1 on the VDAC card.

The SSLD card also has a comparator servo circuit. One of the inputs to this comparator is the digital setpoint signal. The other input, a reference signal, may be either a current or voltage signal. Current signal input points are CSA, CSB and CSC for l-5 ma, 4-20 ma and 10-50 ma signals respectively. The voltage signal input point is REF with a voltage range of 1 to 5 volts or 4 to 8 volts. If the input signal amplitude is less than that of the digital setpoint signal, comparator output PL will be a logic 1. If the input signal amplitude is greater than that of the digital setpoint Signal, comparator output PR will be a logic 1. The servo gain potentiometer R2 are both provided to enable matching the digital setpoint voltage range to the range of the input signal. Comparator circuit deadband adjustment is also provided; PL is adjusted by R3 and PR by R4.

VDAC input signals $\overline{\text{ENAB}}$, $\overline{\text{DN/UP}}$ and CCP are SSLD card output signals. The frequency of output signal CCP is determined by the wiring of, and logic input to the Rate program. There are three periods of time required for travel from "counter full" to "counter empty". They are: (1) 12 minutes - obtained by the use of no jumper, (2) 36 minutes - obtained with TRS (X3) jumpered to CAP (TW), and (3) 48 minutes - obtained by jumpering FA(X4) to CAP (TN). The selected period can be divided by 2, 4, 6 or 8 by applying a logic 0 to signal input line $\overline{\text{HLF}}$ (-2), $\overline{\text{QTR}}$ (-4), SXT (-6) or AT (-8). A logic 0 input to PST provides for full travel in approximately one second. External UP and DN logic signals, when combined with VDAC output signals MM1, MM2 and MM3, comprise the logic signals for counter full and counter empty outputs FULL and EMPTY.

The SSLD setpoint preset circuits allow presetting the VDAC counter to any two pre-selected values. The SSLD inputs for this circuit are PRSTA and PRSTB. The VDAC inputs are PRA, PRB, PRA + PRB and the single shot signal (LOAD).

AIAD Card

(See Figure 3-8) The AIAD card (optional) provides isolation between the process speed setpoint circuit and the SSLD card. The AIAD card accepts the following signals at the designated inputs by making the specified jumpers:

SIGNAL	INPUT CONNECTION	JUMPERS
0 to 5 volt	+D to 0	E to F
0 to 10 volt	+D to 0	E to L
0 to 25 volt	+M to 0	E to F, D to N
1 to 5 mA	+A to 0	E to F, D to A
4 to 20 mA	+B to 0	E to F, D to B
10 to 50 mA	+C to 0	E to F, D to C

The following outputs are possible:

OUTPUT	CONNECTION	JUMPERS
0 to 10 volt	+G to P	G to H, Q to I
1 to 5 mA	+G to Q	Q to I, S to T
4 to 20 mA	+G to Q	Q to J, R to T
10 to 50 mA	+G to Q	Q to K, R to T

ADJUSTMENTS

Input Bias - R70 - 0 to 30% on any scale

Output Bias - R74 - 0 to 2.5 mA (S connected to T)
2.5 to 15 mA (R connected to T)

Gain - R-71 - 0.5 to 2 when used as current-to-current converter.
±15% MIN on any voltage range.

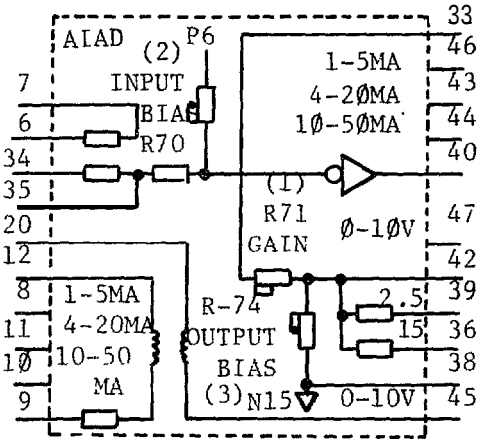


Figure 3-8
AIAD card Elementary

Speed-Load

The speed control is a closed loop system. It modulates variable control electrons (VCE) to control the speed, load and acceleration of the turbine. Various methods of controlling the speed and load are available. The control specification and control panel elementary for a particular job will reflect the method used. The speed/load control requirements for a single-shaft turbine and a two-shaft turbine are different and are therefore covered separately.

Speed-Load; Single Shaft Turbine (HP Shaft). The typical speed-load control for a single shaft turbine consists of cards SHPB, SFPB and SSZB (See Figure 3-9).

SHPB Card., This card contains several circuits used in the speed control loop plus circuitry used elsewhere. Only the circuitry associated with the speed control is covered at this time. (See Figure 3-10).

The SHPB card provides a cross-coupling network so that the speed feedback signal from two magnetic pick-up heads can be combined such that a failure. on either of the sensors will not result in a loss of the feedback signal. Outputs RSZ and SZA are tied together for this function. Pulses on input MPR also result in pulses out Of RSR which are used for zero speed indication.

Short and open detectors are provided for each sensor. A short or open between MPZ and ACOM results in a "1" on FLTZ and illuminates light FLTZ. A short or open between MPR and ACOM results in a "1" on FLTR and illuminates light FLTZ. A short or open between MPR and ACOM results in a "1" on FLTR and illuminates light FLTR.

SFPB Card. (See Figure 3-10). This card provides two functions to the speed control card: (1) It provides the necessary resistor network to operate the system in the droop mode. The resistor network may be connected in such a way that it is always in the circuit or it can be switched in and out of the circuit. If the resistor network is to be permanently connected in a circuit, output SJA is used. If the switch function is desired, SJB is used. When SJB is used, ISOC must be a "1" to have network switched in. (2) The SFPB card also provides a signal that may be used to reprogram the acceleration rate when applicable. A "0" on FSA will give a fixed bias out of ARA.

Adjustments

FSNL - R207
DROOP - R204

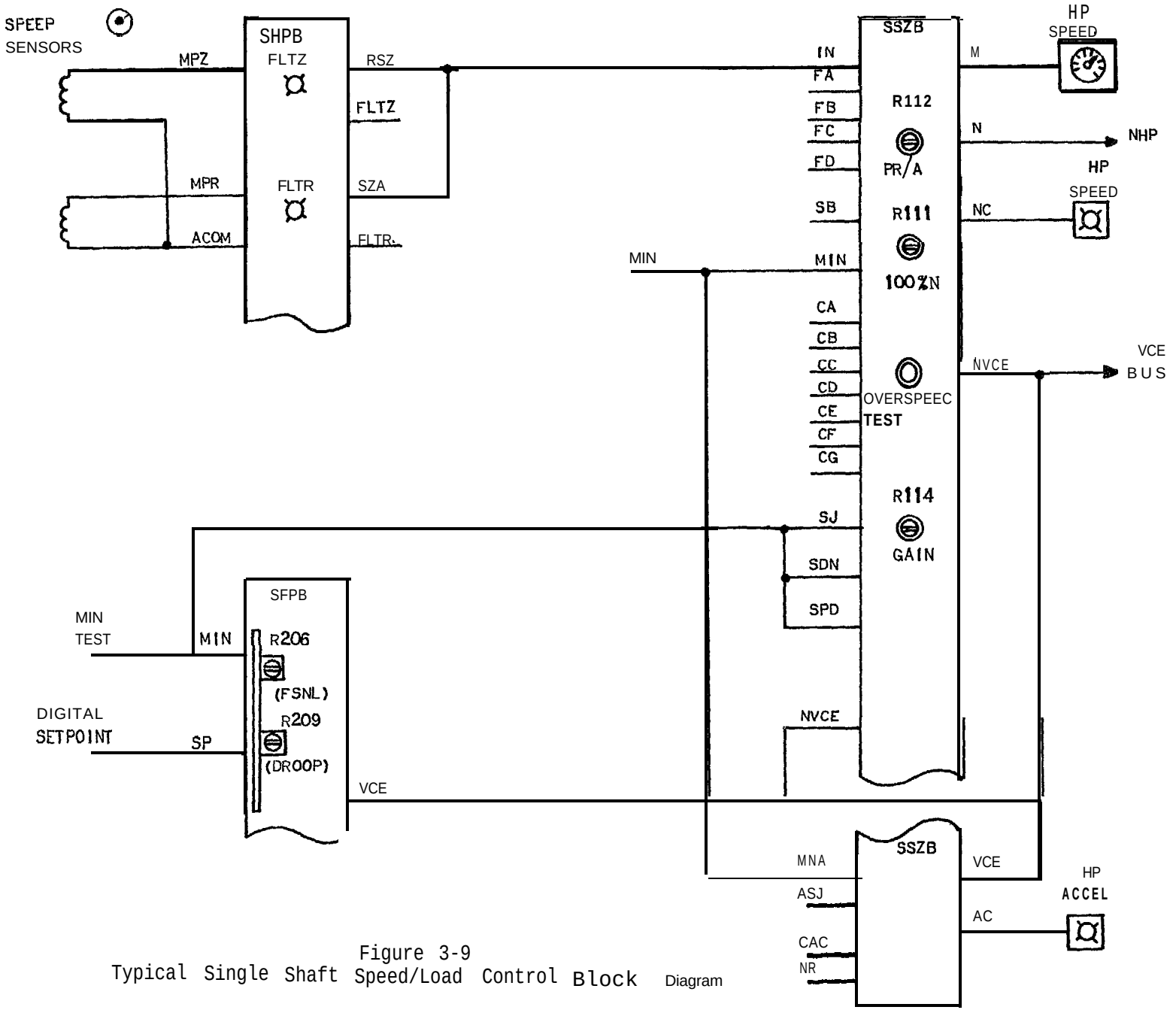


Figure 3-9
Typical Single Shaft Speed/Load Control Block Diagram

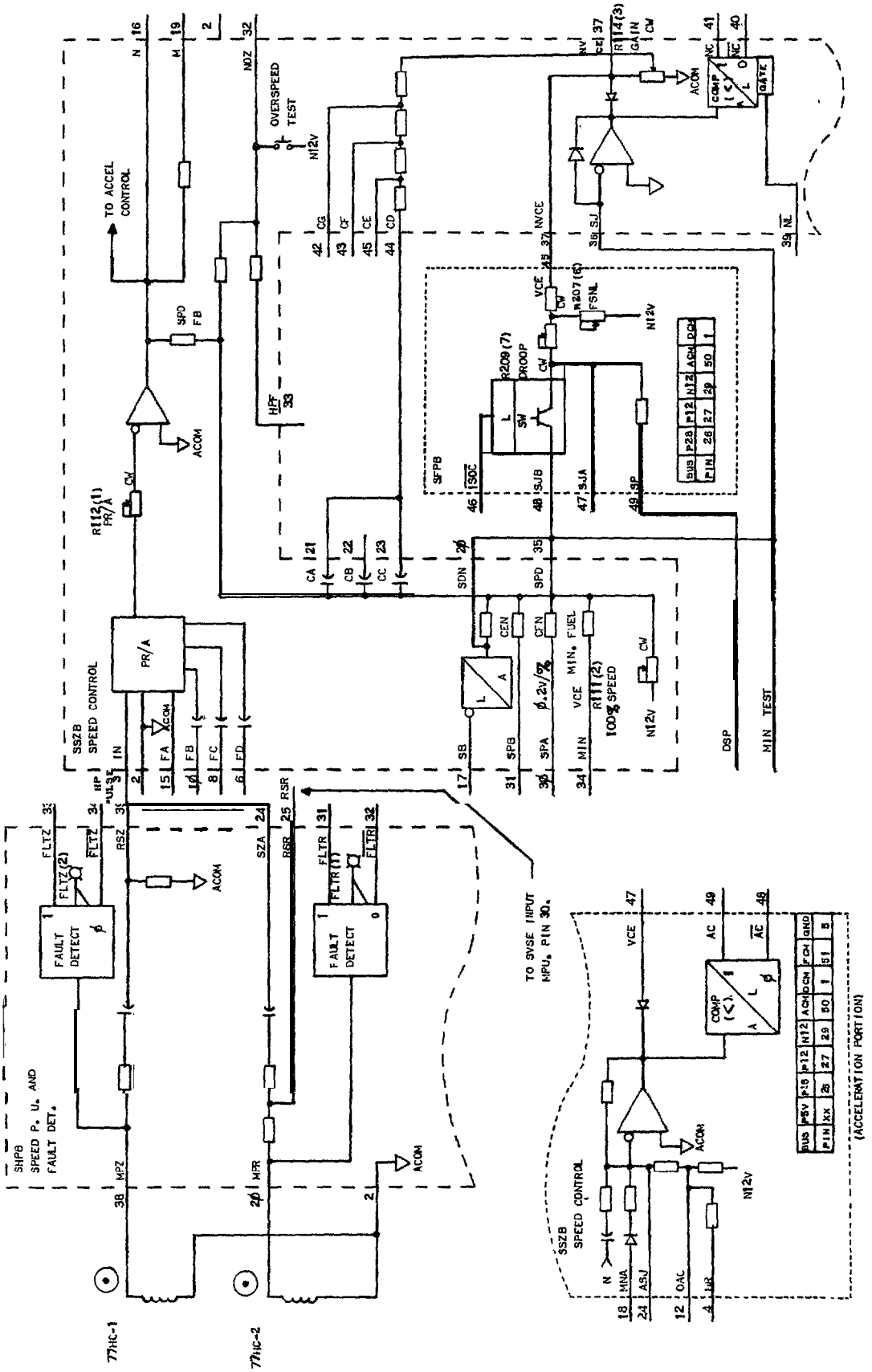


Figure 3-10
Speed Control Cards - Single Shaft Turbine (HP Shaft)

SSZB Card. This card contains a pulse rate to analog (PR/A) Converter which receives an alternating signal at input IN and provides a 0 to 11 volt signal at output N. This speed feedback signal is summed with the reference signal and other signals to modulate NVCE to obtain the desired speed and/or load. Jumpers are made on FA, FB, FC and FD to program the PR/A converter to give the appropriate voltage for various speed machines. (See Figure 3-10.)

A "0" *on* SB puts a positive bias on the SJ and forces NVCE to MIN level if pin SDN is connected to SJ.

An overspeed test pushbutton which feeds a negative bias into SJ and increases NVCE is provided for overspeed testing.

Jumpers are made between either pin CA, CB, CC and either pin CG, CF, CE or CD to select transfer function desired. Potentiometer R114 provides a gain adjustment. Inputs SPA and SPB provide inputs for the digital setpoint signal when the droop resistor network on card SFPB is not used. SPA is used when 2 volt DSP is equal to 1% change in speed, and SPB is used when 1 volt DSP is equal to 1% change in speed.

A differentiator circuit loops at the speed feedback signal N and modulates VCE to limit the accelerations of the turbine. Output AC will be a "1" when VCE is under acceleration control. Connecting HR to ACOM will half the acceleration control rate.

Adjustments:

ADJUSTMENT	POTENTIOMETER	RANGE(% of Rated Speed)
GAIN (PR/A)	R112	NA
GAIN (NVCE)	R114	NA
100%N .	R111	96 to 104

Speed-Load; Two Shaft Turbine (LP Shaft). The LP speed control modulates VCE to control the speed and acceleration of the low pressure shaft. The reference for the speed regulator is the sum of a 100% speed reference and a digital setpoint signal. A typical speed loop consists of cards SLPA and SSZB. (See Figure 3-11).

SLPA Card. (See Figure 3-12.) The SLPA provides a cross-coupling network so that the sensor input signals on MPZ and MPR can be combined by connecting outputs RSZ and SZA together. Each speed sensor is also checked for an open or a short. A short or open on sensor connected to MPZ will generate a "1" on FLTZ and illuminate lamp FLTZ. A short or open on the sensor connected to MPR generates a "1" on FLTR and illuminates lamp FLTR.

SSZB Card. (See Figure 3-12.) This card contains a pulse rate to analog (PR/A) converter which receives an alternating signal at input IN and provides a proportional signal at output N. This speed feedback signal is summed with the reference signal and other input signals to modulate NVCE to obtain the desired speed and/or load. Jumpers are made on FA, FB, FC and FD to program the PR/A converter to provide the appropriate voltage for various machine speeds.

A "0" on SB puts a positive bias into the SJ and forces NVCE to MIN level if pin SDN is connected to SJ. When used for HP set fuel limit and SDN is not connected to SJ, a "0" on SB reprograms the HP fuel limit to 80%.

An overspeed test pushbutton which will feed a negative bias into SJ and increase NVCE is provided for overspeed testing.

Jumpers are made between either pin CA, CB, CC and either pin CG, CF, CE or CD to select transfer function desired. Potentiometer R114 provides a gain adjustment. Inputs SPA and SPB provide inputs for the digital setpoint signal. SPA is used when 0.2 volts DSP is equal to 1% change in speed, and SPB is used when 1 volt DSP is equal to 1% change in speed. A differentiator circuit looks at the speed feedback signal N and modulates VCE to limit the accelerations of the turbine. output AC will be a "1" when VCE is under acceleration control. Connecting HR to ACOM will halve the acceleration control rate.

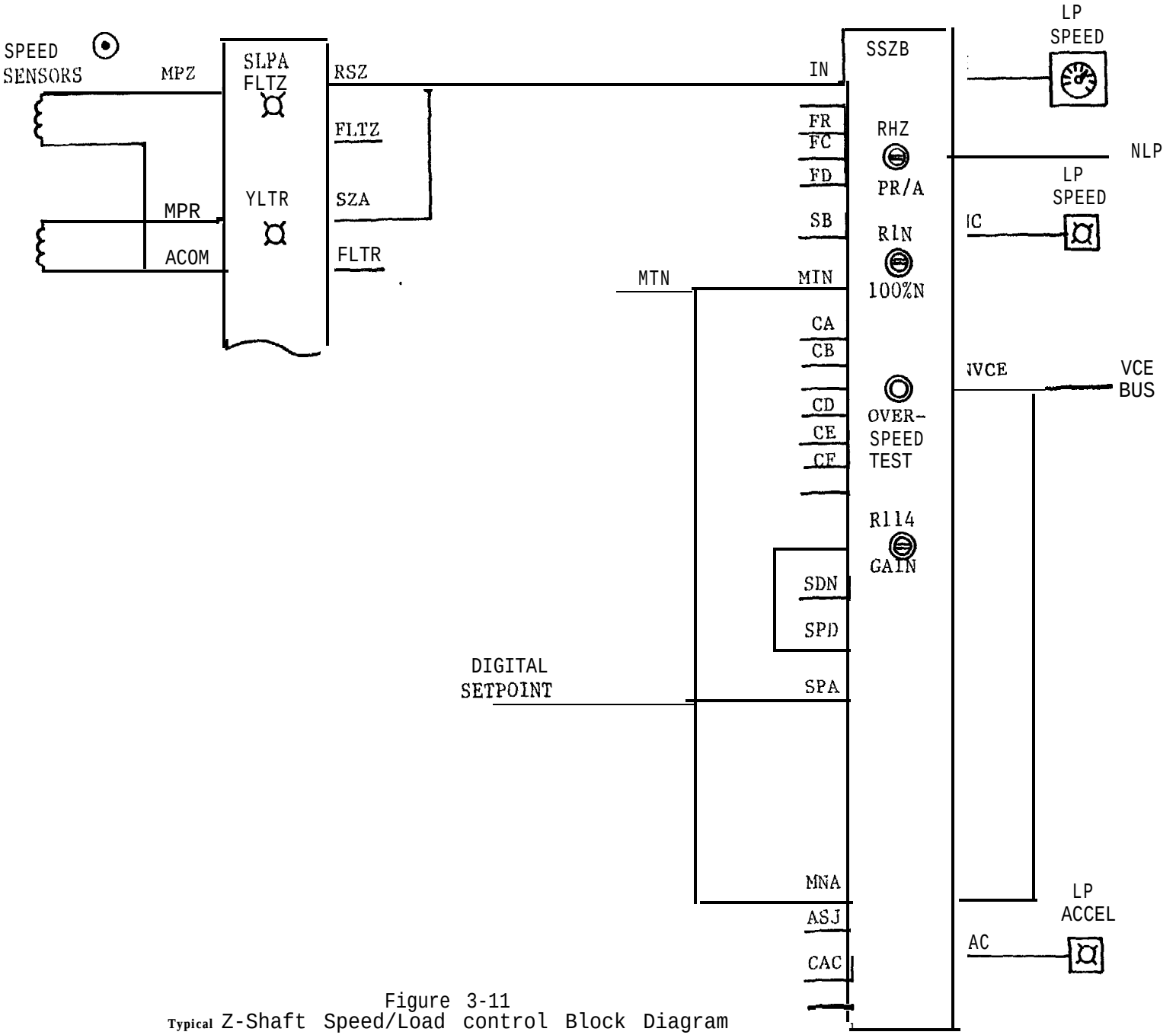


Figure 3-11
Typical Z-Shaft Speed/Load control Block Diagram

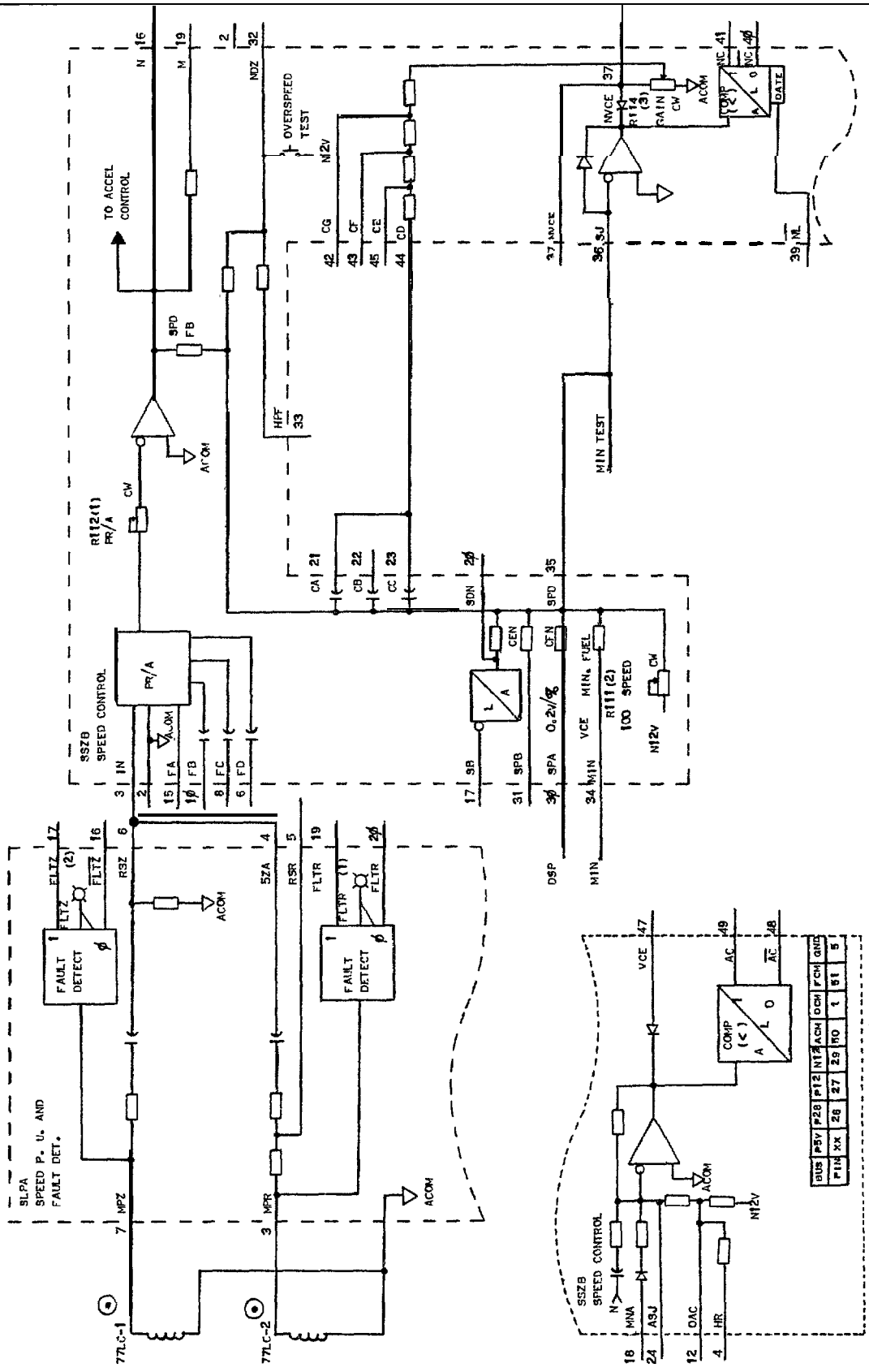


Figure 3-12
LP Speed Control Elementary

Nozzle Control; Two Shaft Turbine-Only

The nozzle control is a closed-loop system which regulates the turbines exhaust temperature by modulating NCE. NCE modulates the nozzle angle which regulates the exhaust temperature by varying the HP shaft speed. The nozzle regulator consists of cards SHPB, SSZB and SNCF (See Figure 3-13.) The nozzle servo-valve control is similar to the gas control servo valve control and will not be covered separately in this publication.

SHPB Card. The SHPB card contains several circuits used in the nozzle control loop plus circuitry used elsewhere. Only the circuitry associated with the nozzle control is covered at this time. (See Figure 3-10 and 3-13)

The SHPB card provides a cross-coupling network so that the speed feedback signal from two magnetic pick-up heads can be combined such that a failure on either of the sensors will not result in a loss of the feedback signal. Outputs RSZ an SZA are tied together for this function. Pulses on input MPR also result in pulses out of RSR which is used for zero speed indication.

Short and open detectors are provided for each sensor. A short or open between MPZ and ACOM results in a "1" on FLTZ and illuminates light FLTZ. A short or open between MPR and ACOM results in a "1" on FLTR and illuminates light FLTR.

SSZB Card. (See Figure 3-14.) The SSZB card contains a pulse rate to analog (PR/A) converter which receives an alternating signal at input IN and provides a proportional signal at output N. Jumpers are made on FA, FB, FC and FD to program the PR/A converter to give the appropriate voltage out for various speed machines.

Input SJ receives the sum of NHP and temperature reference signal from the SNCF card and modulates output NVCE. When SSZB is controlling NCE, output NC will be a "1" provided that permissive TC is a "1".

A differentiator circuit looks at the speed feedback signal N and modulates VCE to limit the accelerations of the turbine. Output AC will be a "1" when VCE is under acceleration control. Connecting HR to ACOM will halve the acceleration control rate. (See Figures 3-10 and 3-13.)

ADJUSTMENTS:

ADJUSTMENT	POTENTIOMETER
Gain (PR/A)	R112
Gain (NVCE)	R114

SNCF Card. (See Figure 3-14.) The SNCF card provides the drive for the NCE bus. The drive is developed similar to VCE on the SSKC card. However, the maximum limit is fixed and not adjustable as VCE is.

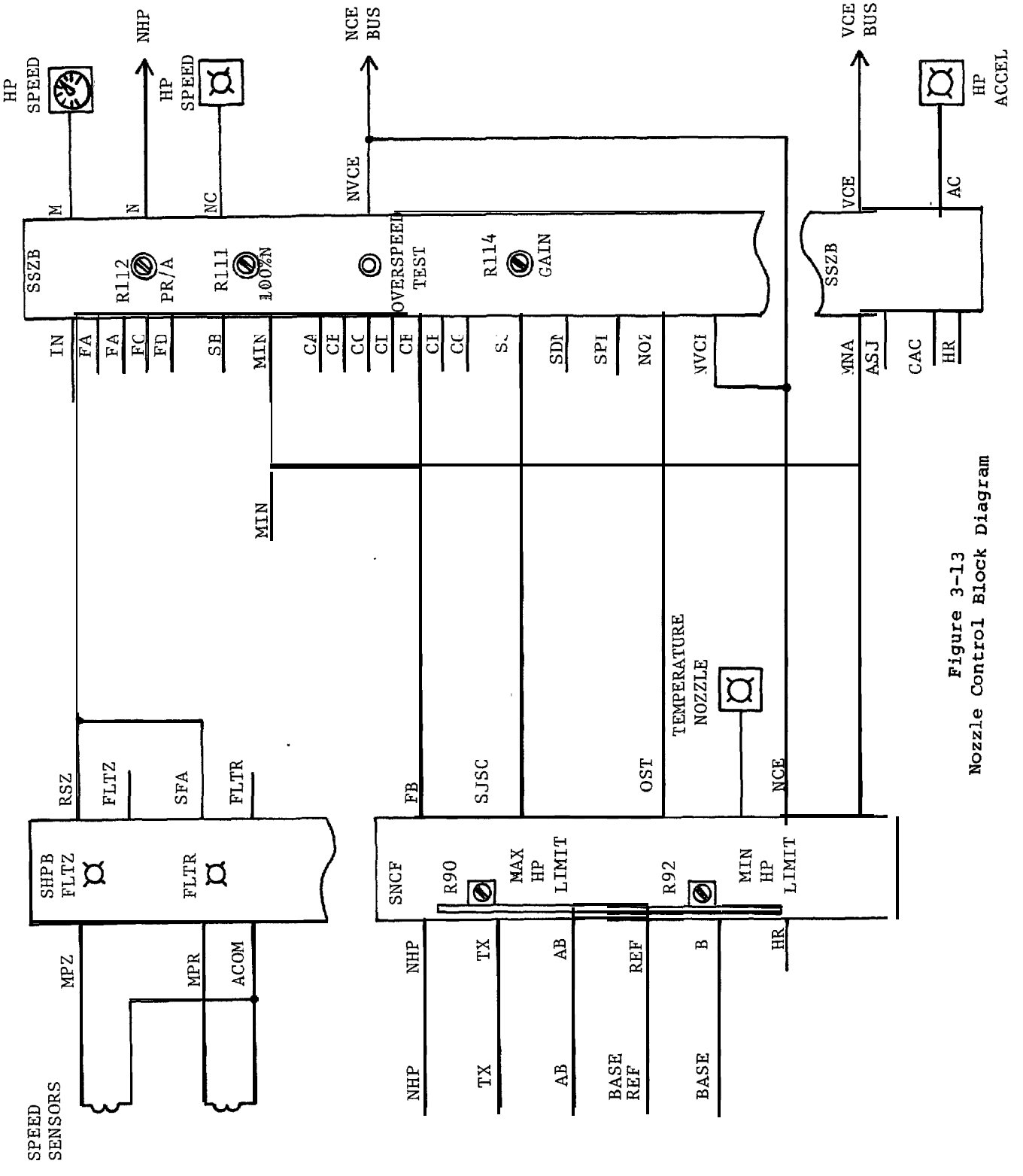


Figure 3-13
Nozzle Control Block Diagram

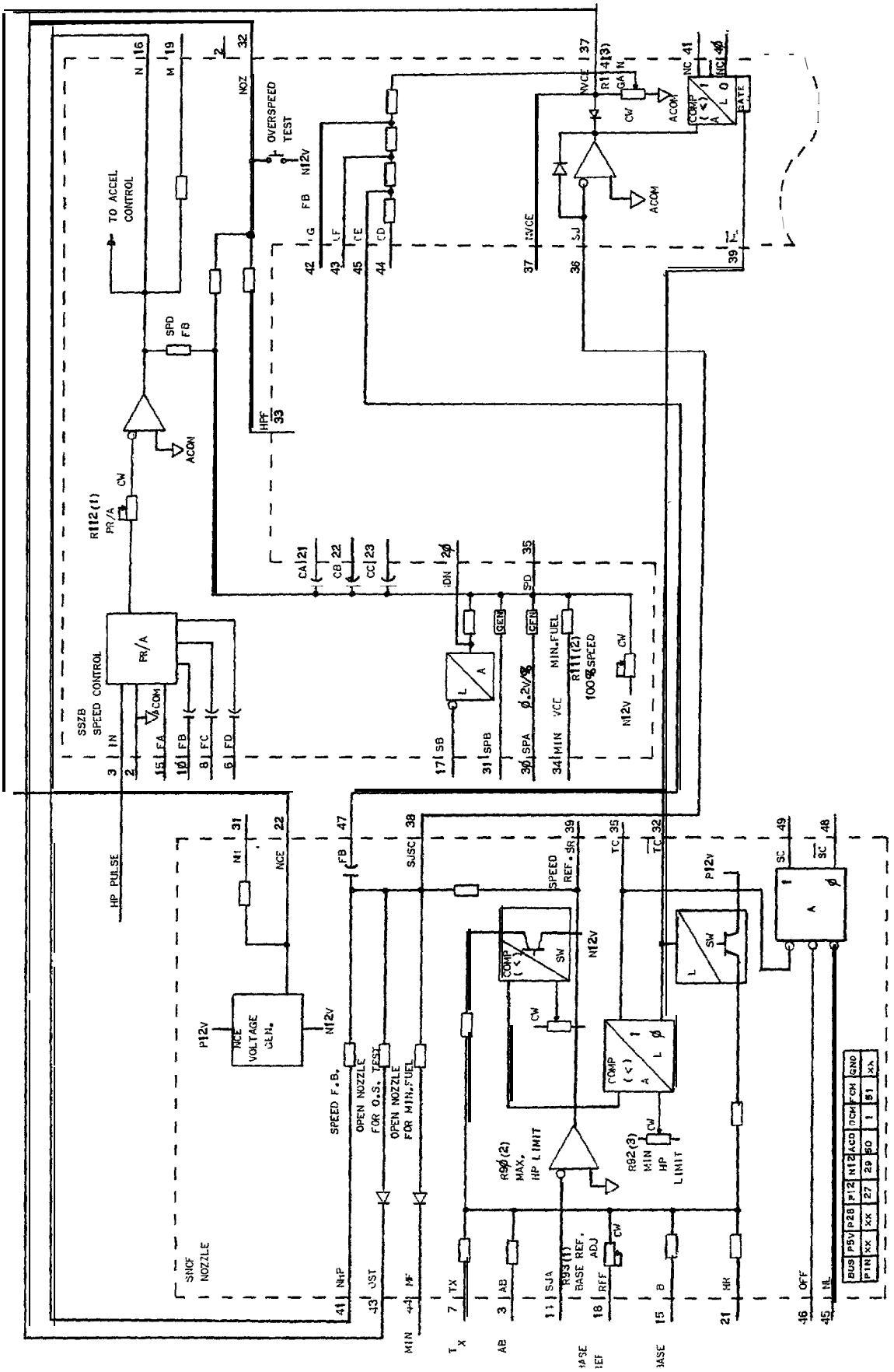


Figure 3-14
Nozzle Control Loop Cards

The SNCF card also receives the HP speed signal on input NHP and sums it with a speed reference and Outputs a signal on SJSC to the regulator operational amplifier on the SSZB card to modulate NCE. The speed reference (temperature error) signal is the sum of the inputs on TX, B and AB. If TX plus AB is less than B, the reference changes in the positive direction to close the nozzle and decrease speed. If TX plus AB is greater than B, the reference changes in the negative direction to open the nozzle and increase HP speed. The temperature operating range can be altered via HR. The speed reference signal is limited to a minimum and maximum value. The nozzle may be forced into the open position by inserting a negative bias at MF. Typically, MF is connected to MIN on SSKC card to open the nozzle at minimum VCE. A negative bias on $\overline{ST}$ will also force the nozzle open. Typically, $\overline{ST}$ is tied to the overspeed test pushbutton on the SSZB card to facilitate overspeed testing of the turbine. Output TC will be a "1" when nozzle is on temperature control.

ADJUSTMENTS:

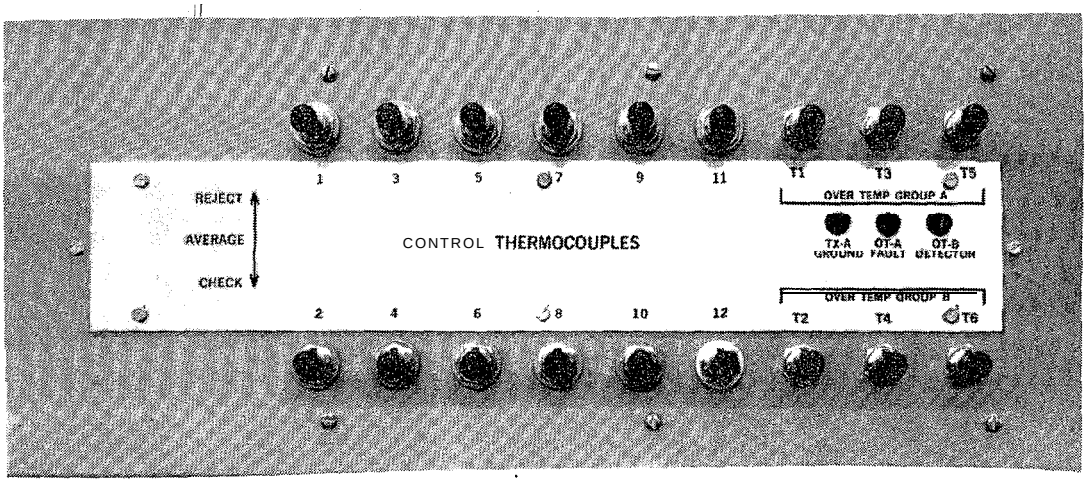
ADJUSTMENTS	POTENTIOMETER	RANGE(% Speed)
Max. Speed	R90	80 to 105
Min. Speed	R92	75 to 100

Temperature Control

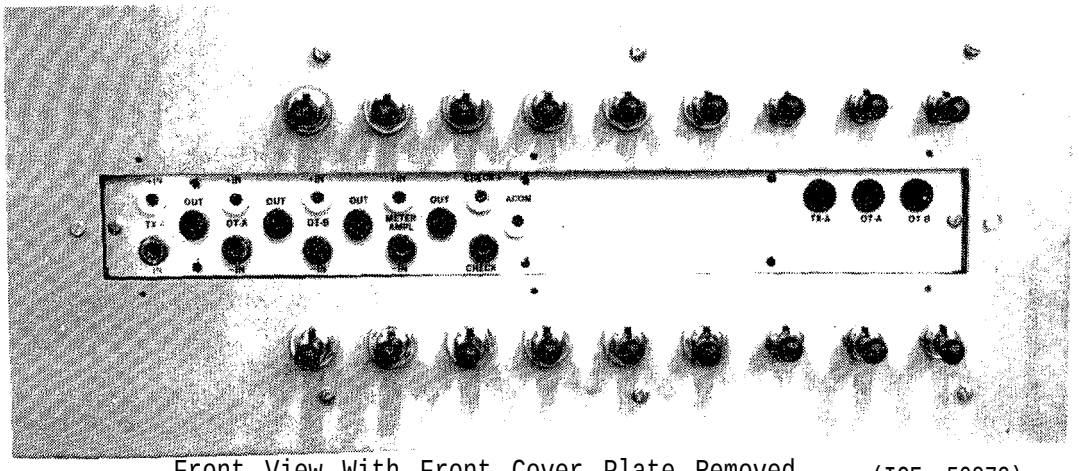
Temperature feedback originates from up to 12 chromel-alumel thermocouples located at selected points of the turbine exhaust duct. Three views of a thermocouple processing module are shown in Figure 3-15. Thermocouple signals originating in the turbine exhaust duct are averaged and cold junction - compensated and then summed with reference signals to modulate VCE. The reference may consist of several hardwired signals and two selectable signals. Included in these signals are base reference, start-up suppression ambient bias (AB), transient overfire, peak load (optional) and peak reserve load (optional). Compensated and uncompensated signals are available for temperature indication on various meters. A typical temperature control loop consists of cards STKK, SPOA, SHPB, SFPB and a temperature averaging module. The SPOA and SFPB cards are optional. A thermocouple ground is detected and compensated for by the SGDE card. (See Figures 3-16, 3-17 and l-18.)

NOTE

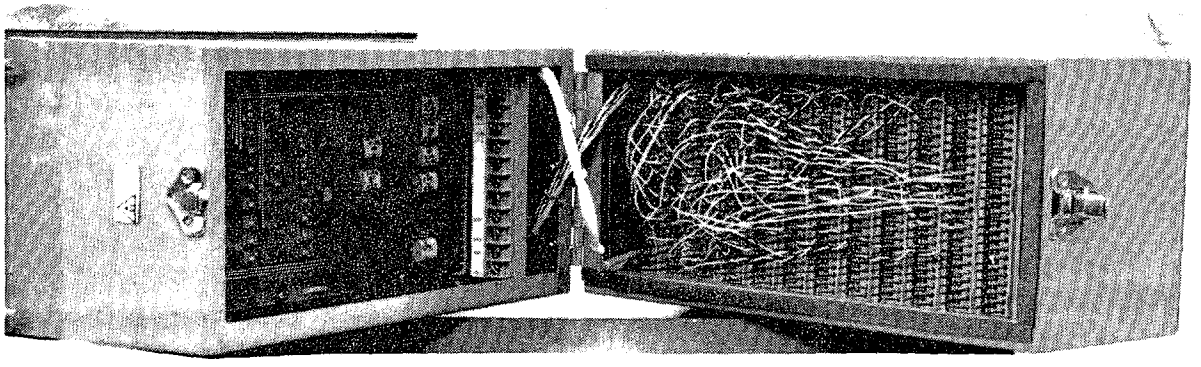
The turbine temperature control and protection function on some control equipment is provided by the optional "Integrated Temperature System" (ITS). In addition to the turbine temperature control and protection function, this system provides various other auxiliary monitoring, control and protection functions. If the control equipment you are involved with has the ITS, disregard the sections of this publication applicable to temperature control and protection and refer to the ITS instruction book.



Front View With Front Cover Plate (ICF 50879)



Front View With Front Cover Plate Removed (ICF 50873)



Rear View With Module Open (ICF 50878)

Figure 3-15. Thermocouple Processing Module.

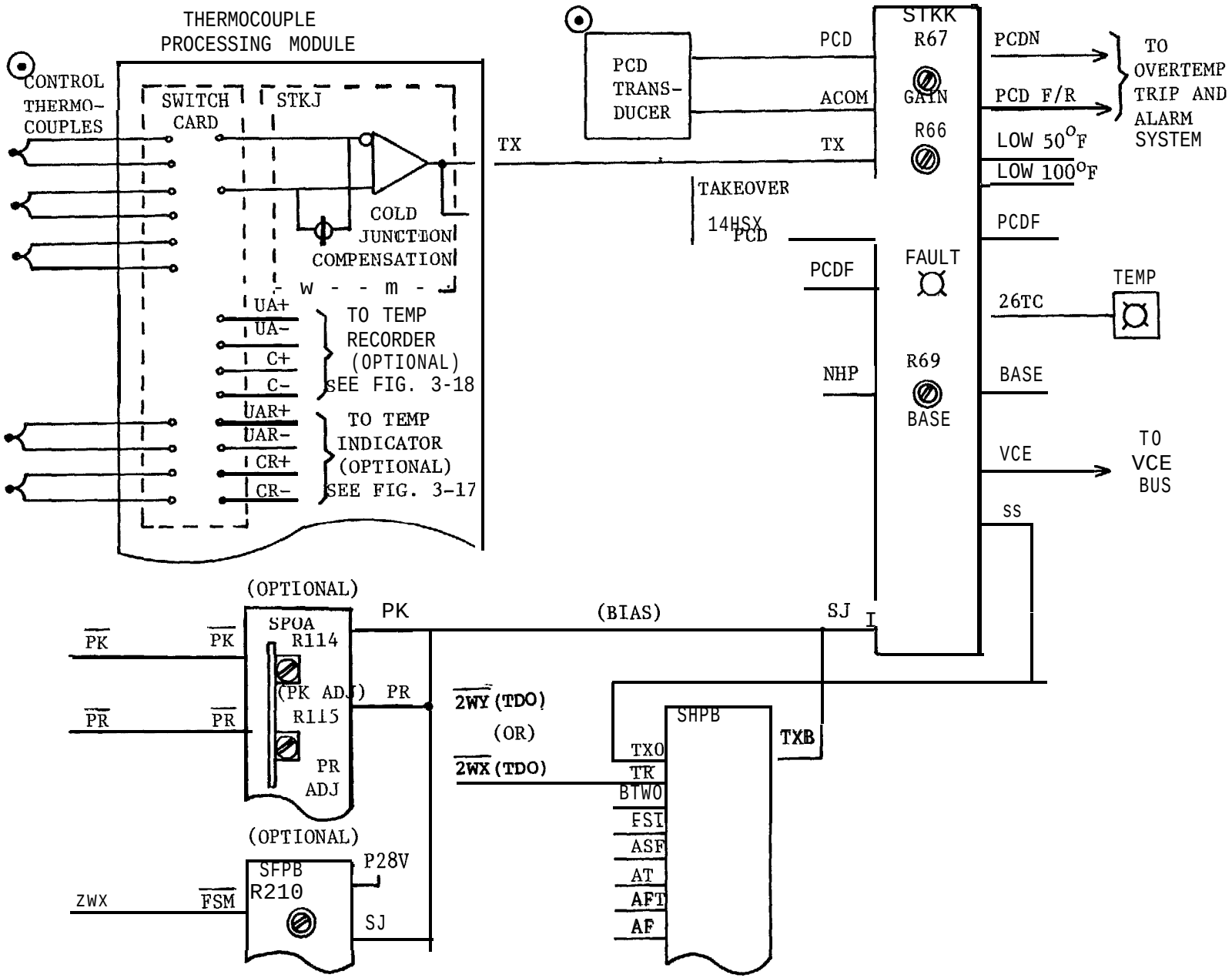


Figure 3-16
Temperature Control Block Diagram

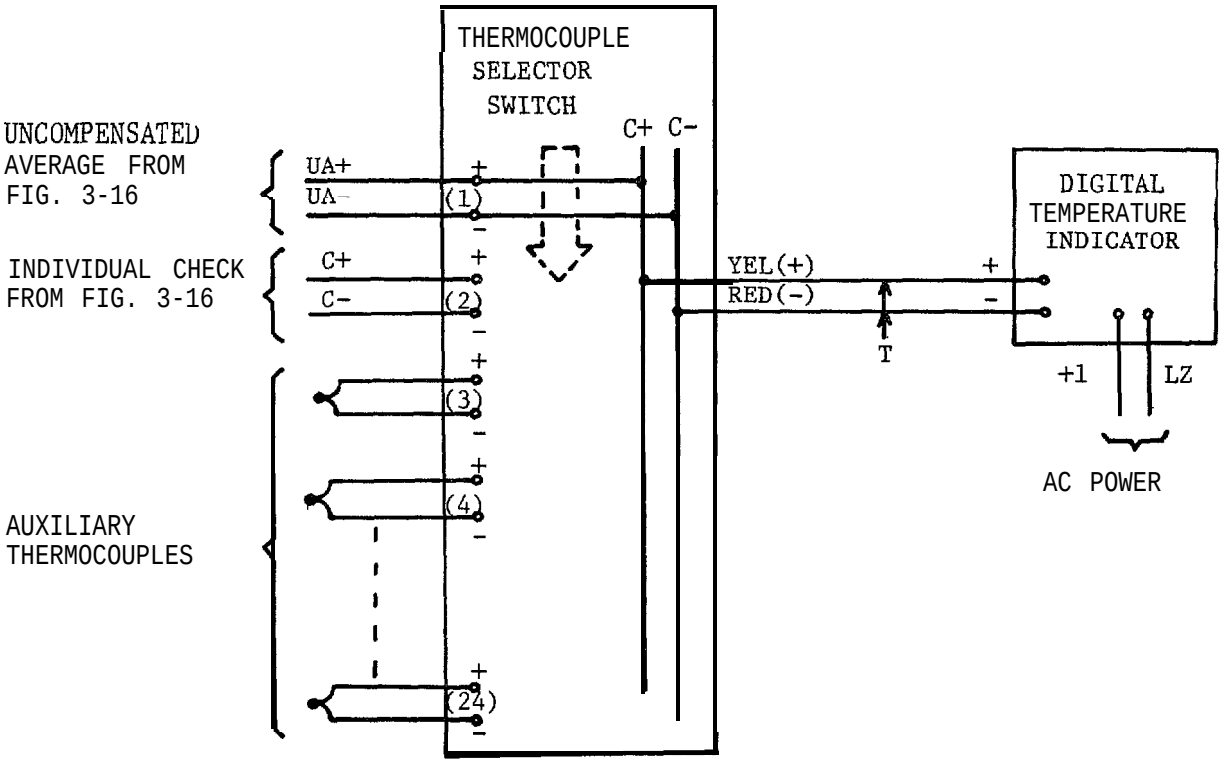


Figure 3-17. Typical Temperature Indicator Block Diagram

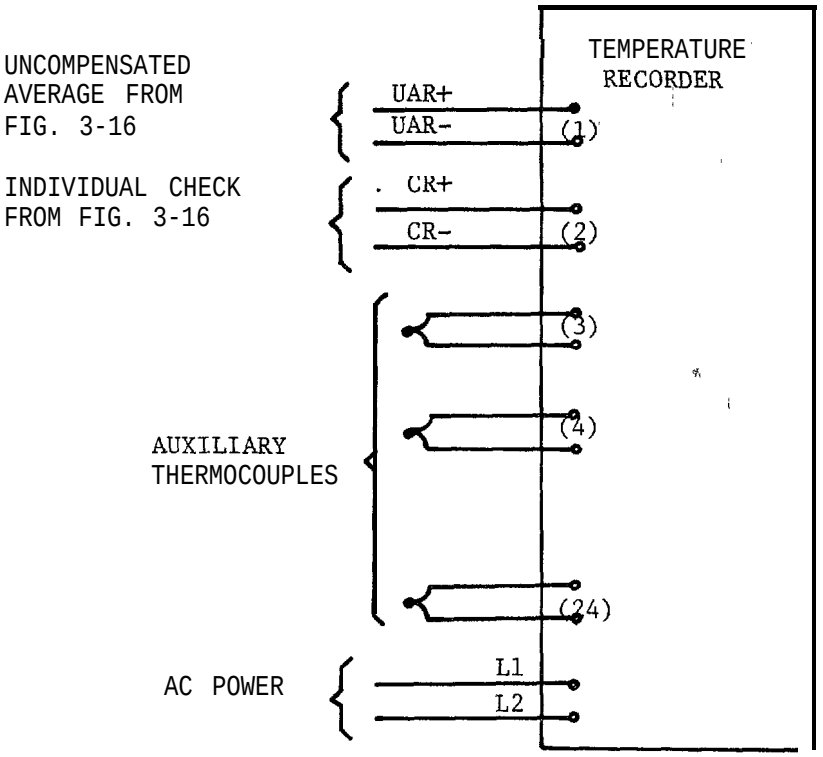


Figure 3-18. Typical Temperature Recorder Block Diagram

Thermocouple Averaging Module. The inputs to the averaging cabinet are produced by twelve chromel-alumel thermocouples mounted in the turbine exhaust duct. These twelve thermocouples are connected in parallel to produce an average signal. A 50 ohm resistor is located in each lead of the thermocouples to swamp out circulating currents due to differences in resistance of the thermocouple wires. Switches are provided to allow a check of each individual thermocouple. The toggle switch corresponding to the thermocouple to be checked must be placed in the CHECK position to enable this function. The CHECK position of each thermocouple switch is tied to a common bus; therefore, if more than one switch is in the CHECK position, their average output will appear on the CHECK bus. Refer to the thermocouple schedule located on the turbine control panel to determine the required thermocouple switch position to connect the digital indicator to the CHECK bus. An individual CHECK signal for connection to a recorder is available at +CR and -CR. A thermocouple can be removed from the average by placing its corresponding switch in the REJECT position.

The averaging cabinet receives the twelve thermocouple inputs from the turbine exhaust, averages these signals, cold junction compensates them, isolates and amplifies them using an STKJ card. A resistance temperature detector is used for the cold junction compensation. A potentiometer is provided to adjust the voltage drop across the RTD, however, this is set up in the factory and should require no field adjustment. Two uncompensated averages are provided. The uncompensated average out of +UA and -UA goes to the thermocouple switch for digital indication. Refer to thermocouple schedule (Figure 3-16) for the required switch position. An uncompensated average output of +UAR and -UAR is available for connecting to a recorder. Calibration and testing jacks are accessible at the front of the module by removing the cover plate.

STKK Card (Refer to Figure 3-19). The STKK card receives the exhaust temperature signal (TX) from the thermocouple processing module. TX is summed with the reference signals and modulates output VCE to control fuel flow. When TX is in control of the VCE bus, output TC is a logic 1 and $\overline{TC}$ is a logic 0.

The base reference for the control is generated on the card and is adjustable using R69. The base reference is also developed on the card. The ambient bias circuit consists of an operational amplifier with an offset and gain adjustment. The input to the ambient bias circuit can be a 0 to 10 volt signal on BVCE, or a 0 to 5 volt signal on PCD. A 10 to 50 milliampere input signal may be used by connecting to PCD and jumpering PCD to CSA/B and CSA or a 4 to 20 milliampere signal by connecting to PCD and jumpering PCD to CSA/B and CSB. The output of the ambient bias amplifier is available at output pin PCDN. Additional reference signals may be connected to input SJ.

Typically, the startup suppression signal and the optional peak and peak reserve signals are connected to SJ. Input HR is available for changing the range of the base adjustment.

ADJUSTMENTS:

TAKEOVER - R66 - MV/V Amplifier Offset Adjustment
GAIN - R67 - MV/V Amplifier Gain Adjustment
BASE - R69 - Base Reference Adjustment

SPOA Card - (Refer to Figure 3-19) (Generator Drive Option). The SPOA card contains the circuitry required to generate the peak and peak reserve bias for the temperature regulator. If both peak and peak reserve biases are desired, output PK and PR are tied together and connected to SJ of STKK. A "0" on input PK results in a bias out of PK that is increased at a controlled rate until the limit set by R114 is reached. A "0" on input PR (independent of input PK) results in a bias out of PR that is increased at a controlled rate until the limit set by R118 is reached. Typically, the sequential logic is designed to generate a "0" on PK whenever PR is "0". This makes PK and PR additive when operating in the peak reserve mode.

ADJUSTMENTS:

PK - R114 - 0 to 150°F above base
PR - R118 - 0 to 100°F above peak

Temperature Suppression During Warm-up (See Figure 3-19). During warm-up, the temperature control bias follows the warm-up temperature, with a back-up temperature allowed. The follower circuit is provided by the SHPB card. An optional max warm-up temperature bias may be provided by the SFPB card.

An exhaust temperature signal, provided by the STKK card, is fed into the SHPB bias circuit. This signal is enabled during warm-up only to bias the temperature control on the STKK card. Therefore, the start-up suppression bias follows the exhaust temperature. When warm-up is completed, the Start-Up suppression will normally be removed at a rate of 5°F/second. (Other rates are available via jumper). This suppression removal starts at the exhaust temperature present at the end of warm-up.

As an option, a maximum bias is provided during warm-up by the SFPB card. Since the temperature bias signals are of a minimum select nature (only the smallest controls the STKK card), the maximum warm-up bias sets a limit for TX during warm-up.

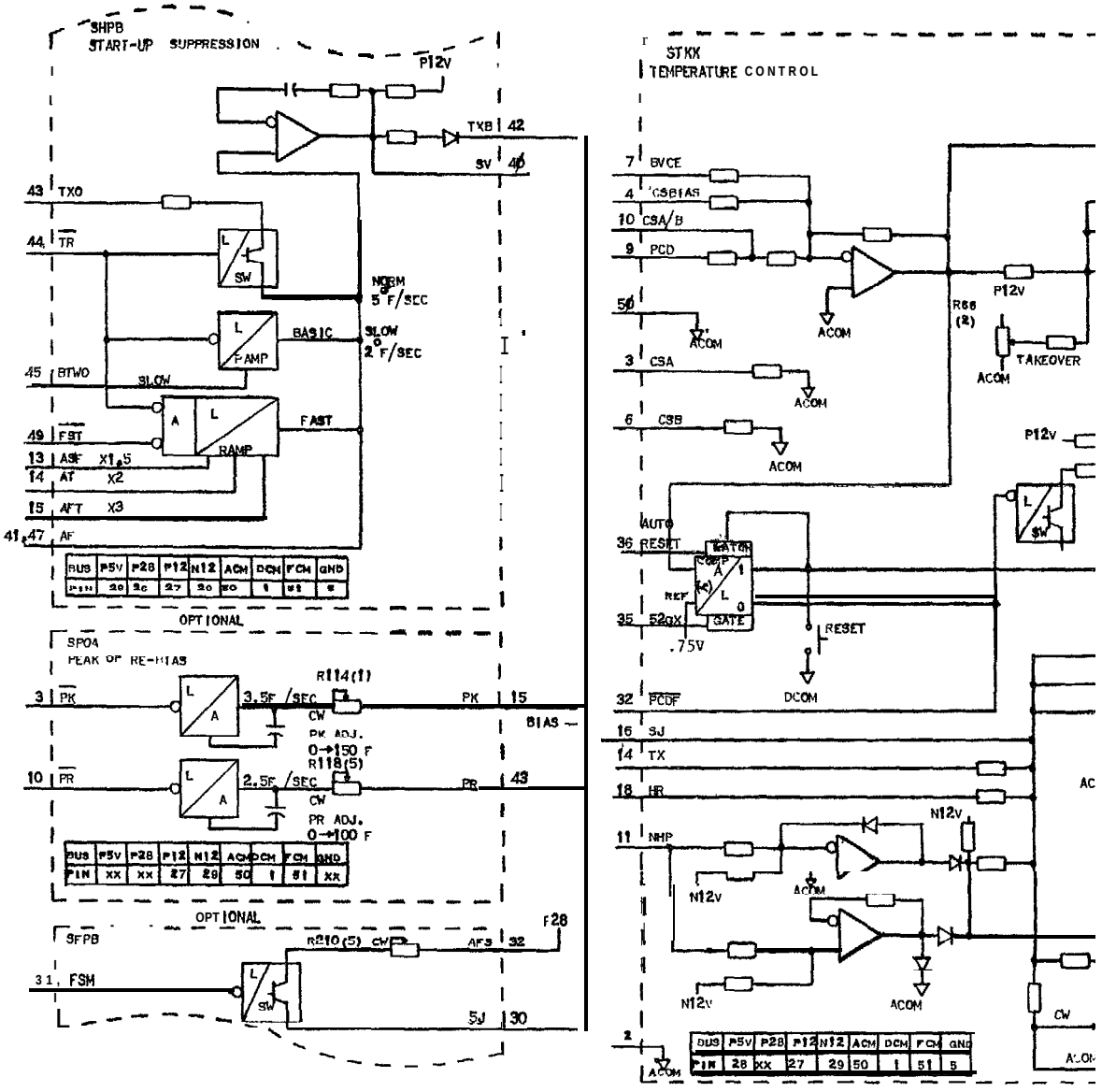


Figure 3-19
Temperature Control Cards

Fuel Splitter (Optional)

A fuel splitter is provided for systems that have both gas and liquid fuel systems. The turbine control specification and control elementary diagram will define the system configuration used for a specific application. The fuel splitter is furnished to permit transfer from one type of fuel to another without interruption of turbine operation.

The fuel splitter receives its command from either the local selector switch or a remotely located switch. A purge period is available when going from full gas to liquid or from full liquid to gas. The fuel mixture can be stopped at any proportion while under local control. The fuel splitter consists of cards SFKL and VRCC. See Figure 3-20.

VRCC Card (See Figure 3-21). The VRCC card contains a 10-bit reversible counter. The operating rate of the counter is determined by the frequency of the pulses coming in on the PUL line. The direction of the count (increasing or decreasing) is controlled by inputs on ST1 and STO. A "1" on ST1 and a "0" on STO will steer the counter upward. A "0" on ST1 and a "1" on STO will steer the counter downward. If both the ST1 and STO lines are at logic "1", all bits will flip on the next count pulse. A "0" on CL will clear all bits.

SKFL Card (See Figure 3-21). The SFKL card generates the steer signals for the VRCC reversible counter on command of the fuel selector switch. A logic "1" on GAS will steer the counter upward (output ST1 = "1") and a "1" on LIQ will steer the counter downward (output STO = "1"). output PUL provides count pulses to VRCC. Pulses are inhibited when any of the following conditions are true:

1. LIQ and GAS are both "0" and "1".
2. Input CARRY is a "1".
3. DGAS is "0" and GAS is "1".
4. DLIQ is "1", NPG is "0" and LIQ is "1".

Inputs DLIQ and DGAS provide the means of allowing a purge time after signals LIQ or GAS are received. NPG provides a way to bypass the purge when transferring from gas to liquid. Input CAP may be used to alter the frequency-of the count pulses on PUL. Output FG is a "0" when fully on gas and FL is a "0" when fully on liquid.

The output of the VRCC counter is the input to the D/A converter on the SFKL card. The signal on input VCE is divided into two signals with their values depending on the output of the D/A converter. A simplified diagram follows. This diagram is used only as an example and does not include all aspects of the actual circuit.

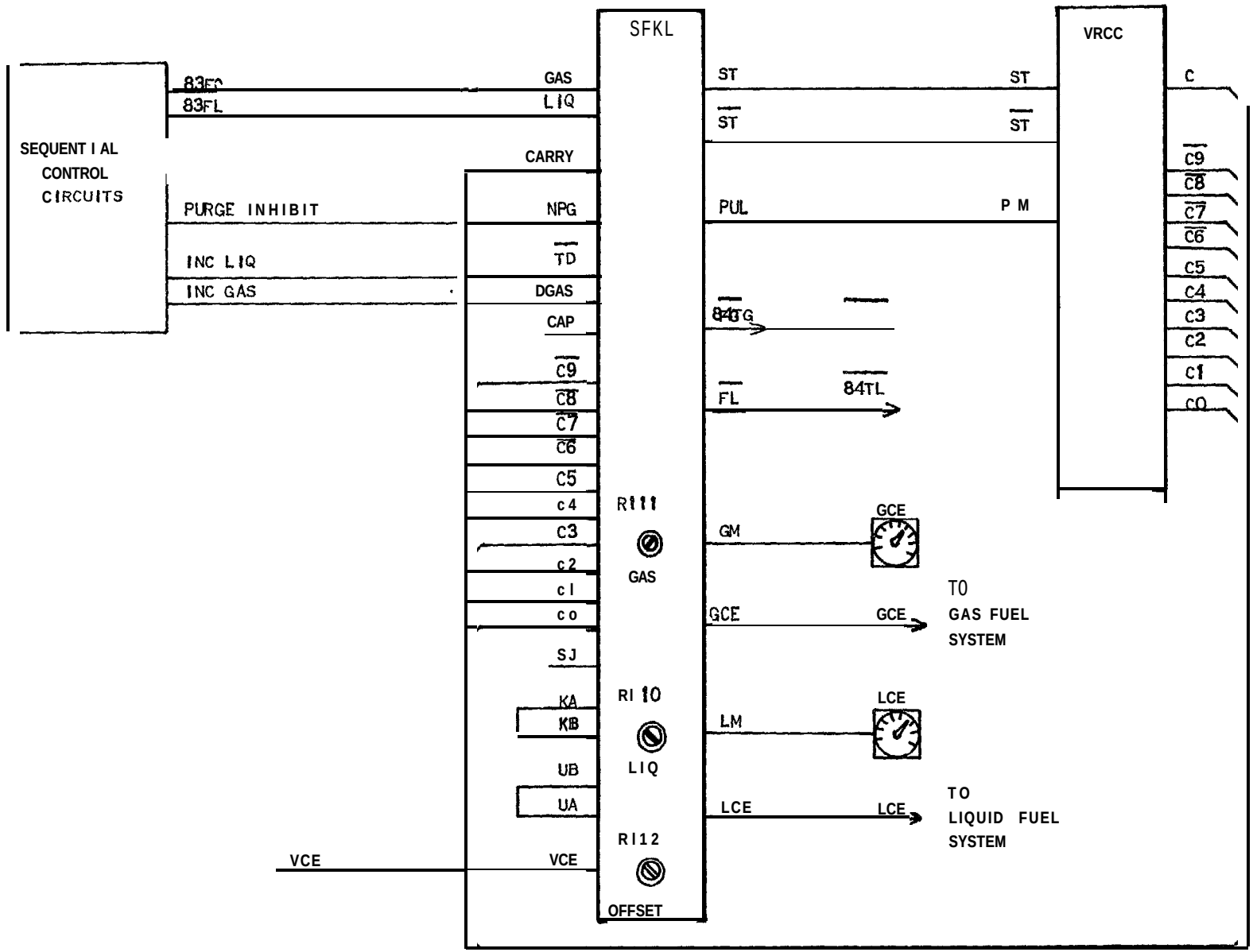


Figure 3-20
Fuel Splitter Block Diagram

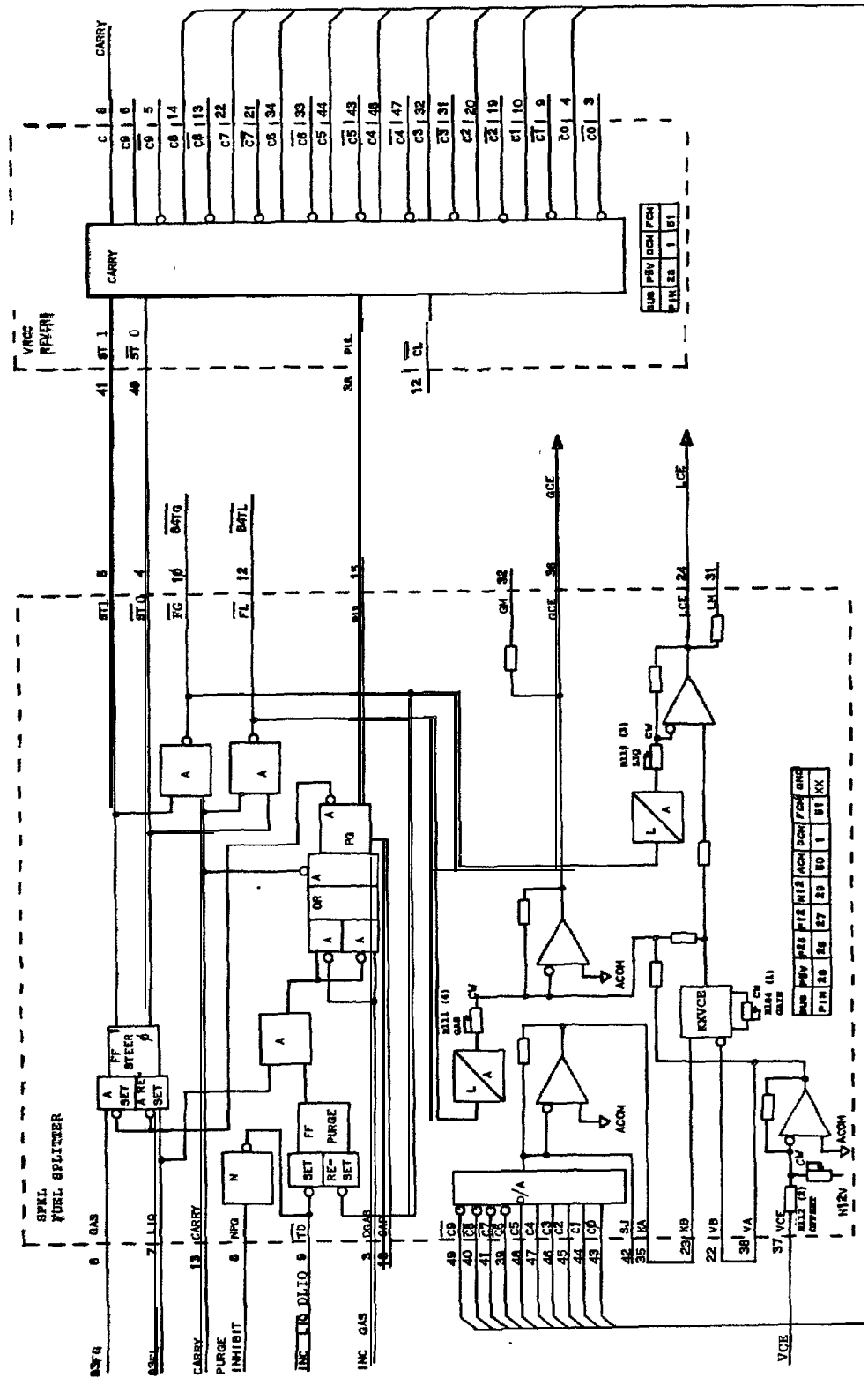


Figure 3-21
Fuel Splitter Cards

For example, the signals VCE, GCE and LCE are each broken into active and inactive signals with the inactive signal being that portion required to crack the valve. All above this is considered to be the active part. The actual circuit is designed such that the active portion of LCE and GCE are always equal to the active portion of VCE.

Looking at the diagram and considering the active portion only, it is seen that

LCE = - (Y) (VCE)

Where Y is proportional to the count in the counter. Also, it is seen that

GCE = VCE -(Y) (VCE).

Therefore, when Y equals 1, LCE is equal to VCE and GCE equals zero. When Y equals zero, LCE equals zero and GCE equals VCE. The counter and D/A converter are connected such that when the counter is full, Y is zero, and when the counter is empty, Y equals 1. For mix operation, Y is somewhere between 0 and 1. Figures 3-22 thru 3-24 illustrate some typical fuel transfers.

ADJUSTMENTS:

ADJUSTMENT	POT	DESCRIPTION
GAS	R111	Adjustment for inactive portion of GCE
LIQ	R110	Adjustment for inactive portion of LCE

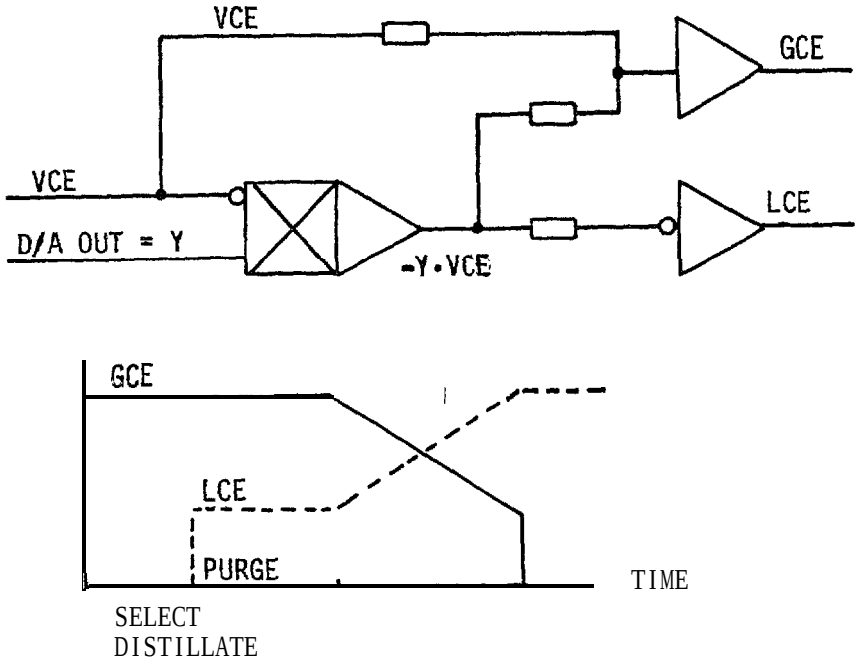


Figure 3-22. Transfer From Full Gas to Full Distillate

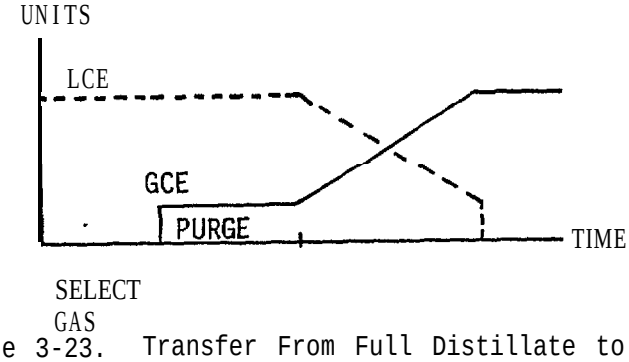


Figure 3-23. Transfer From Full Distillate to Full Gas

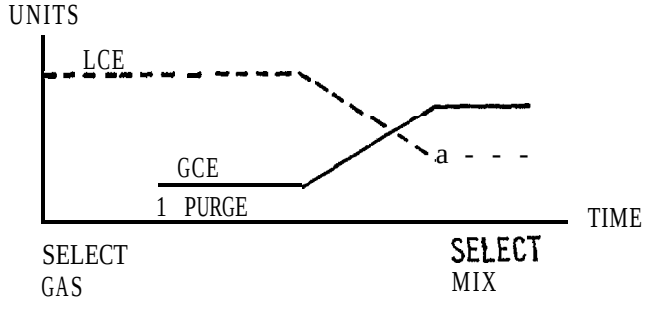


Figure 3-24. Transfer From Full Distillate to Mixture

NOTE

There are two types of liquid fuel systems. One system, uses fuel flow feedback, and is discussed in the following paragraph. The other system, uses fuel flow and fuel valve position feedback, and is used on gas turbines which have a higher fuel flow rate. This system is used to gain stability in the fuel control loop due to higher fuel flow rates. The latter system is discussed in the paragraph entitled "Liquid Fuel Control" (Fuel Flow and Position Feedback System).

Liquid Fuel Control (Fuel Flow Feedback System) (Optional)

The liquid control loop consists of (1) a fuel flow detector card SFUC which receives a frequency proportional to fuel flow rate from the sensor and (2) a regulator card SFKK which compares the feedback signal from SFUC with the product of the signals on inputs VCE and NHP and modulates the servo valve. (See Figure 3-25).

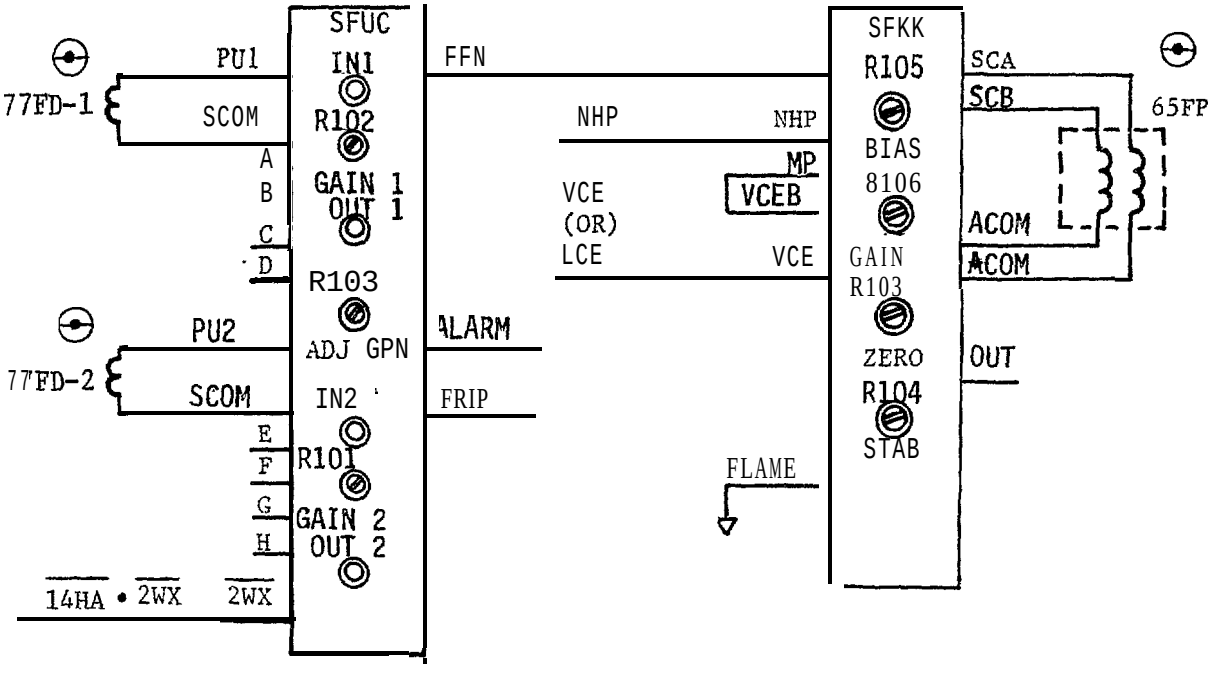


Figure 3-25
Liquid Fuel Servo Block Diagram (Fuel Flow Feedback System)

SFUC Card (See Figure 3-26). The SFUC card contains two PR/A converters. Each of these converters receives a signal from a separate sensor and provides an analog signal at OUT 1 and OUT 2 which is proportional to the frequency of the signal from the sensor. The largest signal from the two PR/A converters is selected and amplified to provide a negative analog signal out of FFN to be used as the fuel flow feedback signal to SFKK regulator.

Different frequency ranges at the input to the PR/A converter can be processed by making appropriate jumpers between A, B, C and D for No. 1 PR/A and E, F, G and H for No. 2 PR/A converter. The following table gives the required jumpers for various ranges:

FREQUENCY RANGES (Hz)	JUMPERS
0 to 4290	No jumpers
0 to 3340	A-D and E-H
0 to 2640	A-C and E-G
0 to 2180	A-B and E-F
0 to 1910	A-B-D and E-F-H
0 to 1660	A-B-C and E-F-G
0 to 1440	A-B-C and E-F-G-H

A comparator looks at the two analog signals from the PR/A converters and output ALARM becomes a "1" if they differ by more than a fixed percentage of the nominal signal. The selected high signal out of the converter is also compared to fuel flow reference and gives a "1" out of TRIP if the feedback signal is greater than the reference and input 2WX is a "1". When input 2WX goes to a "0", output TRIP is inhibited.

ADJUSTMENTS:

ADJUSTMENT	POTENTIOMETER
GAIN 1	R102
GAIN 2	R101
TRIP	R103

SFKK Card. (See Figure 3-26). The SFKK card contains the fuel regulator amplifier, a multiplier, and servo-valve fault detector. The reference to the regulator amplifier is the product of the signals on inputs MP and NBP. Typically, one reference input is received at VCE and an offset is added to remove the inactive portion of the signal. The offset signal is inverted and appears on pin VCEB. VCEB is connected to MP to provide one input to the multiplier. The regulator sums the reference signal from the multiplier with the feedback signal on FFN and modulates the servo valve.

ADJUSTMENTS:

ADJUSTMENT	POT	RANGE (volts D.C.)
ZERO	R103	1.3 to 2.6
STAB	R104	
BIAS	R105	
GAIN	R106	

Liquid Fuel Control (Fuel Flow and Position Feedback System).

The liquid fuel control system consists of outer and inner control loops. The outer loop is for flow control. It consists of an SFUC card (providing fuel flow feedback derived from magnetic pickups monitoring flow feedback derived from magnetic pickups monitoring flow divider speed) (See Figure 3-27) and an SFKK card which sums the flow reference signal (function of VCE and turbine speed) with the flow feedback to generate a flow error signal. The inner loop is for bypass position control. It consists of an SSVF card (Providing valve position feedback) (See Figure 3-28) and an SSVG card which sums the position reference and position feedback and provides forward gain to drive the liquid fuel servo valve.

Interface (outer loop forward gain) between the loops is provided by two circuits on an AOAB card. The AOAB (D) circuit integrates the flow error to generate the position reference signal, thus forming an integral flow control loop. The other circuit, AOAH (A), provides a fast ramp or initialization effect to a pre-defined valve position when VCE goes to the "FIRE" level or when changing from complete gas to liquid to minimize the effects of integrator wind-up. Figures 3-30 and 3-31 show block diagrams which include AOAH card circuitry.

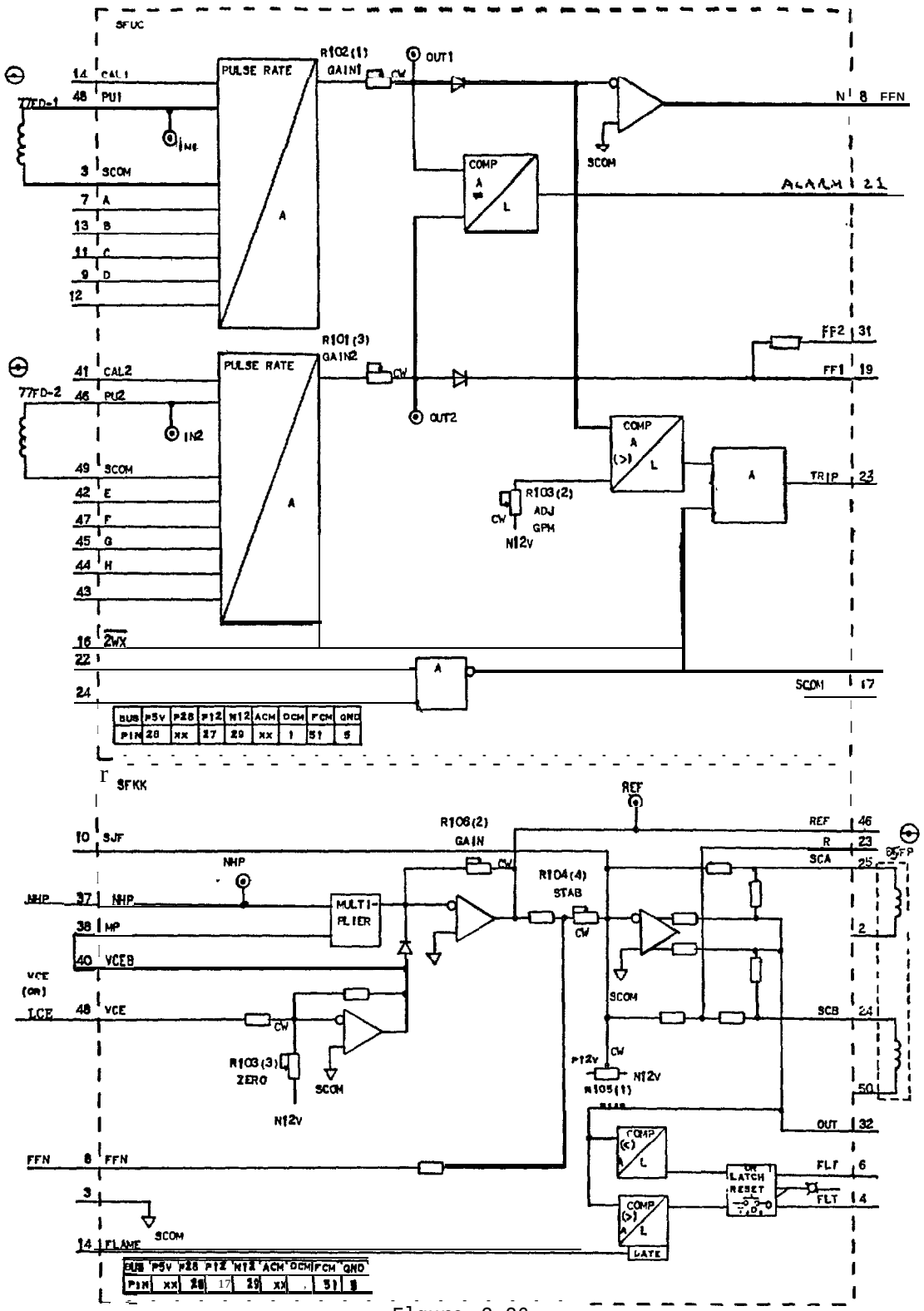


Figure 3-26
Liquid Fuel Control Cards
(Fuel Flow Feedback System)

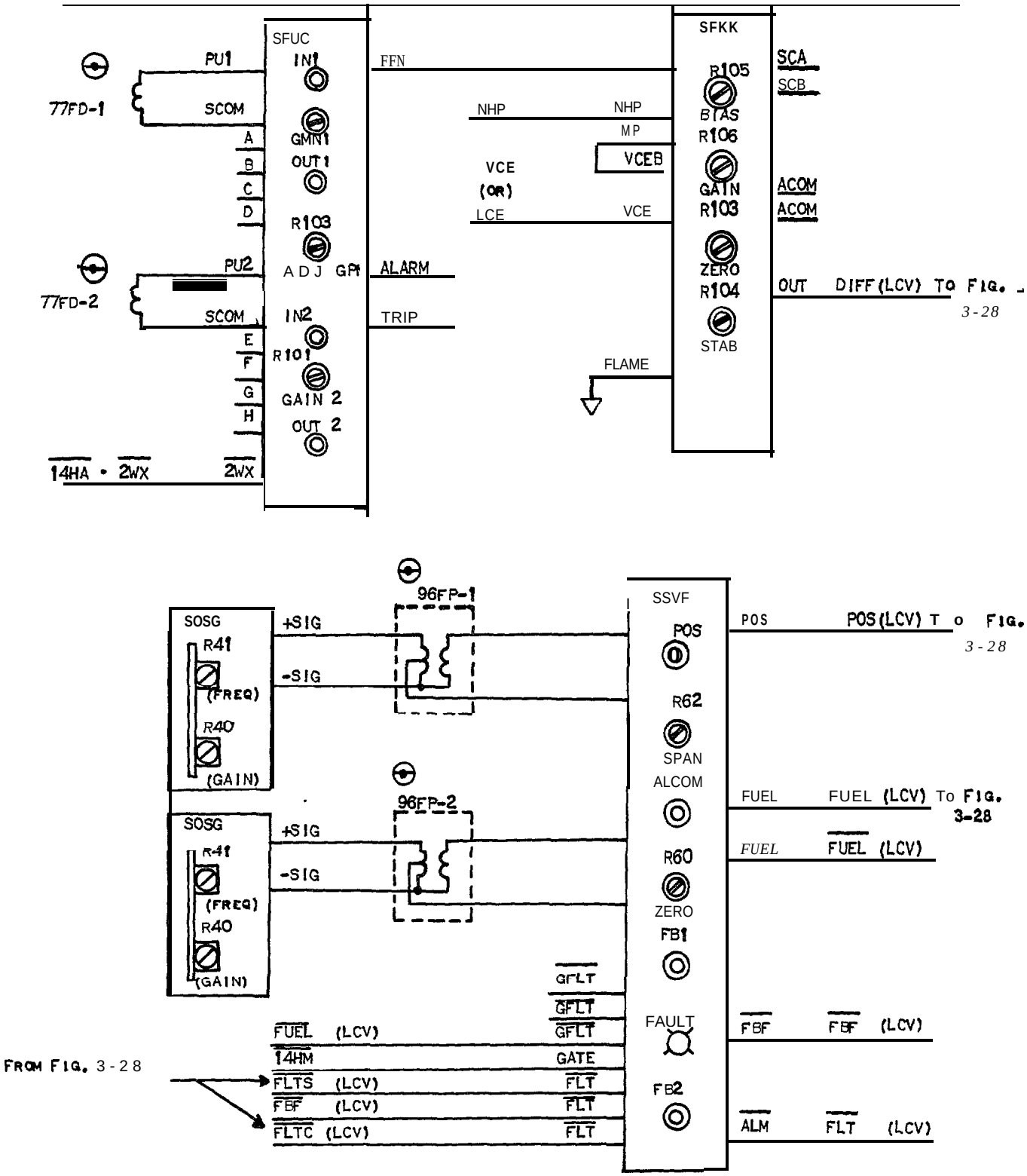


Figure 3-27
Liquid Fuel Servo Block Diagram (Part 1)
(Fuel Flow and Position Feedback System)

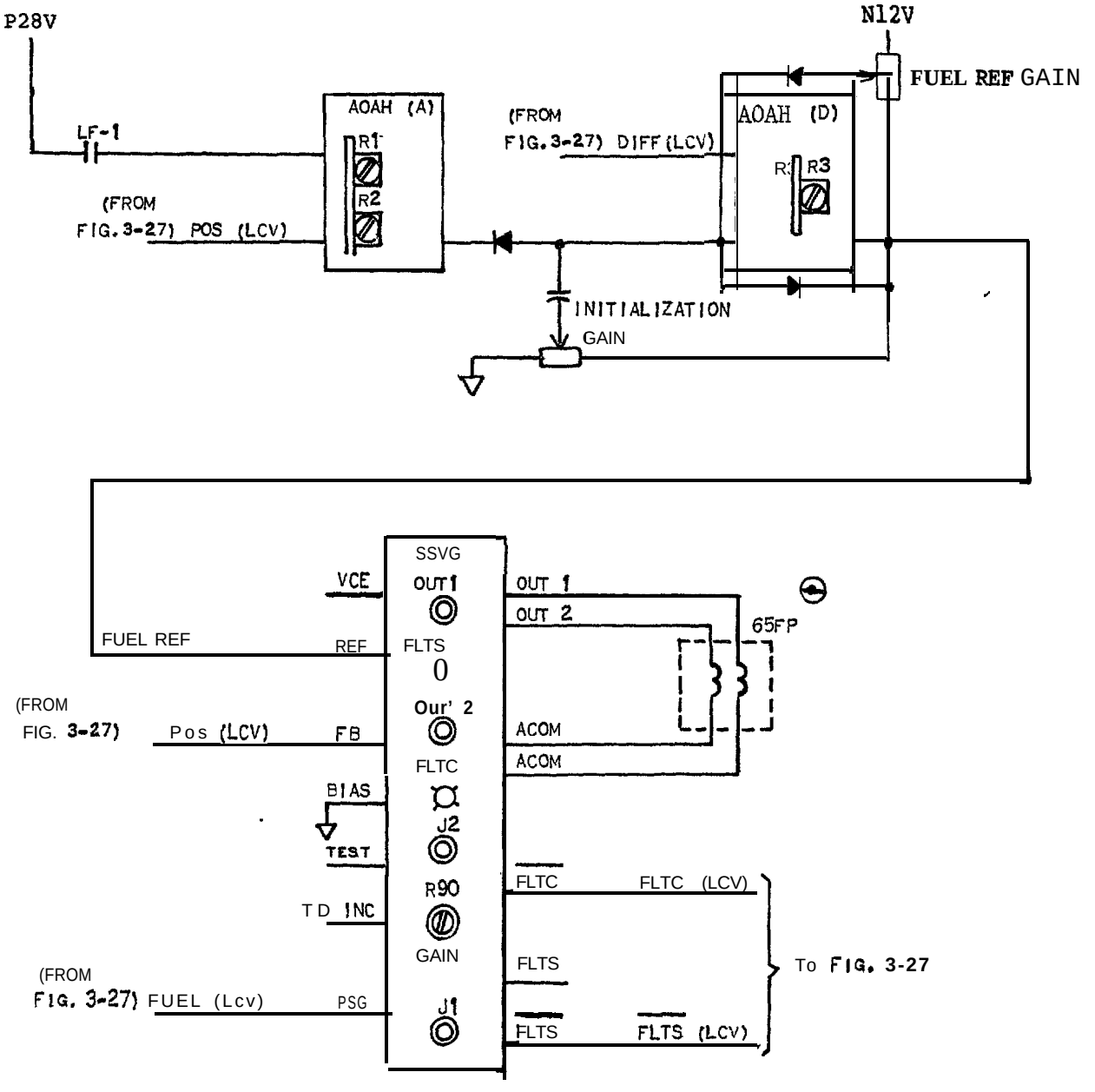


Figure 3-28
Liquid Fuel Servo Block Diagram (Part 2)

SFUC Card (See Figure 3-29). The SFUC card contains two PR/A converters. Each of these converters receives a signal from a separate sensor and provides an analog signal at OUT 1 and OUT 2 respectively which is proportional to the frequency of the signal from the sensor. The largest signal from the two PR/A converters is selected and amplified to provide a negative analog signal out of FFN to be used as the fuel flow feedback signal to the SFKK regulator.

Different frequency ranges at the input to the PR/A converter can be processed by making the appropriate jumpers between A, B, C and D for the first PR/A converter and E, F, G and H for the second PR/A converter. The following table gives required jumpers for various ranges:

FREQUENCY RANGES (Hz)	JUMPERS
0 to 4290	No jumpers
0 to 3340	A-D and E-H
0 to 2640	A-C and E-F
0 to 2180	A-B and E-F
0 to 1910	A-B-D and E-F-H
0 to 1660	A-B-C and E-F-G
0 to 1440	A-B-C and E-F-G-H

A comparator looks at the two analog signals from the PR/A converters and output ALARM becomes a "1" if they differ by more than a fixed percentage of the nominal signal. The selected high signal out of the converter is also compared to the fuel flow reference and gives a "1" out of TRIP if the feedback signal is greater than the reference and input 2WX is a "1". When input 2WX goes to a "0", output TRIP is inhibited.

ADJUSTMENTS:

ADJUSTMENT	POT
GAIN 1	R102
GAIN 2	R101
TRIP	R103

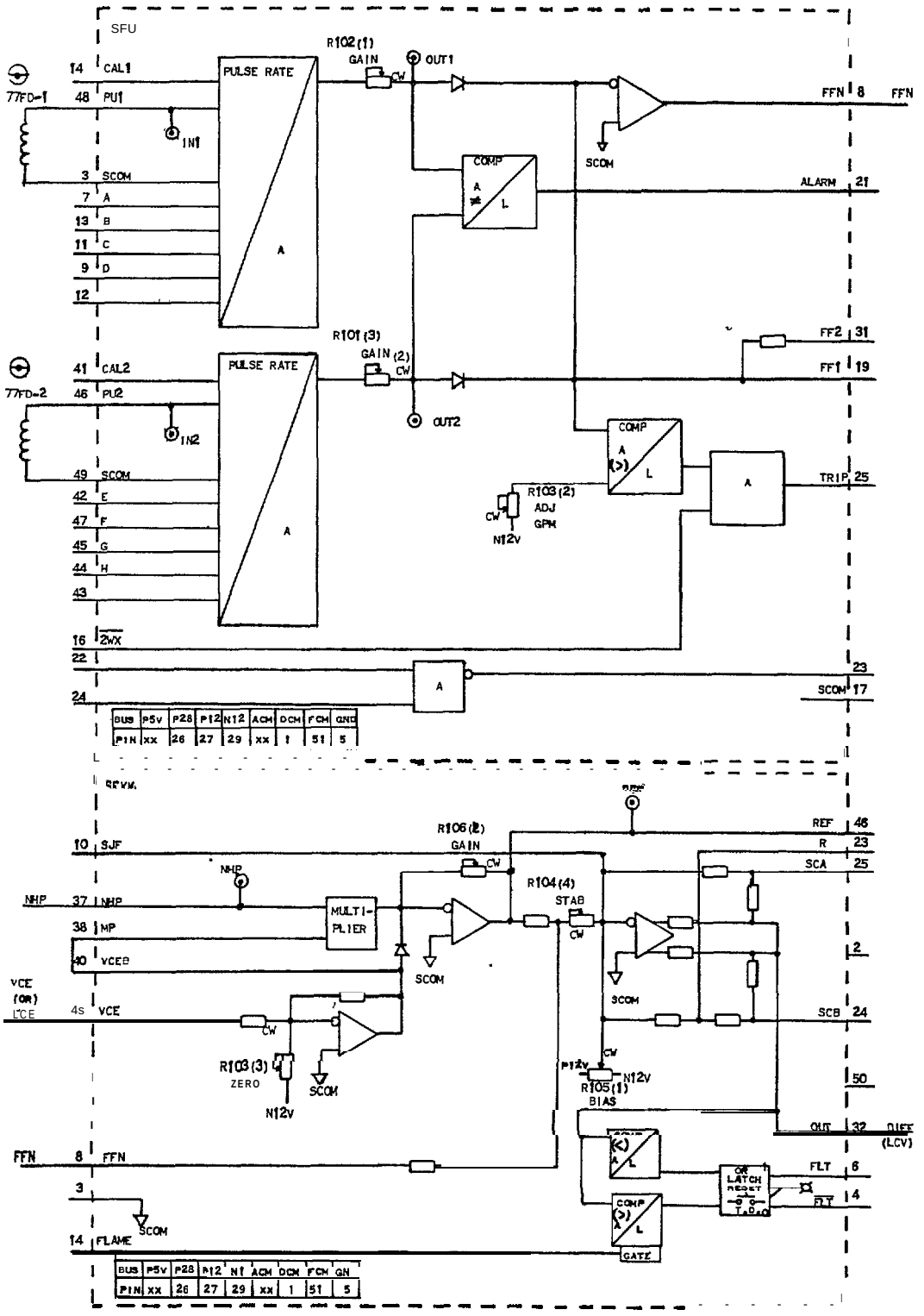


Figure 3-29
Liquid Fuel Control Cards (Part 1)
(Fuel Flow and Position Feedback System)

SFKK Card. (See Figure 3-29). The SFKK card contains the fuel regulator amplifier, a multiplier, and servo valve fault detector. The reference to the regulator amplifier is the product of the signals on inputs MP and NHP. Typically, one reference input is received at VCE and an offset is added to remove the inactive portion of the signal. The offset signal is inverted and appears on pin VCEB. VCEB is connected to MP to provide one input to the multiplier. The regulator sums the reference signal from the multiplier with the feedback signal on FFN and provides a reference to the servo card SSVG.

ADJUSTMENTS:

ADJUSTMENT	POT	RANGE (volts D.C.)
ZERO	R103	1.3 to 2.6
STAB	R104	
SPAN	R105	

SOSG Card (See Figure 3-35). The SOSG card contains an adjustable oscillator to supply a signal for the LVDTs. One SOSG card can drive a 200 milliampere RMS load at 6 to 8 volts. This will drive up to 4 LVDTs.

SSVF Card (See Figure 3-30). The SSVF card receives its input(s) signal from the LVDT and detects the peak value of the signal(s) and amplifies it to provide a negative feedback signal on POS for the SSVG regulator card. If either of the signals out of the peak detectors goes below a set threshold, output OFBF becomes a "1". A minimum no fuel signal of 0.5 volts RMS is required from the LVDTs to prevent the alarm signal on OFBF. A fuel detector circuit is also provided to give a "1" on output FUEL when the feedback signal reaches a set threshold.

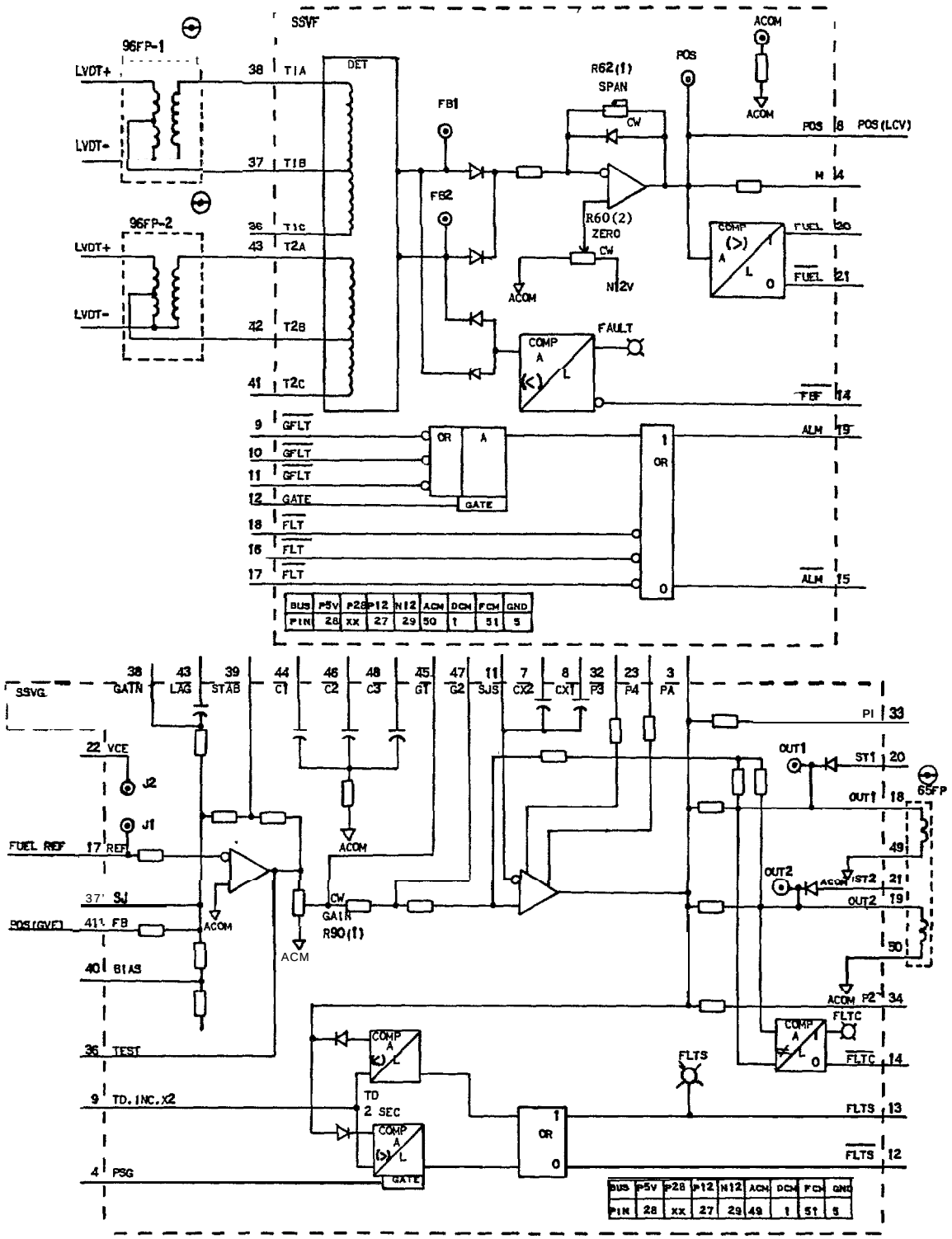


Figure 3-30
Liquid Fuel Control Cards (Part 2)
(Fuel Flow and Position Feedback System)

SSVG Card (See Figure 3-30). The SSVG card sums the feedback signal on FB with the reference signal on REF to modulate the servo valve. A signal on input VCE can be used as the reference by jumpering J2 to J1 an removing the signal on REF. A negative bias equivalent to 2 volts on REF is also summed with the reference to hold the gas control valve closed until the reference is greater than 2 volts. A voltage may be put on ST1 top impose a bias directly on the servo valve.

The card contains a comparator circuit which gives a "1" on FLTS and illuminates the lamp FLTS if the absolute value of the output voltage remains above 3.5 volts for an extended period of time. A comparator circuit is also provided to detect an open or shorted servo valve coil. Output OFLTC becomes a "0" in the event of an open or shorted coil.

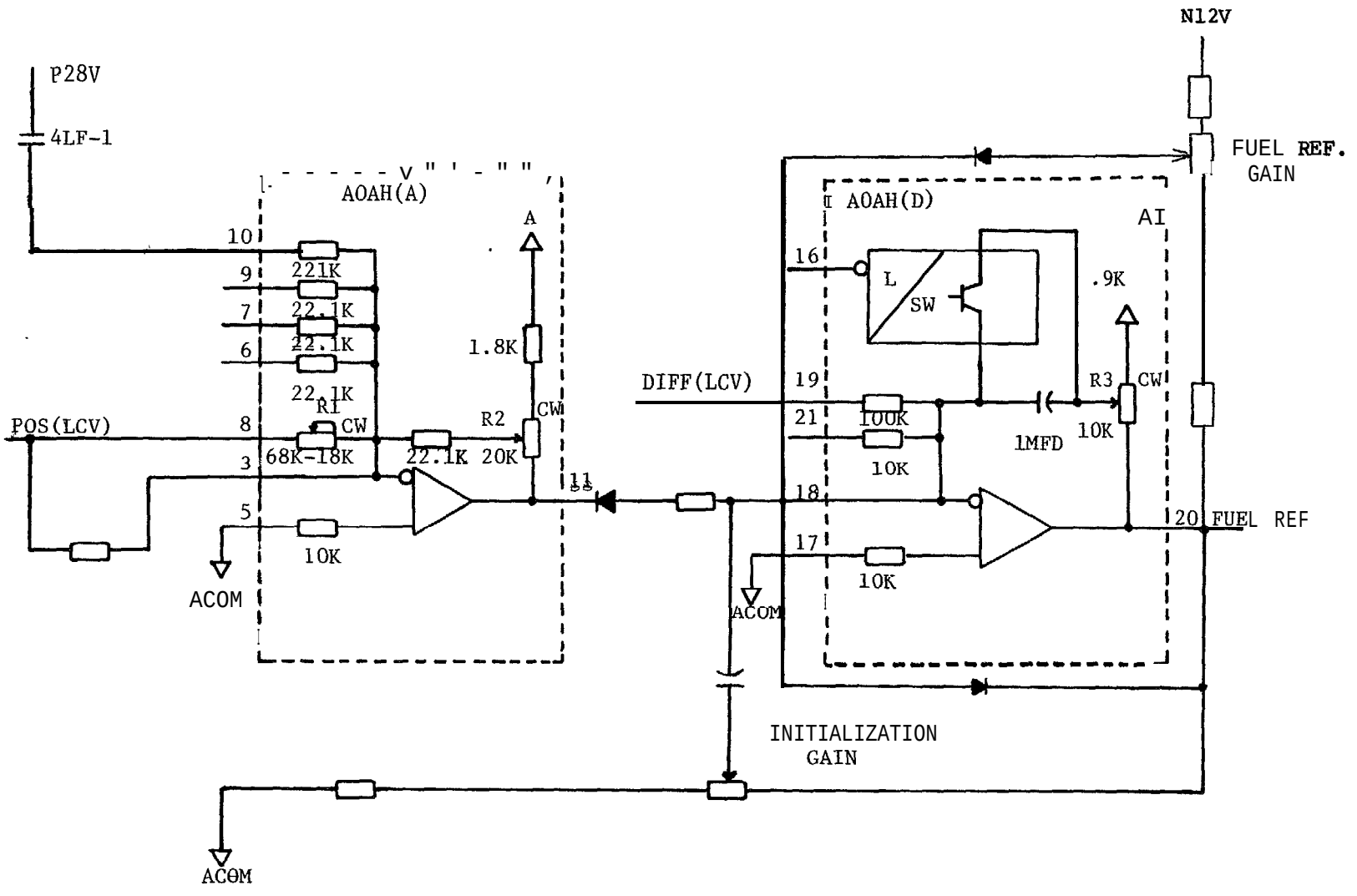


Figure 3-31
Liquid Fuel Control Cards (Fuel Flow and Position Feedback System)

Gas Control Valve (Optional)

A typical diagram of the gas control valve loop is shown in Figure 3-32. The SSVG regulator card receives a feedback signal from SSVF and compares it to a fuel reference signal to modulate the servo valve. The feedback is developed using a linear, variable displacement transformer (LVDT) whose coupling co-efficiency is a function of the control valve position. An oscillator of approximately 3 KHZ provides the input for the LVDT. The amplitude of LVDT secondary varies with the position of the control valve. Two LVDTs and oscillators may be provided for increased reliability. An alarm is generated if the feedback circuit fails. (See Figure 3-33.)

SOSG Card (See Figure 3-35). The SOSG card contains an adjustable oscillator to supply a signal for the LVDTs. One SOSG card can drive a 200 mA RMS load at 6 to 8 volts. (This will drive up to 4 LVDTs).

ADJUSTMENTS:

ADJUSTMENT	POT	RANGE
GAIN	R40	4.5 to 8.5 volts RMS
FREQ.	R41	2350 to 3359 Hz

SSVF Card. (See Figure 3-32.) The SSVF card receives its input signal from LVDT and detects the peak value of the signal and amplifies it to provide a negative feedback signal on POS for the SSVG regulator card. If either of the signals out of the peak detectors goes below a set threshold, output FBF becomes a "1". A minimum no fuel signal of 0.5 volts RMS is required from the LVDTs to prevent the alarm signal on FBF. A fuel detector circuit is also provided to give a "1" on output FUEL when the feedback signal reaches a set threshold.

ADJUSTMENTS:

ADJUSTMENT	POT
FB ZERO	R60
FB SPAN	R62

SSVG Card (See Figure 3-32). The SSVG card sums the feedback signal on FB with the reference signal on REF to modulate the servo valve. A signal on input VCE can be used as the reference by jumpering J2 to J1 and removing the signal on REF. A negative bias equivalent to 2 volts on REF is also summed with the reference to hold the gas control valve closed until the reference is greater than 2 volts. A voltage may be put on ST1 and ST2 to impose a bias directly on the gas control valve.

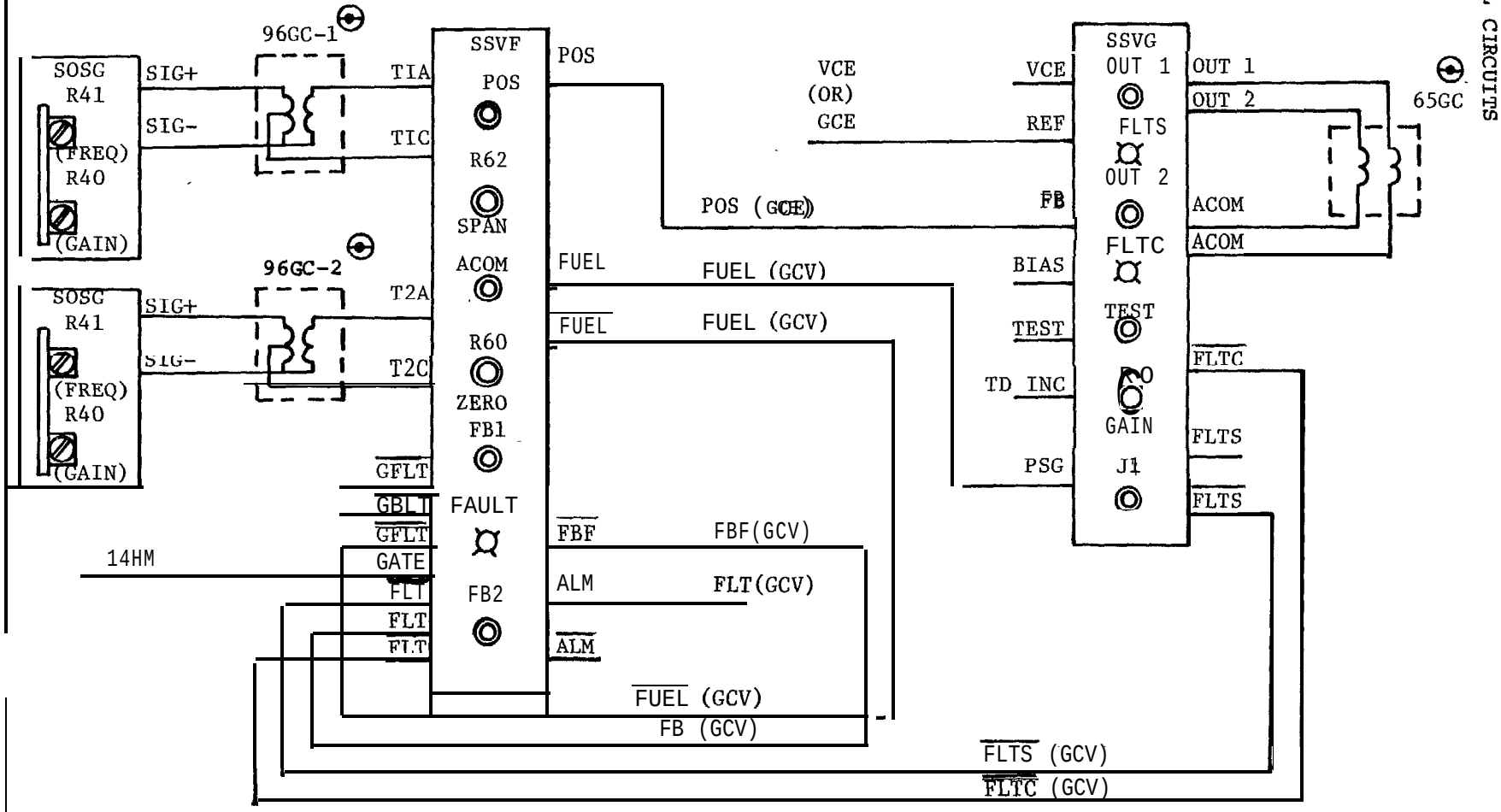


Figure 3-32
Gas Control Valve Block Diagram

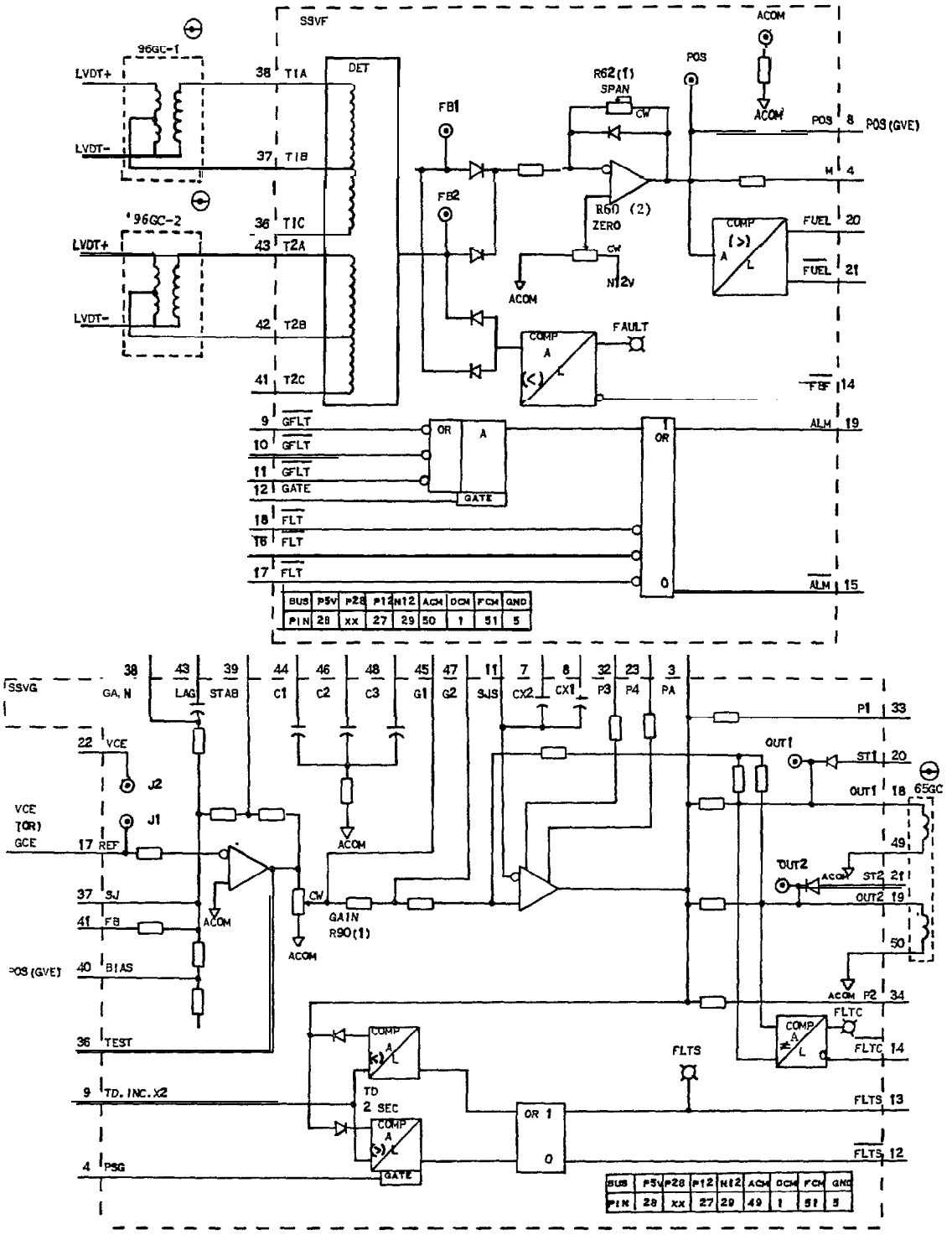


Figure 3-33
Gas Fuel Control Cards

The card contains a comparator circuit which gives a "1" on FLTS and illuminates lamp FLTS if the absolute value of the output voltage remains above 3.5 volts for an extended time period. A comparator circuit is also provided to detect an open or shorted servo valve coil. output FLTC becomes a "0" in the event of an open or shorted coil.

ADJUSTMENTS:

ADJUSTMENT	POT
GAIN	R90

STOP/SPEED RATIO VALVE (Optional)

When a gas fuel system is used, a stop/speed ratio valve is required to maintain the pressure upstream of the gas control valve at a value that is a function of turbine speed. (See Figure 3-34.)

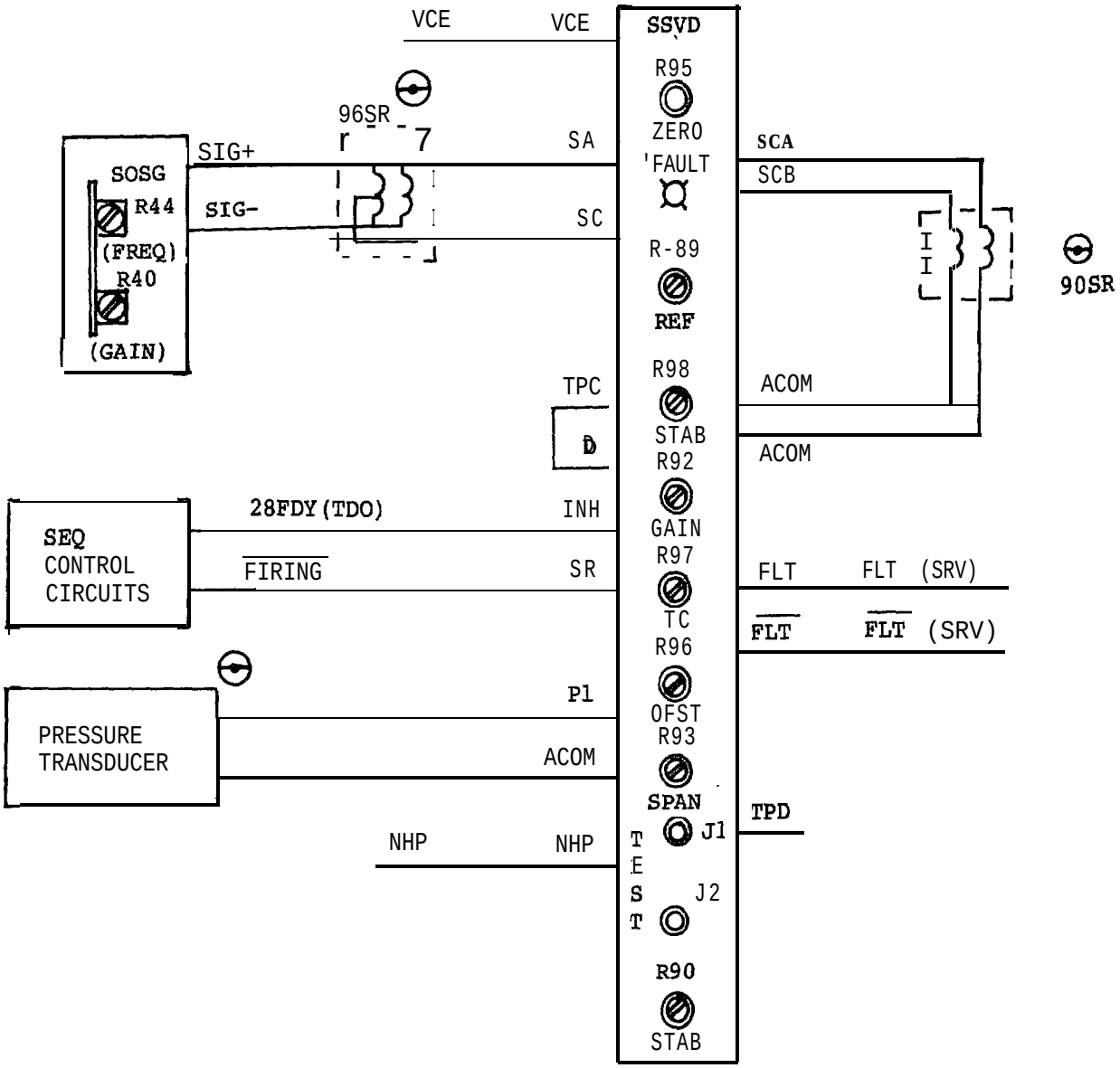


Figure 3-34
Speed Ratio Valve Block Diagram

SSVD Card (See Figure 3-35). The SSVD card receives a signal on P1 that is proportional to pressure. This signal is compared with the speed signal on NHP and provides an output on TPD. An offset and gain is provided on the NHP signal. The signal on TPD is the reference for the servo valve regulator on the card. A "1" on input SR will force TPD to zero volts.

The feedback signal for the regulator is derived using an LVDT as was done in the gas control valve. Only one LVDT is used in this case.

Alarm output FLT will be a "1" and the FAULT lamp will be illuminated for the following conditions:

(1) If TPC goes more positive than -1.2 volts indicating loss of feedback.

(2) If input D is not between -1.2 and -2.8 and input INH is a "0".

(3) when two SSVDs are used to drive the servo valve, C of each card can be tied to SCB of the other card to detect a faulty valve or circuit. For this case, an alarm will occur when the signals on SCB and C are of opposite sense. The alarm will be indicated on the card that SCB is more positive than C.

ADJUSTMENTS:

ADJUSTMENT	POT
ZERO	R95
GAIN	R92
PRESS SPAN	R93
NHP OFFSET	R96
STAB	R90

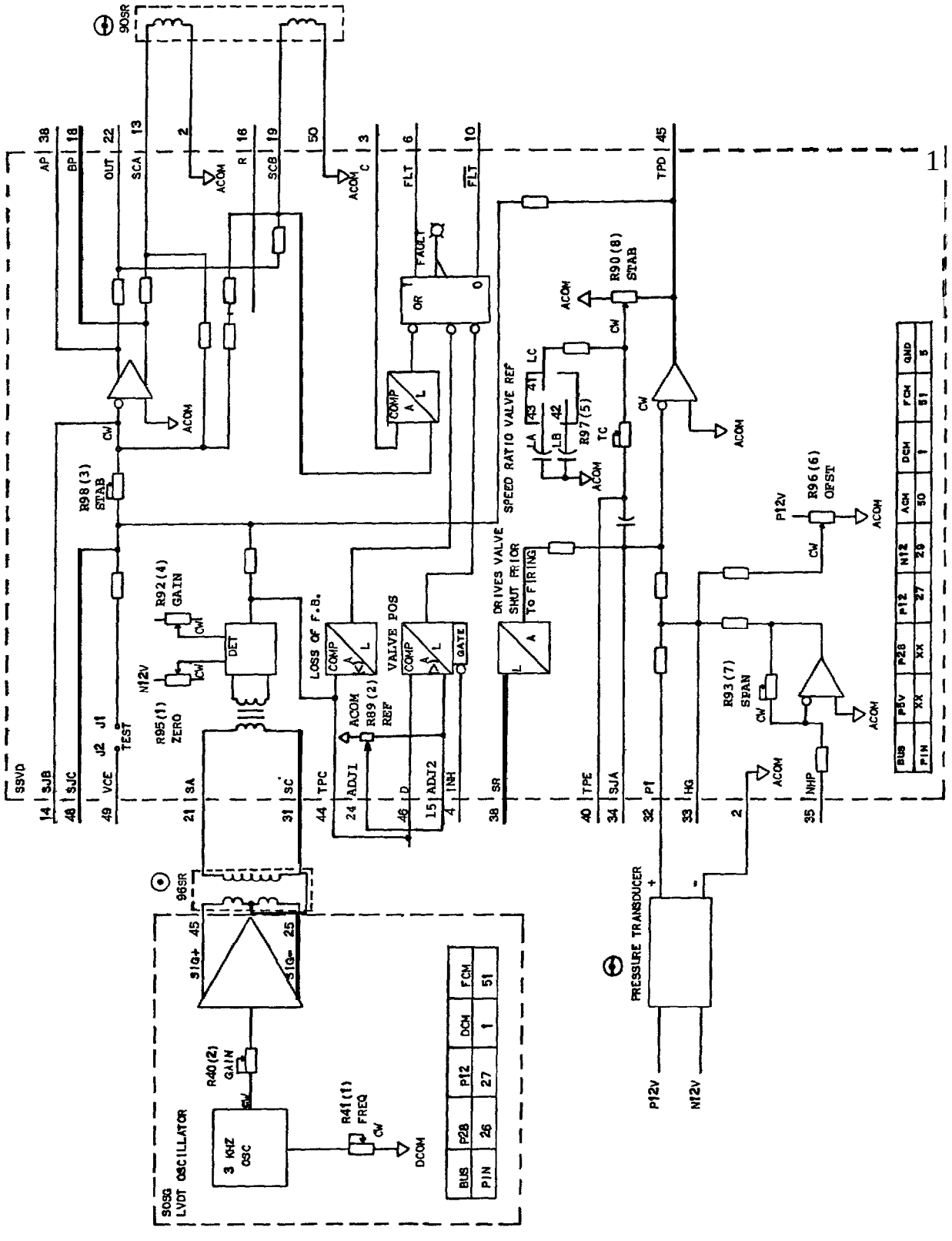


Figure 3-35
Speed Ratio Valve Control Cards

Digital Setpoint

The digital setpoint produces a programmable analog signal (DSP) used as a speed or load signal to the speed control card. The system consists of cards VDAD, SSLE and associated control signals. (See Figure 3-36).

VDAD Card (See Figure 3-7) The VDAD card contains a 12-bit reversable counter, a digital-to-analog (D/A) converter and twelve light emitting diodes (LEDs) to indicate the output of the counter. A logic 0 on the ENAB input permits the counter to count down or up as a function of the logic state of the DN/UP input. With ENAB at logic 0 and DN/UP at logic 0, the counter counts up. The CCP input is a pulse rate signal which sets the rate at which the counter counts. CCP is a pulse train signal which causes the counter to increment/decrement in response to the steering logic. When the counter is empty and a down count is present, or when the counter is full and an up-count is present, output signals MM1, MM2 and MM3 will all be a logic 1.

SAOUT is the amplifier output from the D/A converter. The following full scale voltage output ranges are as follows:

1. Nominal 0 to 8V; no jumper between SJA and SJB.
2. Nominal -4V to +4V; jumper SJA to SJB.

Binary inputs 1MSB and 1B2 thru 1B12, along with signal load, provide the VDAD card with a "preset" capability. To set and clear desired bits, the following three actions are taken simultaneously: (1) apply a logic 1 to the bits to be set, (2) a logic zero to the bits to be cleared, and (3) apply a single shot signal (⊓) to the LOAD line.

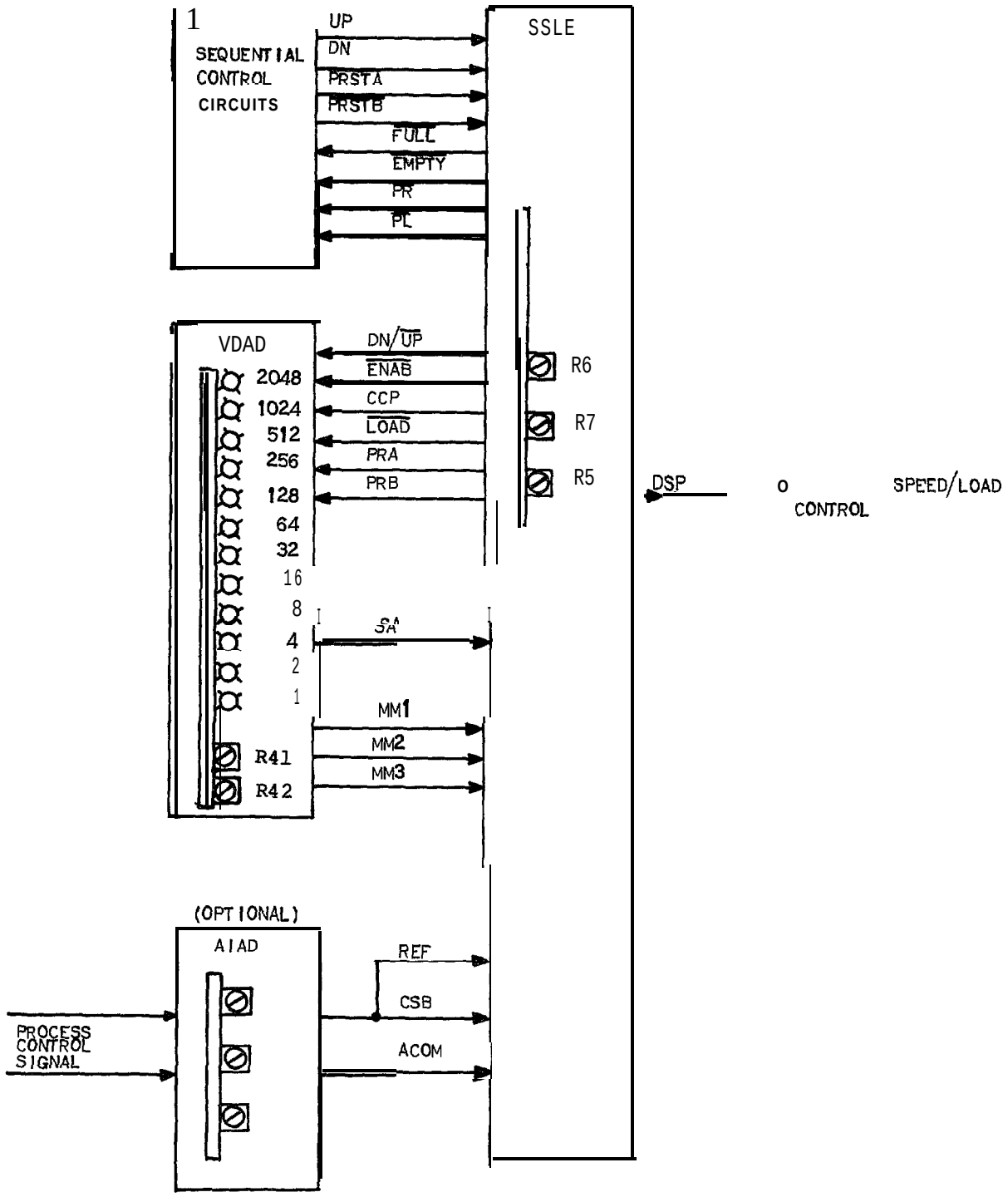


Figure 3-36
Digital Setpoint Block Diagram

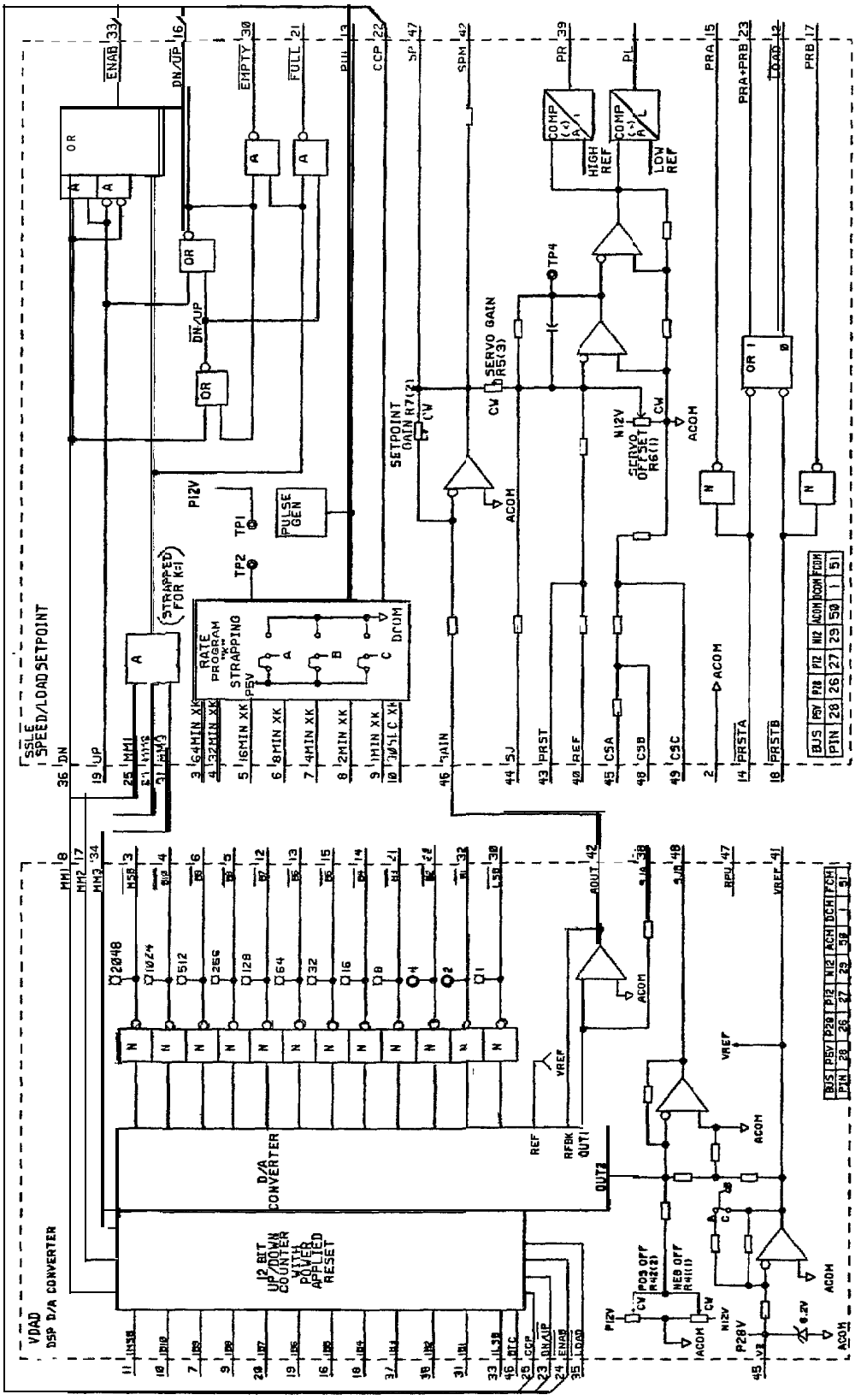


Figure 3-37
Digital **Setpoint** Cards

SSLE Card (See Figure 3-371. The SSLE card is used in conjunction with the Previously described VDAD card to provide the digital **setpoint** signal for speed/load control. The D/A output signal from the VDAD card is fed into the SSLE card and amplified to produce the digital **setpoint signal.** The op amp on the SSLE card provides signal inversion and potentiometer R7 provides for signal gain adjustment. Signal offset is provided by **R41/R42** on the VDAD card.

The SSLE card also has a comparator servo circuit. One of the inputs to this comparator is the digital **setpoint** signal. The other input, a reference signal, may be either a current or voltage signal. Current signal input points are CSA, CSB and CSC for l-5 ma, 4-20 ma and **10-50** ma signals respectively. The voltage signal input point is REF with a voltage range of 1 to 5 volts or 4 to 8 volts. If the input signal amplitude is less than that of the digital **setpoint** signal, comparator output PL will be a logic 1. If the input signal amplitude is greater than that of the digital **setpoint** signal, comparator output PR will be a logic 1. The servo gain potentiometer R5 is provided to enable matching the digital **setpoint** voltage range to the range of the input signal. The comparator circuit **deadband** adjustment is also provided by **R6.**

VDAD input signals ENAB, **DN/UP** and CCP are SSLE card output signals. The frequency of output signal CCP is determined by the wiring of and logic input to the Rate program. A logic high **"1"** will add the selected ramp time. There are eight periods of time obtainable for travel from "counter full" to "counter empty". They are: (1) 64 minutes, (2) 32 minutes, (3) 16 minutes, (4) 8 minutes (5) 4 minutes, (6) 2 minutes, (7) 1 minute, (8) 30 seconds. Different selected periods can be obtained through sequencing logic which add the ramp time together. Pins "3-10" which are not used should be tied low.

External UP and DN logic signals, when combined with VDAD output signals **MM1, MM2,** and **MM3,** comprise the logic signals for counter full and counter empty outputs FULL and EMPTY.

The SSLE **setpoint** preset circuits allow presetting the **VDAD** counter to any two pre-selected values. The SSLE inputs for this circuit are PRSTA and PRSTB. **The VDAD** inputs are PRA, PRB, PRA + PRB and the single shot signal **(LOAD).**

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PROTECTION SYSTEM

"4" Circuit

The focal point of the protection system is the VMPA card which generates the "4" signal and the shutdown bias to the servo valves. (Due to the magnitude of its importance, a detailed description will be presented.) The symbol for the VMPA card is shown in Figure 4-l. To eliminate some confusion, a "*" is placed near those pins not normally used. A typical protection system illustrating the "4" bus is shown in Figure 4-2.

The VMPA card serves the dual function of monitoring input signals as permissives to start the turbine and as protective signals to trip the turbine. The "4" signal outputs are the function of a flip-flop which is set upon receipt of a start command when all permissives have been satisfied. With the flip-flop set, the "4" signal outputs permit the turbine to be started and operated. The flip-flop is reset when an abnormal operating condition is detected by protective monitoring devices. Resetting the flip-flop trips the turbine.

The VMPA card functions, inputs and outputs, are discussed below. The discussion is referenced to Figure 4-l by numbers shown inside a circle.

Sum "1" Gate. The sum "1" gate (refer to ① on symbol) generates the "start-check" permissive leg to set the flip-flop. The permissive is satisfied when all inputs to the sum "1" gate are "1"s. Once the flip-flop is set, this permissive is no longer required.

An AND expander point, SAE, is provided to expand the sum "1" gate by the use of LGXA cards.

Inputs: 14X, $\overline{14HM}$, 12HT-1, 12HT-2, 12HT-3, 12LT-1, 12LT-2, 12LT-3, 14HSX (signal inverted), SAE.

Output: $\overline{STCK}$ - goes to logic "0" when all inputs to sum "1" gate are "1"s.

Majority Logic. Two majority logic elements are used - one for HP overspeed ② , and another for LP overspeed ③ . The output of the HP logic is a "0"as long as two of the three inputs are "0". However, when two of the three inputs become "1"s, the output becomes a"1" indicating the unit should be tripped. The LP logic is ANDed with a speed signal OR circuit which bypasses the majority logic output during start-up. When in the run mode, the majority logic is active and operates in the same manner as the HP logic. The output of the LP logic is "0" for a safe condition and a "1" for a trip.

HP Logic Inputs: 12HT-1, 12HT-2, 12HT-3.

HP Logic Output: Internal output to the start/run gate ⑥.

LP Logic Inputs: 12LT-1, 12LT-2, 12LT-3, 14LS, 14HSX.

LP Logic Output: 12LT (connected to LT input of the start/run gate ⑥.

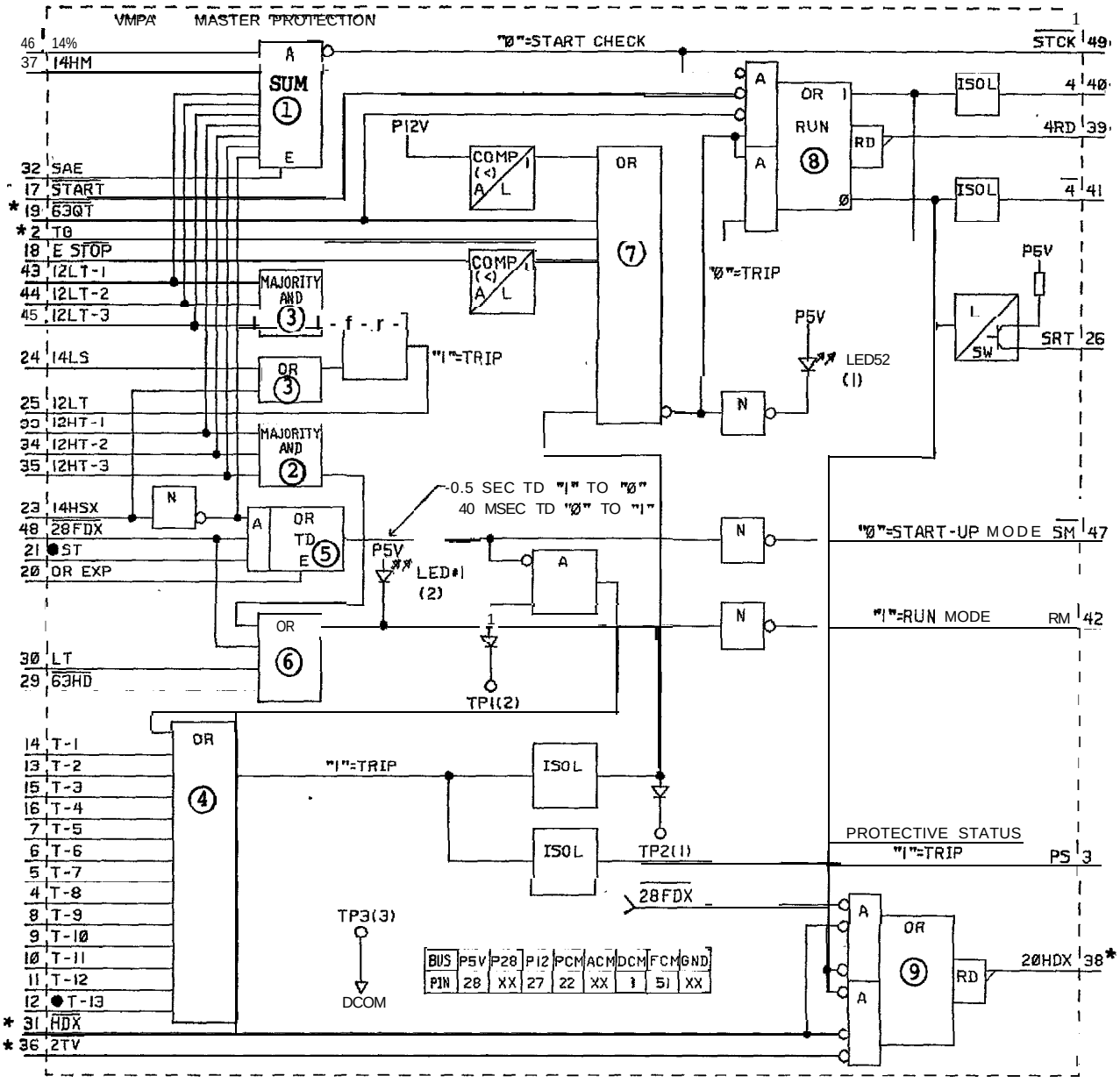


Figure 4-l
VMPA Card Elementary

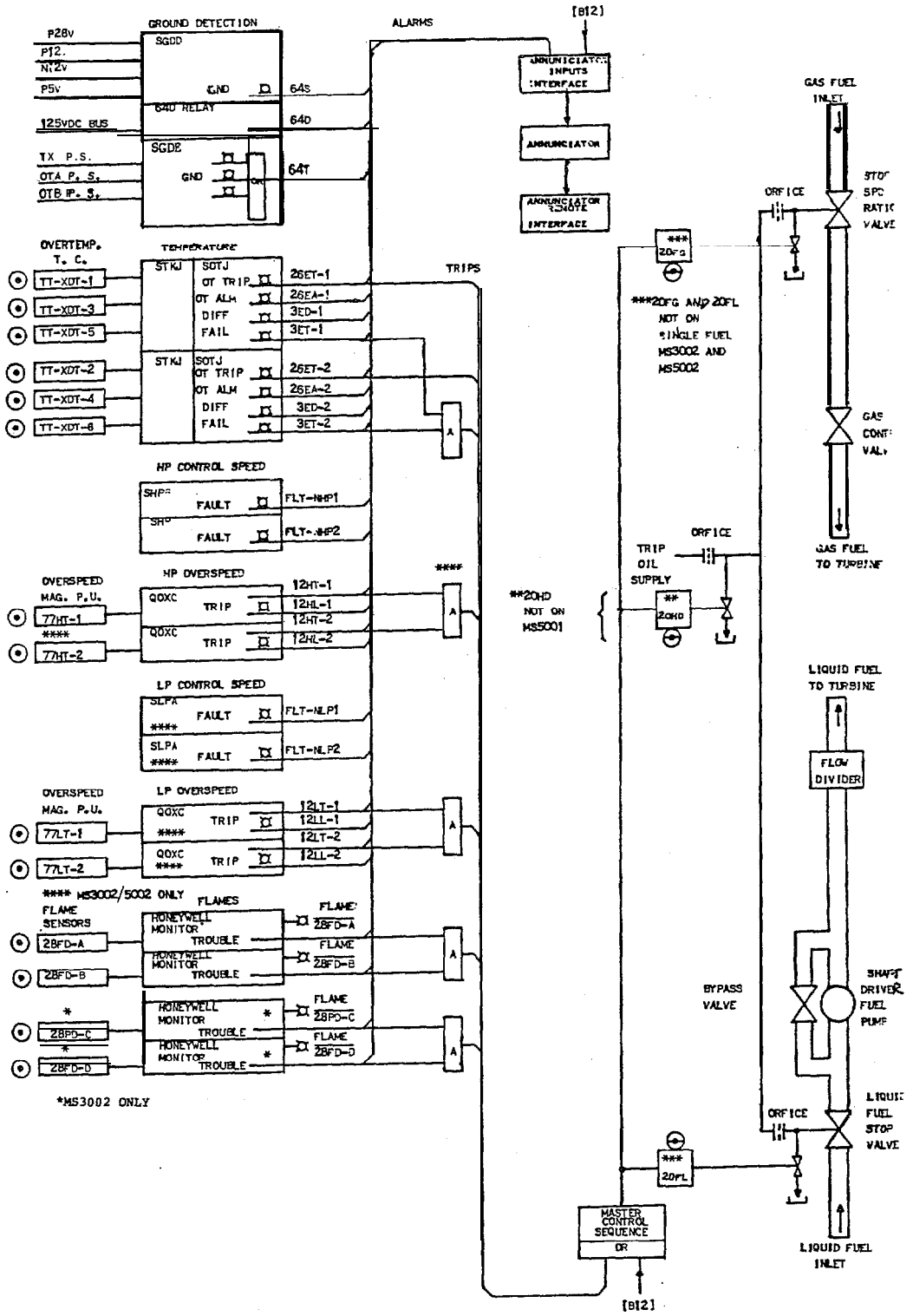


Figure 4-2
Typical Protective Block Diagram (Part 1)

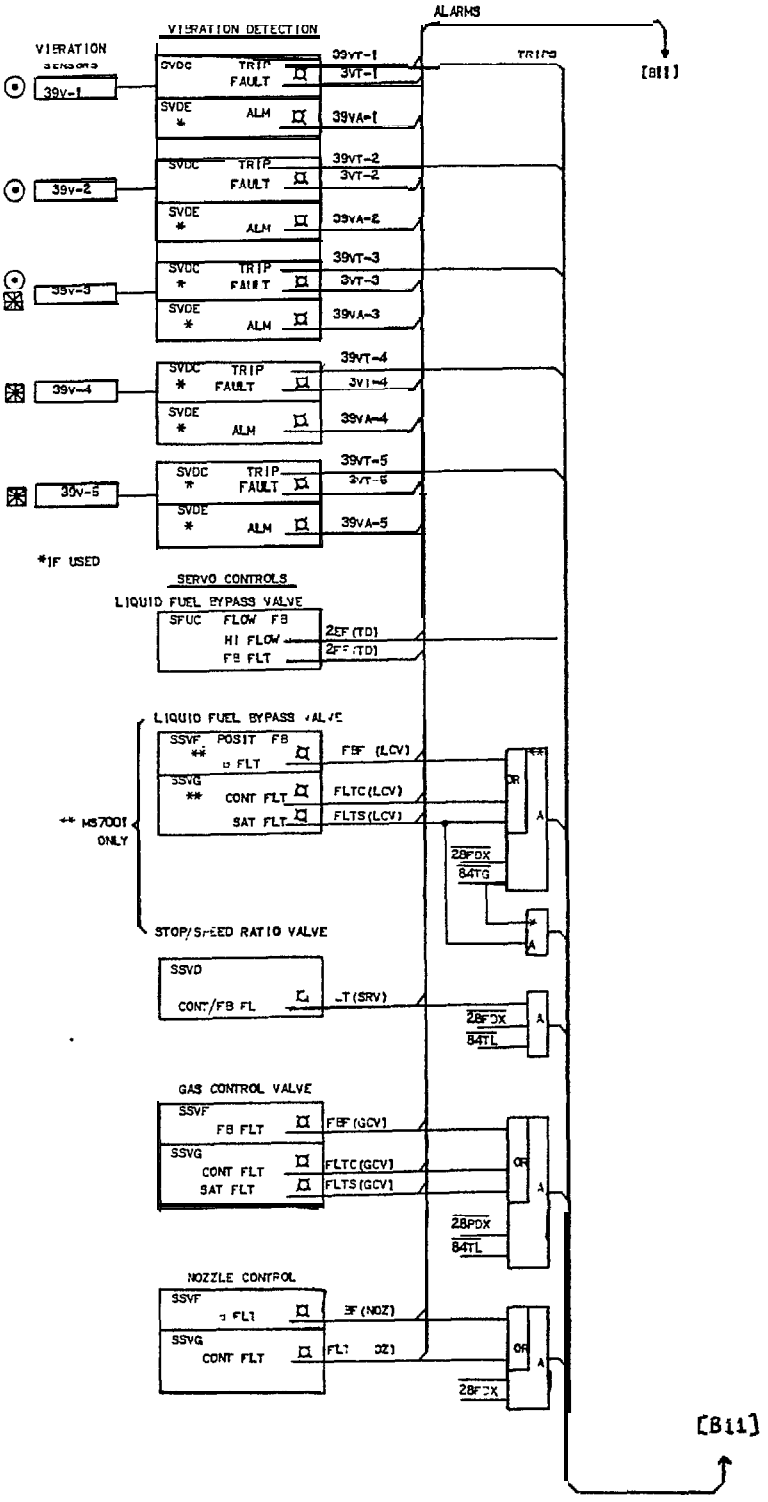


Figure 4-2
Typical Protective Block Diagram (Part 2)

OR "1" Gate The OR "1" gate (4) monitors trip protective devices. The gate output; a "0" when all inputs are "0"s. This "0" output is a permissive to operate condition. If any input goes to a "1" state, the output becomes a "1" and trips the turbine. Therefore, any unused input must be tied to common to allow the turbine to run. Approximately 0.7 milliseconds after the trip signal is initiated, the trip bus becomes a "1" to trip the turbine. Therefore, any unused input must be tied to common to allow the turbine to run. Approximately 0.7 milliseconds after the trip signal is initiated, the trip bus becomes a "1" to trip the turbine.

Inputs: T01, T02, T03, T04, T05, T06, T07, T08, T09, T10, T11, T12, T13; and internally the output of the Start/Run gate (5) and (6).

output: Internally tied to an isolator whose output is an input to the MASTER trip gate (7).

Start/Run Gate. The start/run gate consists of a gate for startup (5) and a gate for running (6) . The outputs of these two gates are ANDed to supply a signal which is an input to the OR "1" gate previously described.

The start-up portion is an AND gate with OR expansion capabilities. This gate must be satisfied prior to and during the start-up sequence. The OR expander portion is used to maintain the output during the firing period. This gate is de-energized when its inputs change state, normally upon detection of flame (28FDX). The output has a 0.5 second time delay on transmission from a "1" to a "0" to allow the run gate to be energized and maintain turbine operation. The output also has a 40 millisecond time delay on transition from a "0" to a "1" to insure tripping the turbine on loss of flame.

Inputs: $\overline{14HSX}$ from internal inverter, $\overline{28FDX}$, ST, and OR EXP.

outputs: Internal connection to AND gate with run portion, and inverted output $\overline{SM}$ for indication of start-up mode satisfied.

The run mode portion of the start/run gate provides monitoring of protective trip signals which must be bypassed during startup. This run gate is an OR "1" gate used for monitoring HP overspeeds, LP overspeeds, and hydraulic dump pressure. These signals are monitored for permissible state after flame has been established. A trip state on any one of the inputs, including loss of flame, will trip the turbine.

Inputs: Internal HP overspeed signal from majority logic, LT normally connected to output 12LT from LP majority logic, 63HD, 28FDX.

outputs: Internal connection to AND gate with start-up portion, and inverted output RM for indication of run mode satisfied.

Master Trip Gate. The master trip gate (7) is an OR gate which monitors various trip conditions. Detection of a trip condition by the OR "1" gate (4) will trigger the master trip gate. Loss of power as detected by internal comparators also causes a trip condition. Operating the emergency stop pushbutton results in a trip condition by disconnecting 28V power from input "ESTOP", which triggers a voltage comparator circuit connected to the master trip gate. Two additional external inputs, 63QT and TG, are also ORed into the master trip gate. All external input signals which are not used must be connected to DCOM because any logic "1" input will trigger the master trip gate.

Inputs: Isolated output of the OR "1" gate (4), output of P1 12V comparator circuit, output of P28V comparator circuit (P28V connected via external input "ESTOP"), 63QT, and TG.

outputs: Inverted output is an input to both AND gates of the MASTER flip flop (8).

Master Flip Flop. The state of the master flip flop (8) determines whether the unit may run or is tripped. Setting the flip flop permits the unit to operate. . Resetting the flip flop trips the turbine.

The flip flop is set by a four-input gate. This gate sets the flip flop only when the start-up checks are satisfied, input 63QT is a logic "0", the master trip gate (7) is dropped out (no trip conditions exist), and a start signal has been received.

The master flip flop is reset when the output of the master trip gate indicates detection of a trip condition as described in the foregoing paragraph entitled "Master Trip Gate".

Inputs: $\overline{\text{START}}$, INTERNAL OUTPUT OF THE SUM "1" gate (1), $\overline{\text{63QT}}$, and internal output of the master trip gate@.

outputs: 4, $\overline{4}$ - Logic outputs for sequencing.

4RD - Relay drive output to energize a relay when the master flip flop is set.

SRT - Servo trip bias to drive control valves to their proper shutdown state when the turbine is tripped.

Hydraulic Pump Circuit (not normally used). The hydraulic solenoid may be controlled by this driver output through a relay, (9). Two AND functions are ORed. The lower AND provides an output prior to firing, and the top AND provides an output during the run mode. All inputs on the ANDs must be "0"s to enable. Both AND circuits are controlled by the master flip-flop output and the $\overline{\text{HDX}}$ input.

Overspeed

Electronic over-speed protection is provided for each turbine shaft in addition to a mechanical overspeed protection system. Generally only one channel per shaft of electronic overspeed protection is provided for utility drive units, whereas mechanical drive units normally have two channels per shaft of electronic overspeed protection for improved operating reliability. Both channels must be in the trip state to trip the unit. A block diagram of a typical utility drive HP shaft protection circuit is shown in Figure 4-3.

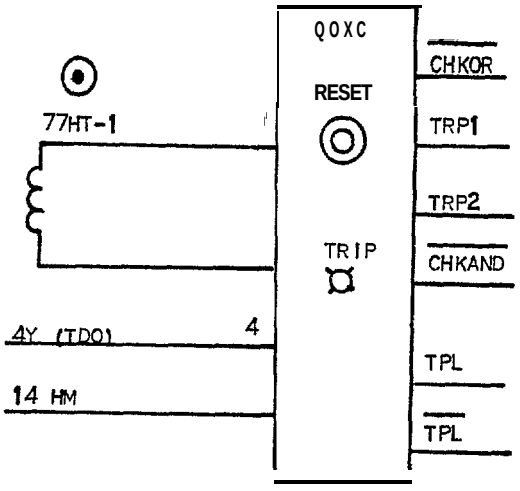


Figure 4-3
Single Channel Overspeed Trip Block Diagram

QOXC Card (See Figure 4-4). The QOXC card receives a frequency proportional to the speed of the turbine between pins IN and ACOM. Outputs TRP1 and TRP2 will be a "1" for the following conditions:

(1) Input speed is 10% above rated speed.

(2) Input speed is below 9% of rated speed when turbine is accelerating.

(3) Input speed is below 4% of rated speed upon deceleration of turbine.

Output $\overline{\text{CHKOR}}$ is a "0" if either TRP1 or TRP2 is a "1". Output $\overline{\text{CHKAND}}$ is a "0" if both TRP1 and TRP2 are "1". However, the primary purpose of the circuit is to generate a "1" on outputs TRP1 and TRP2 when the turbine is operating over a predetermined speed. A latching circuit is provided to hold the trip indication signal on TPL. Output TPL will latch in the "1" state if either TRP1 or TRP2 is a "1" and inputs 14HM and 4 are "1". The pushbutton on the card front must be depressed to clear the latch.

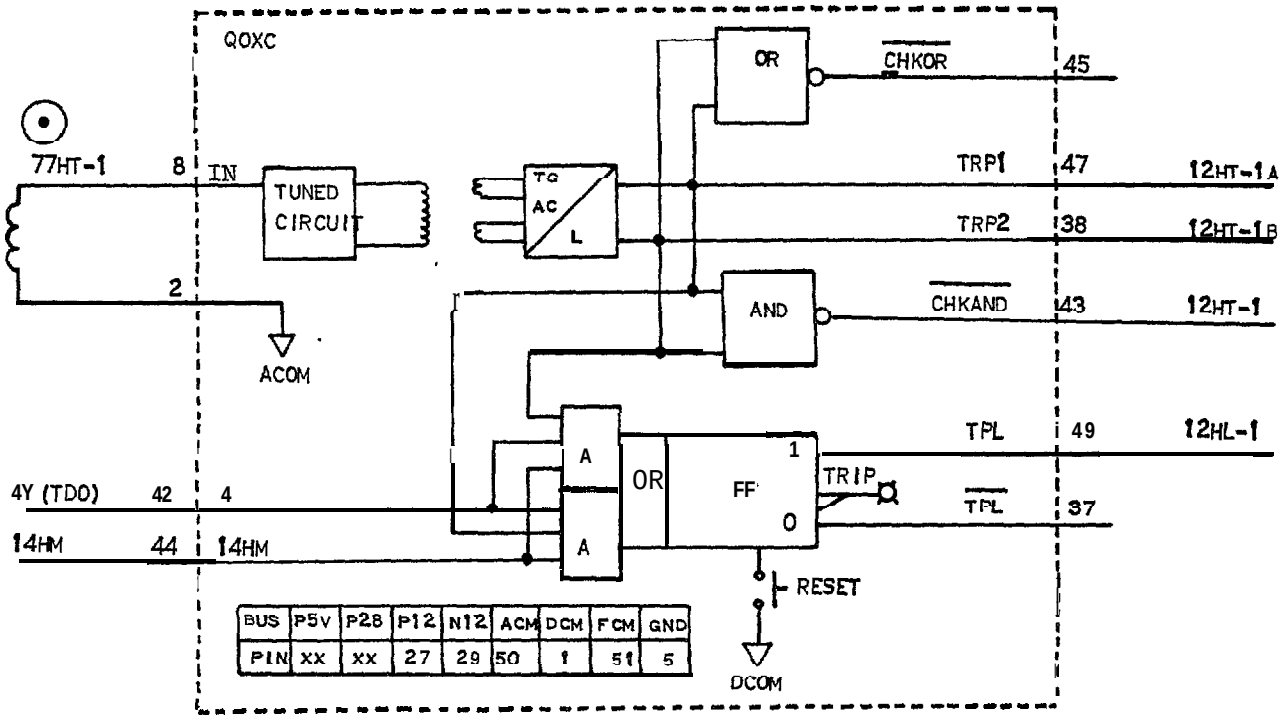


Figure 4-4
Overspeed Trip Card

Overtemperature Trip and Alarm

The overtemperature protection system consists of two separate channels. Each channel looks at the average value from three thermocouples located in the turbine exhaust duct. The overtemperature system includes two SOTJ cards, the temperature averaging cabinet and optionally an SPOA card. The averaging cabinet is the same one used in the temperature control loop and will not be discussed here. A block diagram of the protection circuitry is shown in Figure 4-5.

SOTJ Card. (See Figure 4-6). There are two SOTJ cards used for the overtemperature trip- and alarm function. One card is used for each overtemperature channel. Since the function of both cards is identical, only one is described here.

The SOTJ card provides the following five functions:

1. Overtemperature Trip Function: Generates a latch signal when the exhaust temperature exceeds a preset value. The circuit sums the exhaust temperature with ambient compensation (PCDN), pressure sensor failed (PCDF) and peak reserve bias if used. When the total exceeds the allowable value the autput will trigger and latch. It can only be reset with PB1.

2. Overtemperature Alarm: Indicates that the turbine is approaching the overtemperature trip point. This is adjustable by R2.

3. A Differential Alarm: Compares the two channels. If an adjustable difference of between 20°F and 100°F exists, an alarm is generated. This is adjustable by R3.

4. System Failure Trip: Provides a latched output when all of the overtemperature thermocouples read "cold."

5. A Transient Load Capability: Allows the turbine to take on extra transient loads between the 100 speed condition. This function is not used on single shaft machines-but is used on two shaft machines.

SPOA Card (See Figure 4-6). If the optional peak reserve function is provided, the SPOA card is used to re-bias the overtemperature trip and alarm points for this function. The bias is adjustable from 0°F to 100°F by R115 for one channel and R116 for the other.

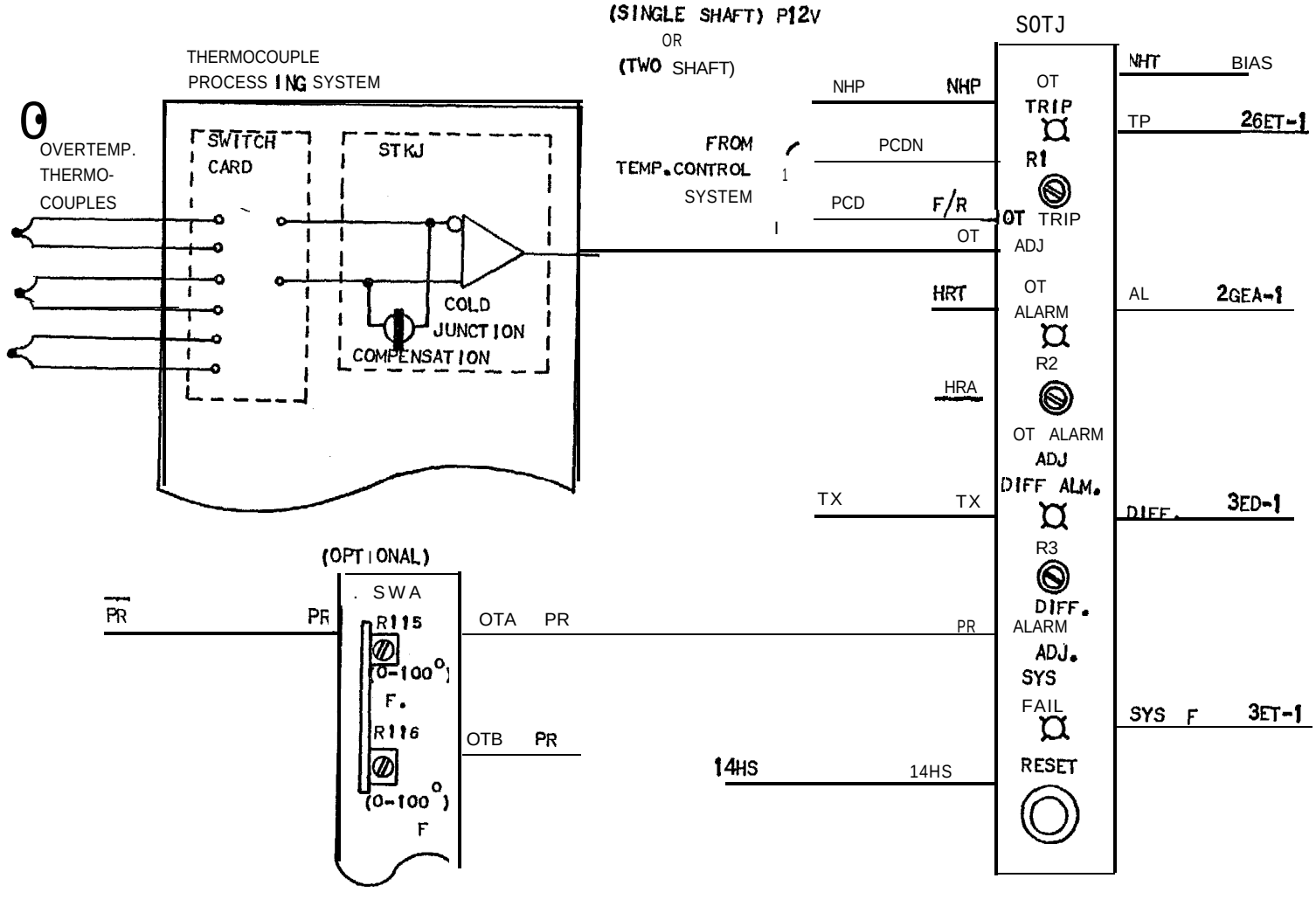


Figure 4-5
Overtemperature Alarm and Trip Block Diagram (One Channel)

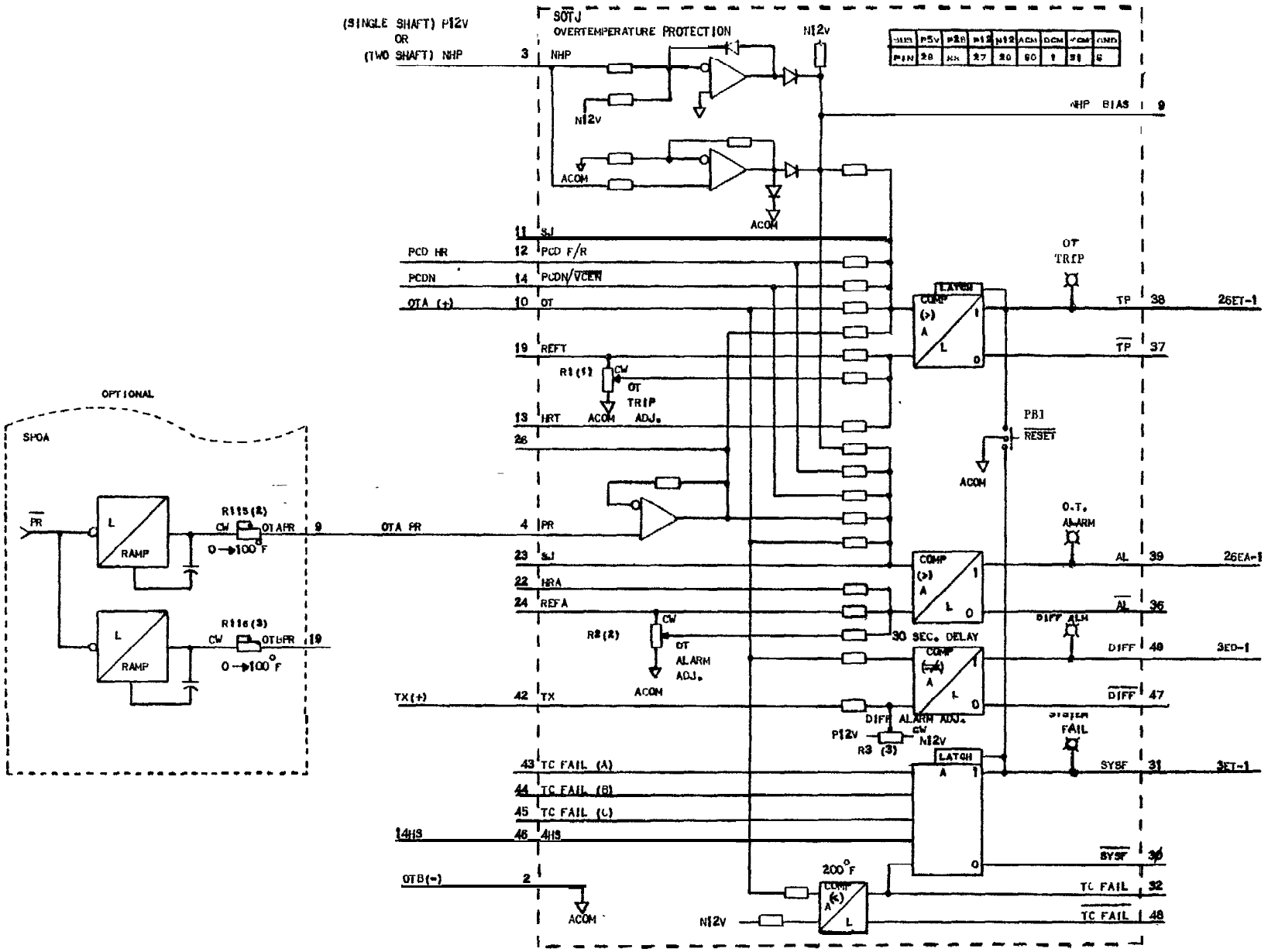


Figure 4-6
Overtemperature Alarm and Trip Cards (One Channel)

Vibration Protection

A vibration protection system may consist of up to five trip and alarm (optional) channels. A trip on any channel will trip the turbine. A trip is latching and must be reset by pushbutton or remote signal. Velocity vibration sensors provide the input signals. Each channel has circuitry to test for an open or shorted sensor. A typical block diagram is shown in Figure 4-7.

SVDC Card (See Figure 4-8). The SVDC card receives the signal from the sensor between pins VIB and ACOM. The peak value of this signal is detected and compared to a reference set by Rl. If the peak value of the signal exceeds the reference a "1" is latched up on output T and a light is illuminated. The latching circuit may be reset by the local reset pushbutton or a "1" on input RST. The setpoint reference may be monitored between TP3 and TP4. The peak value (VlB LEVEL) of the input signal may be monitored between TPl and TP2. If the sensitivity of the sensor used is 0.2 volts peak for 1 inch/sec. pin GAIN is connected to ACOM to double the gain of the detector.

A sensor fault detector is provided to detect a shorted or open sensor. This is a latching circuit and must be reset by local pushbutton or remote signal. The trouble light is illuminated and output "A" becomes a "1" for a faulty sensor. The following table lists the impedance required for no fault conditions:

JUMPERS	NO FAULT IMPEDANCE
None	2.9K to 50K ohms
Pin 9 - P12V	80 to 800 ohms
Pin 6 - 12v	160 to 1.5K ohms

ADJUSTMENTS:

ADJUSTMENT	POTENTIOMETER	R A N G E
SET PT .	Rl	0.25 to 1.6 in./sec.

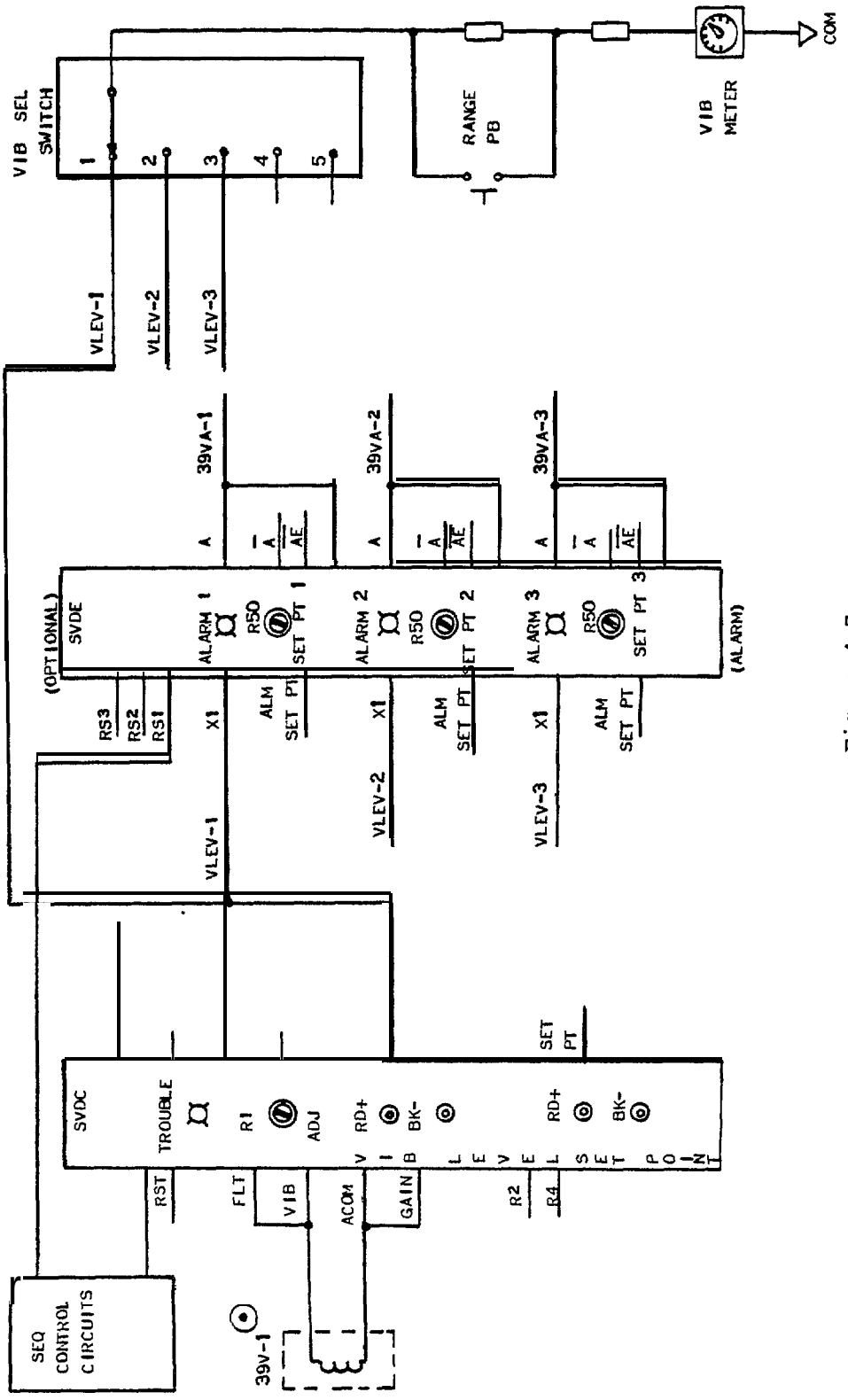


Figure 4-7
Vibration Protection Block Diagram (Alarm and Trip)

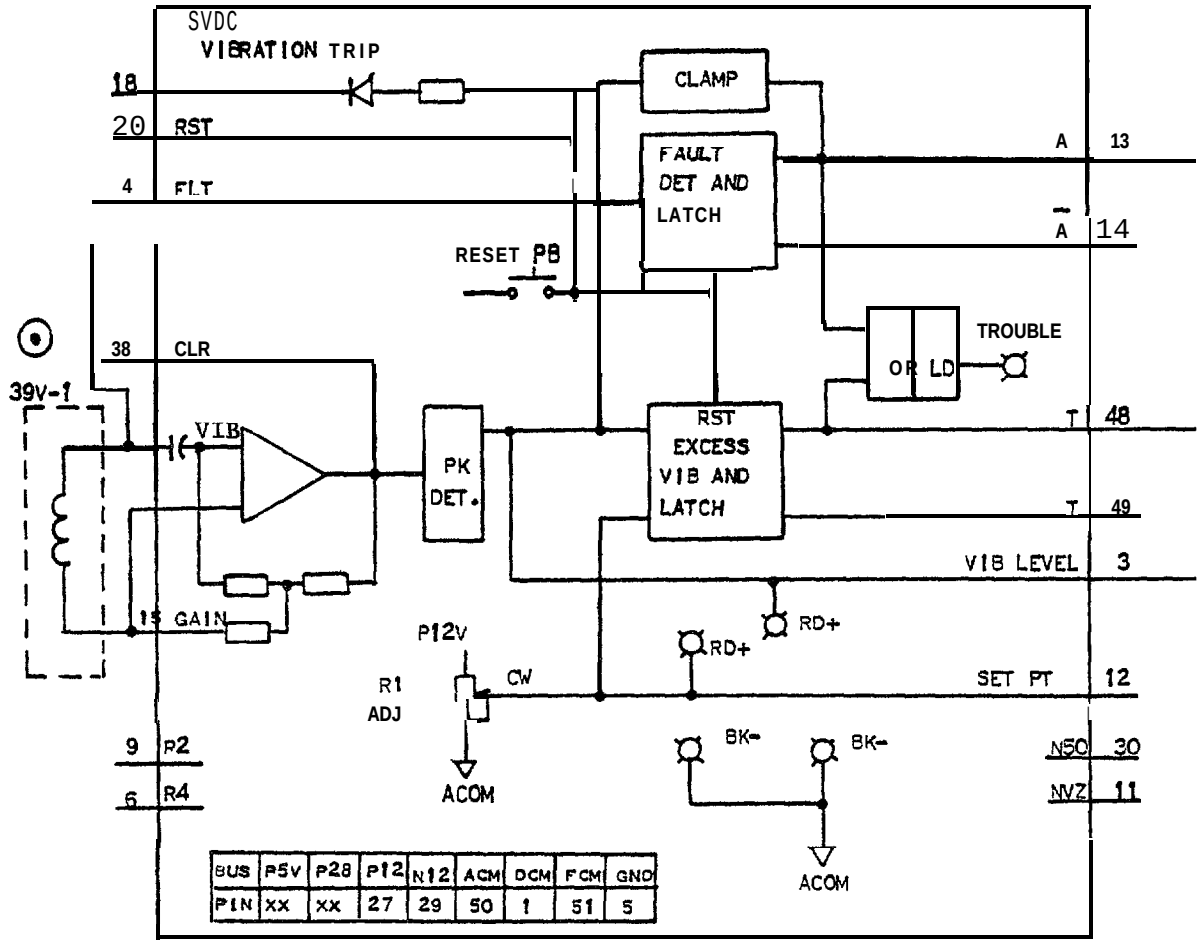


Figure 4-8
Vibration Protection Card

SVDE Card (Optional) (See Figure 4-9). The SVDE card contains three identical circuits. Only one of these circuits will be discussed. Typically, the input to the card is the 0 to 8 volts vibration level from the SVDC card and compares it to a setpoint reference. Input xl is used for a 0 to 8 volt signal. An input signal of 0 to 16 volts may be used by connecting J1 to J2 and receiving the signal on X2.

If the input signal reaches or exceeds the setpoint level, output A will become a "1" and the alarm light will illuminate. If output L is tied to A, the alarm will latch up until reset via reset pushbutton on card front, P12 volts on RS2 or "0" on RSl. If L is left open, the latch will reset when the input signal decreases 2% below alarm setpoint level. The setpoint level may be monitored at ALARM SETPT using a high input impedance meter.

ADJUSTMENTS:

ADJUSTMENT	POTENTIOMETER	RANGE
ALARM SETPT	R50	0 to 8 volts on Xl 0 to 16 volts on X2

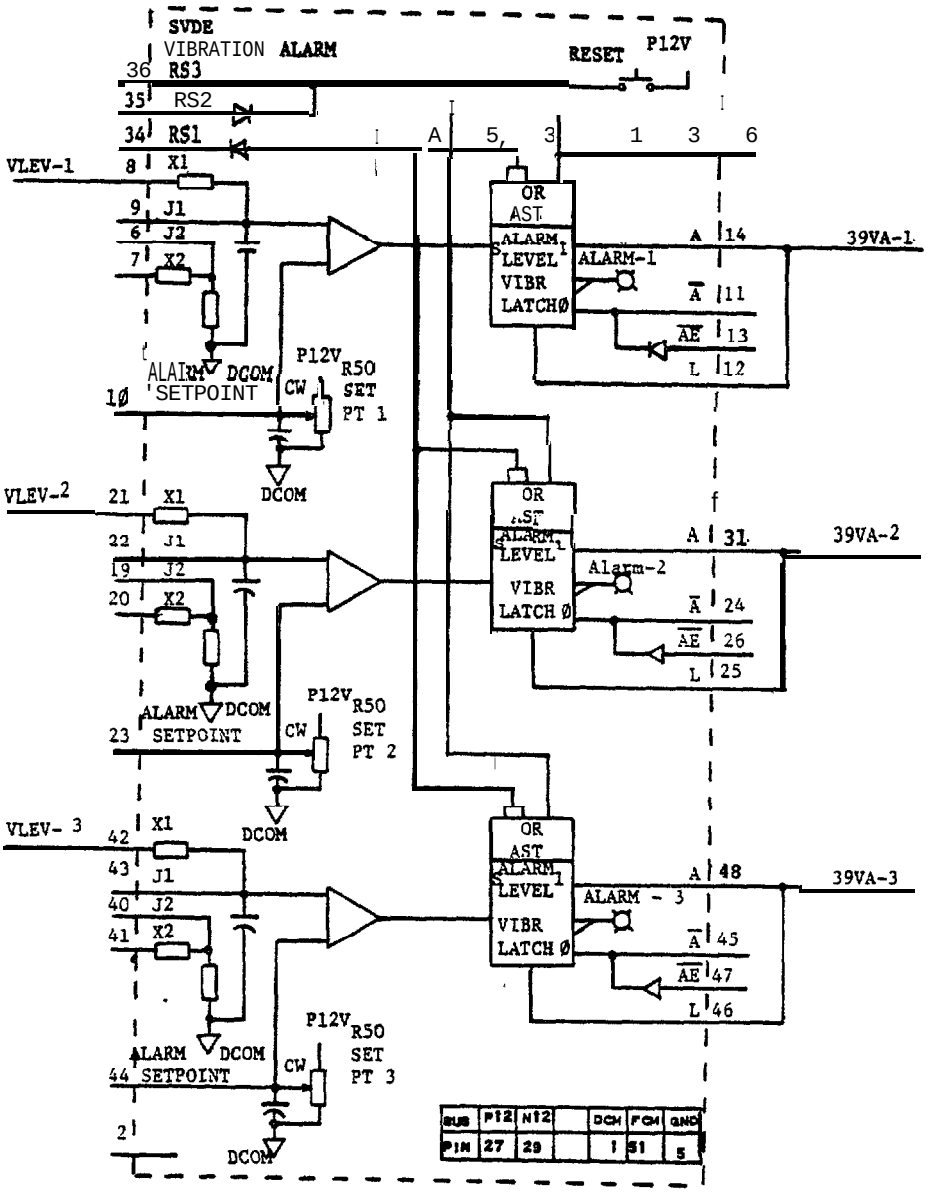


Figure 4-9
Vibration Alarm Card (Optional)

Flame Detector

A typical flame detection system is shown in Figure 4-10. Each channel of the flame detection system receives an input from a separate ultra-violet scanner. The Honeywell system is described in the Honeywell publication in Volume 11 of the instruction book.

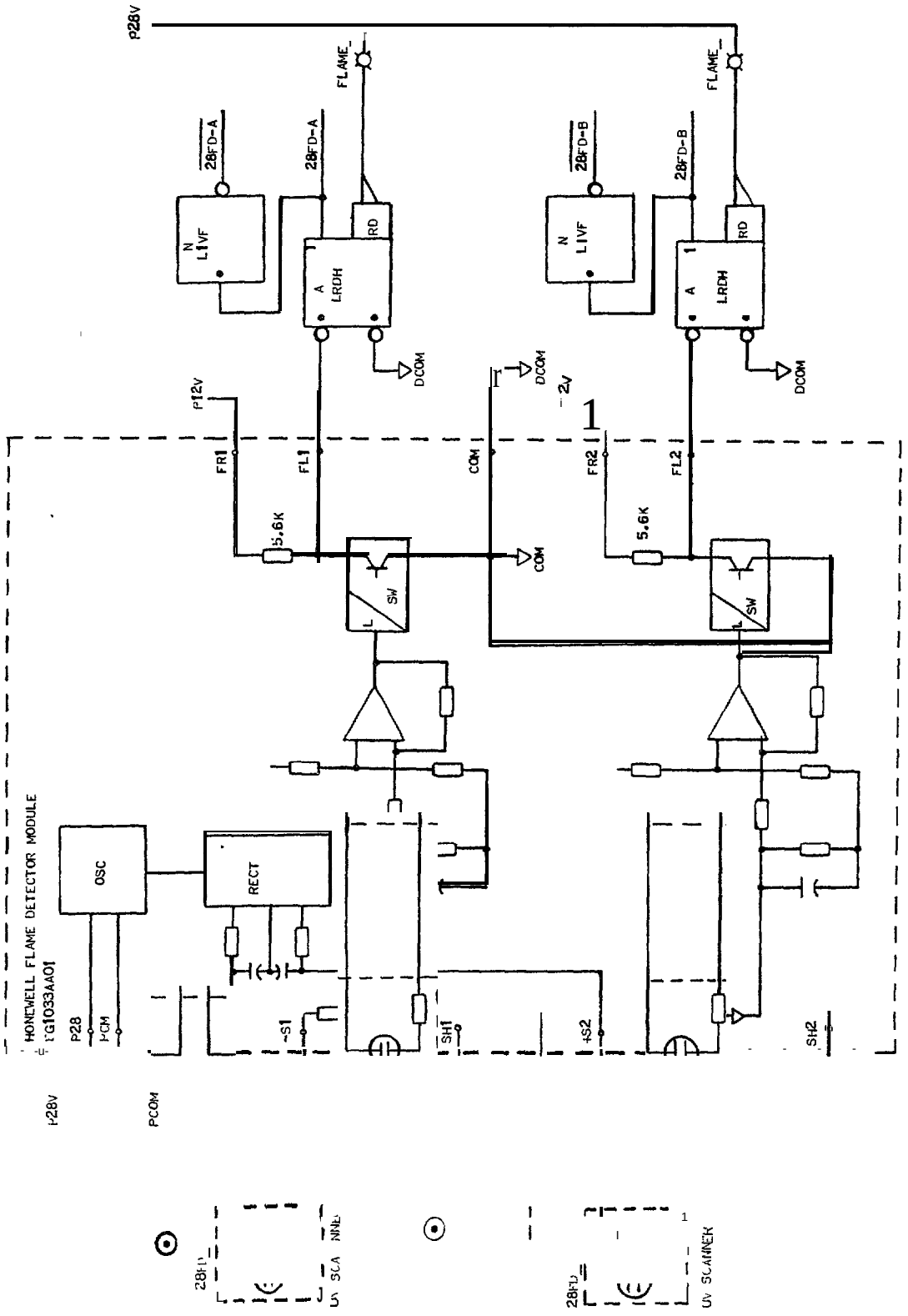


Figure 4-10
Honeywell Flame Detection System
(Two Channel-Typical)

Ground Detector

The turbine control power supply, thermocouple processing module power supply and the 125 VDC battery supply are floating, or ungrounded systems. Ground detector circuits are provided for the indication of a ground on the various voltage busses. Detected grounds are annunciated. If the Integrated Temperature System is used, the turbine control power supply and the Integrated Temperature System power supply are grounded but the 125 VDC battery supply remains a floating system. Therefore, ground detection circuits are provided only on the 125 VDC battery supply system.

Turbine Control Power Supply Detector (See Figure 4-11). The SGDD card provides ground detection for the turbine control power supply. The following table indicates impedance to ground for a fault condition:

P28	31K-51K ohms
P12	13.1K-21.7K ohms
P5.3	3.5K-7.5K ohms
ACOM	6K-10K ohms
N12	170K-270K ohms

The SGDD card also provides an analog output signal "OUT" which is an input to the thermocouple amplifier cards (STKJ) which are in the thermocouple processing module. The purpose of this analog signal is to cancel any error that may be induced in the STKJ circuits in the event that a bus and a thermocouple leakage to ground occurs. Offset and gain adjustments are provided by trimpot R40 and R25 respectively.

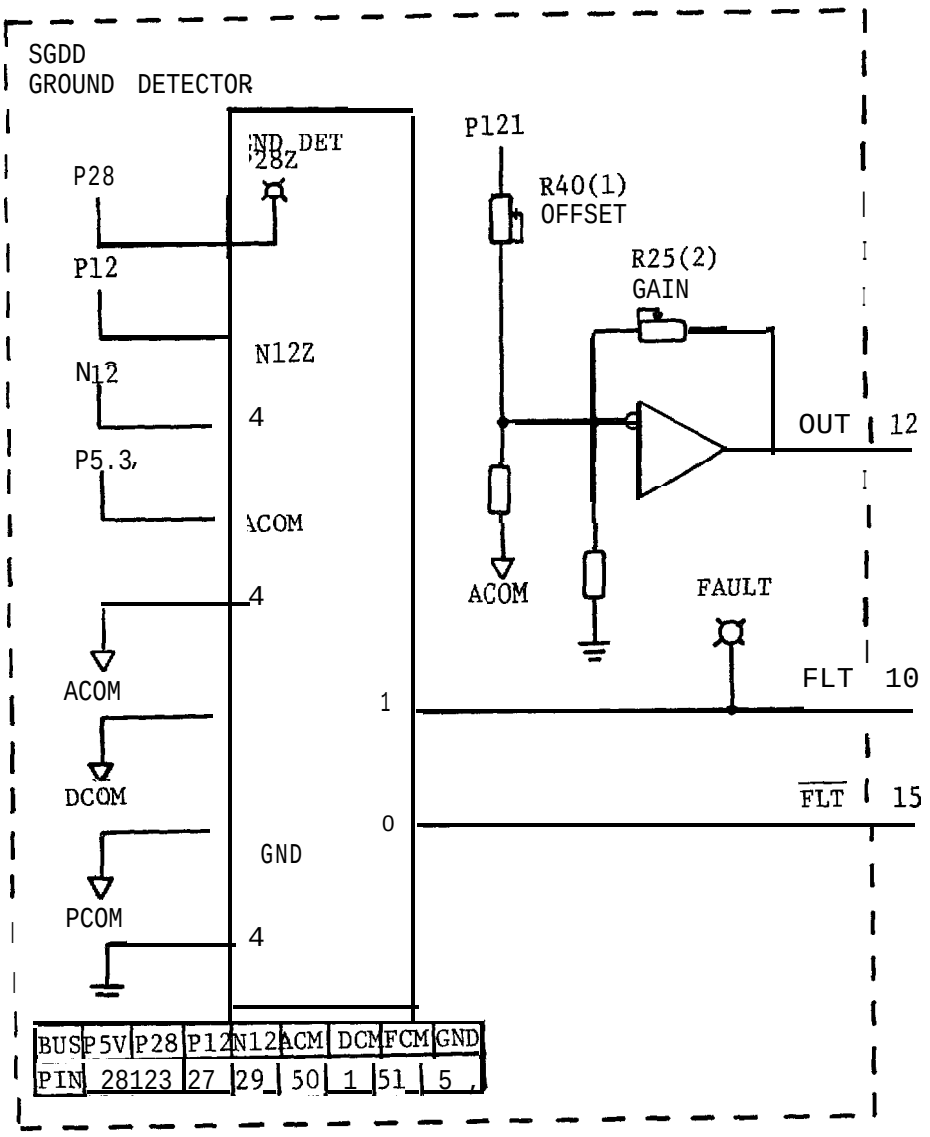


Figure 4-11
Turbine Control Power Supply Ground Detector

Thermocouple Ground Detector (See Figure 4-12). The SGDE card, mounted in the thermocouple processing module, provides thermocouple ground detection. The card provides four independent ground detector circuits. One circuit is used in conjunction with the control thermocouples, two circuits are used in conjunction with the protection thermocouple channels A and B. The fourth circuit may be used in conjunction with an optional third protection thermocouple channel. A thermocouple leakage to ground, 12K to 36K ohms, will be indicated as a fault.

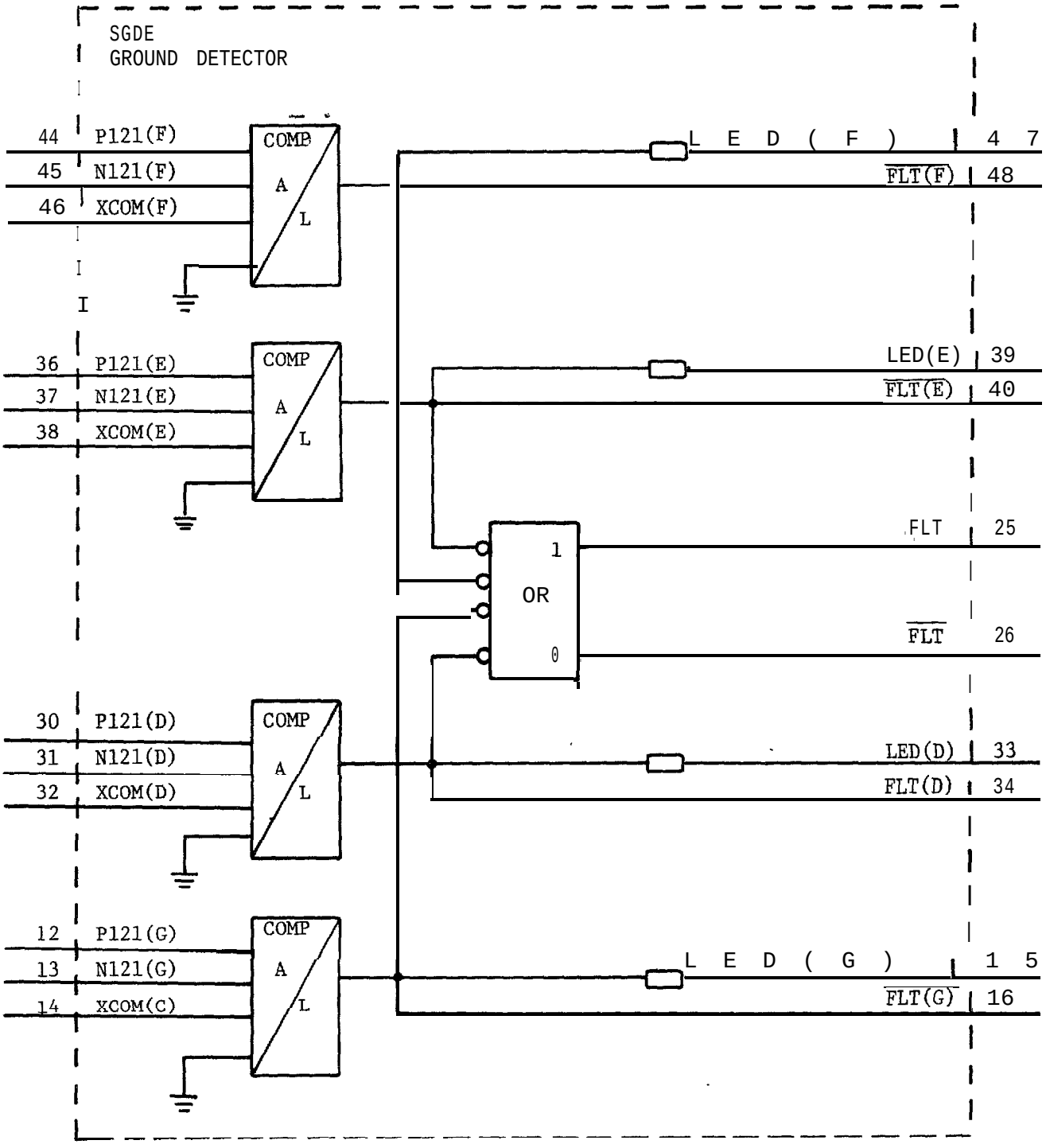


Figure 4-12
Thermocouple Processing Module System Ground Detector

125VDC Battery Supply Ground Detector (See Figure 4-13). A dual coil Sigma relay and associated circuitry provides ground detection for the 125 VDC battery supply system. Leakage impedance of either bus to ground of approximately 12K ohms or less will be indicated as a fault condition.

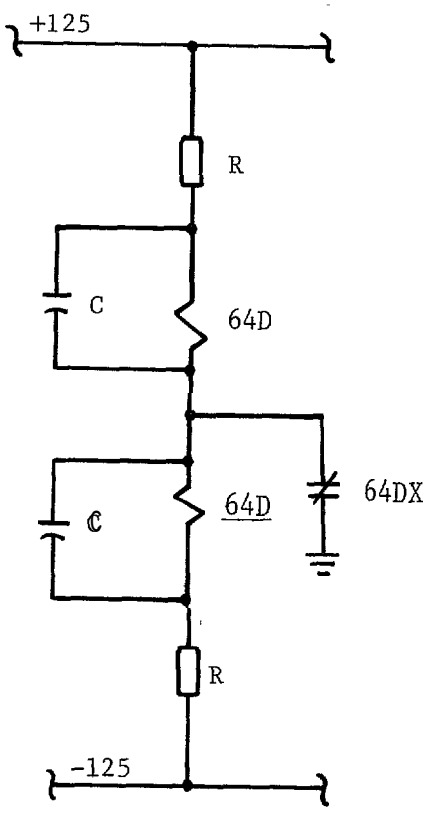


Figure 4-13
125vDC Supply Ground Detector

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POWER SUPPLY SYSTEM

A block diagram of the power supply system is shown in Figure 5-1. The system is broken down into the following functional assemblies:

(a) Source Selection
(b) DC Bulk Supply
(c) Final Regulator

General Description

Source Selection (Optional). The source selectors allow the DC bulk supply to use power from one of as many as three sources. As illustrated in Figure 5-2, these sources are:

(a) DC Battery
(b) Commercial AC Power
(c) Inductor Alternator

The standard configuration is to use one source, namely the DC battery. In this case, no source selection modules are needed; the DC bulk sourc- is connected directly to the battery. The source selectors are available as options to satisfy special case requirements. The operation of the source selectors is almost obvious when reference is made to Figure 5-3. Source selector IC3603A166A has two possible inputs, DC battery and/or AC commercial power. The battery input has:

(1) fusing, 20 amperes

(2) filtering for transient voltage suppression

(3) a lamp (DSI) for a power on indication

(4) a diode to isolate the battery source from other sources

The AC commercial input has:

(1) fusing, 20 amperes

(2) transformer isolation with taps

(3) rectification with filtering

(4) a lamp (DSZ) for a power on indication

(5) a diode to isolate the source from other sources

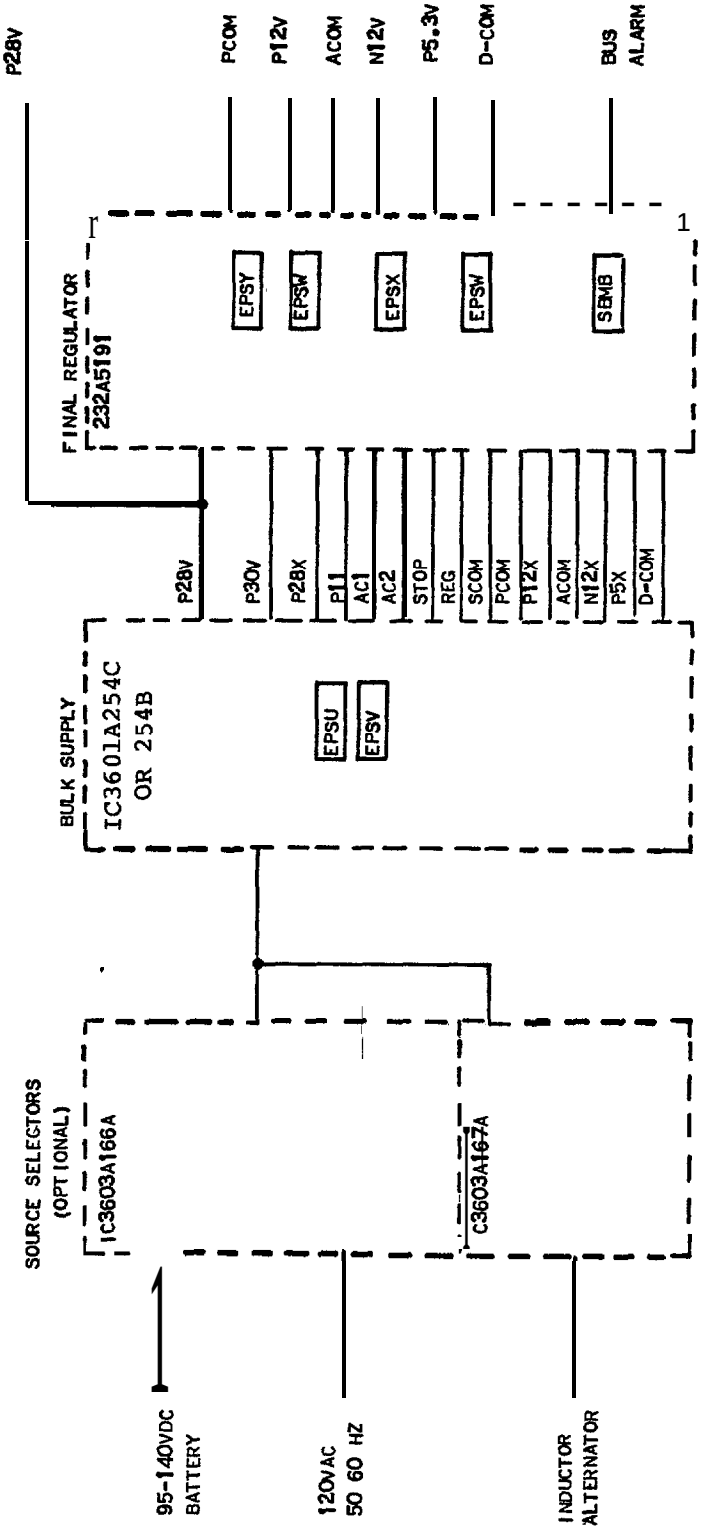


Figure 5-1
Mark II Power Supply Configuration

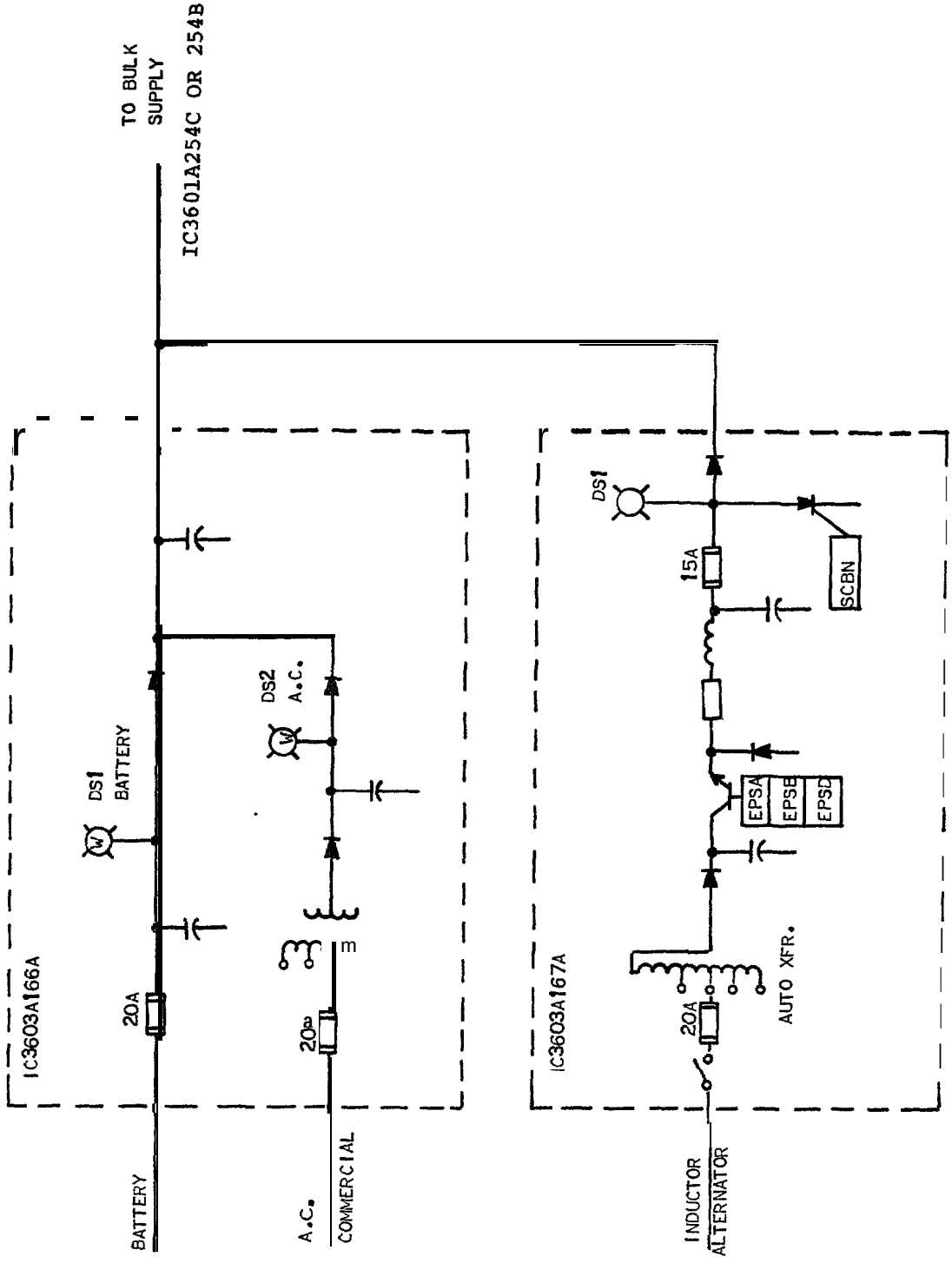


Figure 5-2
Source Selector for Mark II Power Supply (When Used)

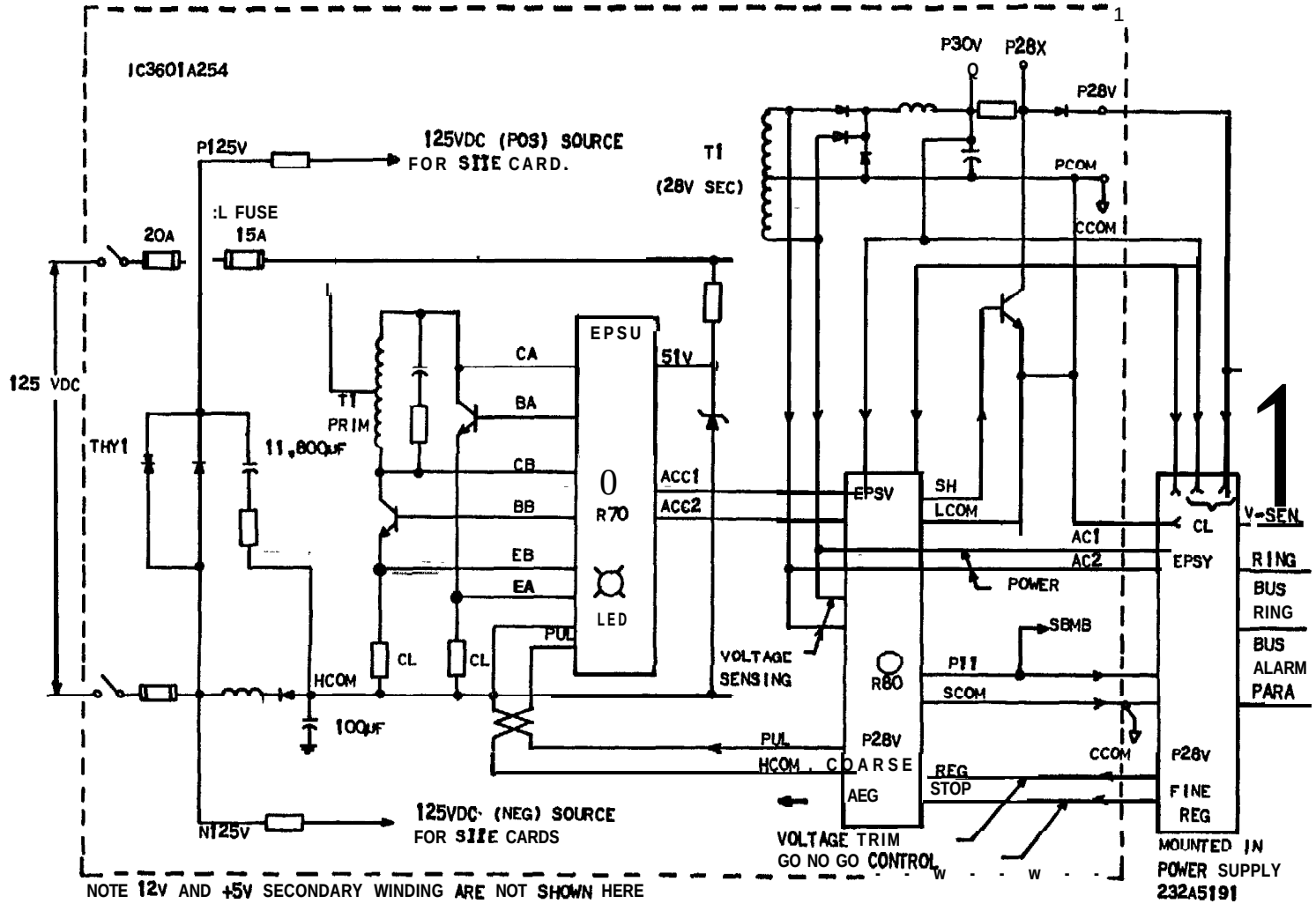


Figure 5-3
DC! Bulk Power Supply

The AC commercial input has:

(1) fusing, 20 amperes

(2) transformer isolation with taps

(3) rectification with filtering

(4) a lamp (DSZ) for a power on indication

(5) a diode to isolate the source from other sources

Source Selector IC3603A167A is used only for an inductor alternator source. This module performs the following:

(1) fusing, 20 amperes

(2) autotransformer with taps

(3) rectification and filtering

(4) pre-regulation to 120 volts DC

(5) current limit

(6) fusing, 15 amperes

(7) lamp for power on indication

(8) crowbar, to short circuit output if the voltage exceeds 135 volts DC

The source with the highest voltage automatically takes over and supplies the current. .

DC Bulk Supply. Basically, the DC bulk supply performs the following:

(1) accepts a nominal 125 volts DC voltage

(2) converts to AC (350 to 450 hertz)

(3) transforms voltage to appropriate levels to feed the P28V, P12V, N12V and P5V power supply circuits.

(4) provides isolation from the primary (i.e. the 125 volt DC circuitry).

(5) rectifies and filters the above voltages

(6) feeds the P28V bus directly from the DC bulk supply

(7) feeds the P12V, N12V and P5V buses via the final regulator as illustrated in Figure 5-4

(8) provides a buffered 125 VDC source to power IC3606SIIE cards

The main features of the DC bulk supply are illustrated in Figure 5-3. Working from left to right in Figure 5-3:

(1) on-off switch

(2) fusing, 20 amperes

(3) thyrector for spike suppression

(4) inductor to isolate switching spikes (generated in bulk supply) from getting back onto the source bus

(5) 11,800 microfarads of capacitance to allow the power supplies to ride through a 65 millisecond source voltage dip

(6) isolating diode to prevent a voltage dip on the source from discharging the capacitor of step (5)

(7) 100 microfarad capacitor to provide AC coupling to ground

(8) fusing, 15 amperes, current limit (fast) type

(9) transformer; the 125 volts DC (nominal) is applied to the center of the primary, with each end alternately switched to the return 125 volts DC, via switching transistors. The secondaries are rectified and filtered to provide DC voltages to the final regulator and the P28V bus (refer to Figure 5-4).

(10) The EPSU card provides control of the switching transistors of step (9) and provides control power to EPSV. The EPSU is, in turn, controlled via pulses from the EPSV card.

(11) The EPSV card is a course P28V voltage regulator; it senses the output voltage on the P28V secondary winding, compares it to a reference and sends out control pulses to EPSU (step 10). EPSV also provides a shunt load (power transistor) to ensure continuous current in the filter inductor(minimum of three amperes).

A photograph of the bulk supply is shown in Figure 5-5.

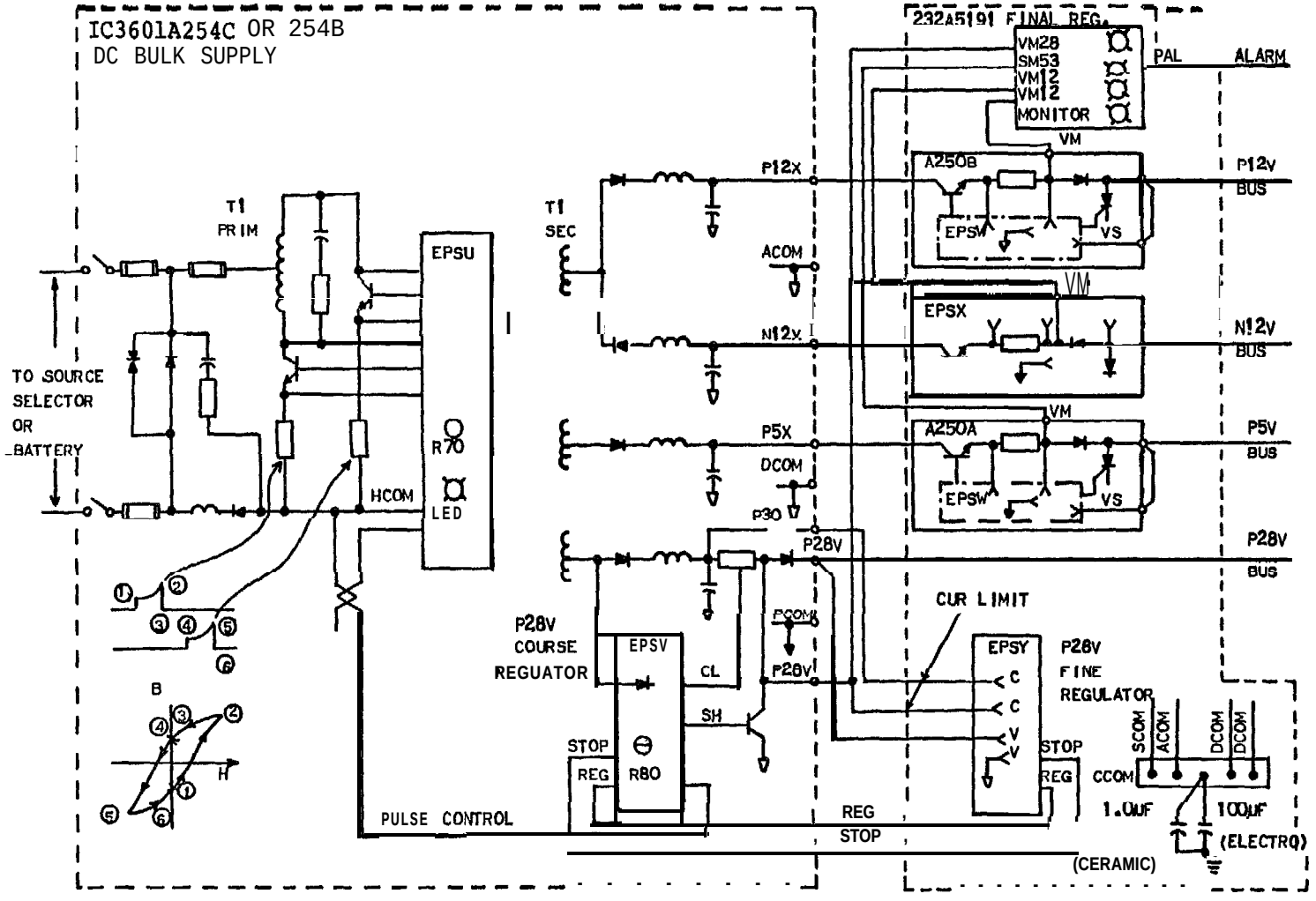


Figure 5-4
DC Bulk Supply and Filter Regulator

Final Regulator. The functions of the final regulator are illustrated in Figure 5-4 and are as follows:

(1) Module IC3601A250B (including EPSW):

a. steps down a coarsely regulated voltage, P12X, to a finely regulated voltage P12V.

b. provides current limit

c. provides current and voltage jacks for measuring purposes

d. provides overvoltage protection by means of a crowbar circuit (i.e. by firing an SCR).

(2) Module IC3601A250A: same as step (1) except P5V bus is controlled instead of the P12V bus.

(3) Card EPSX: same as step (1) except N12V bus is controlled instead of the P12V bus, and overvoltage protection is via a diode instead of an SCR (provides protection against being pulled positive).

(4) Card EPSY:

a. provides an outside, fine voltage, control loop on the P28V bus, i.e. it trims the coarse voltage regulator EPSV.

b. provides current limit via a STOP signal to EPSV.

c. provides current and voltage jacks for measuring purposes.

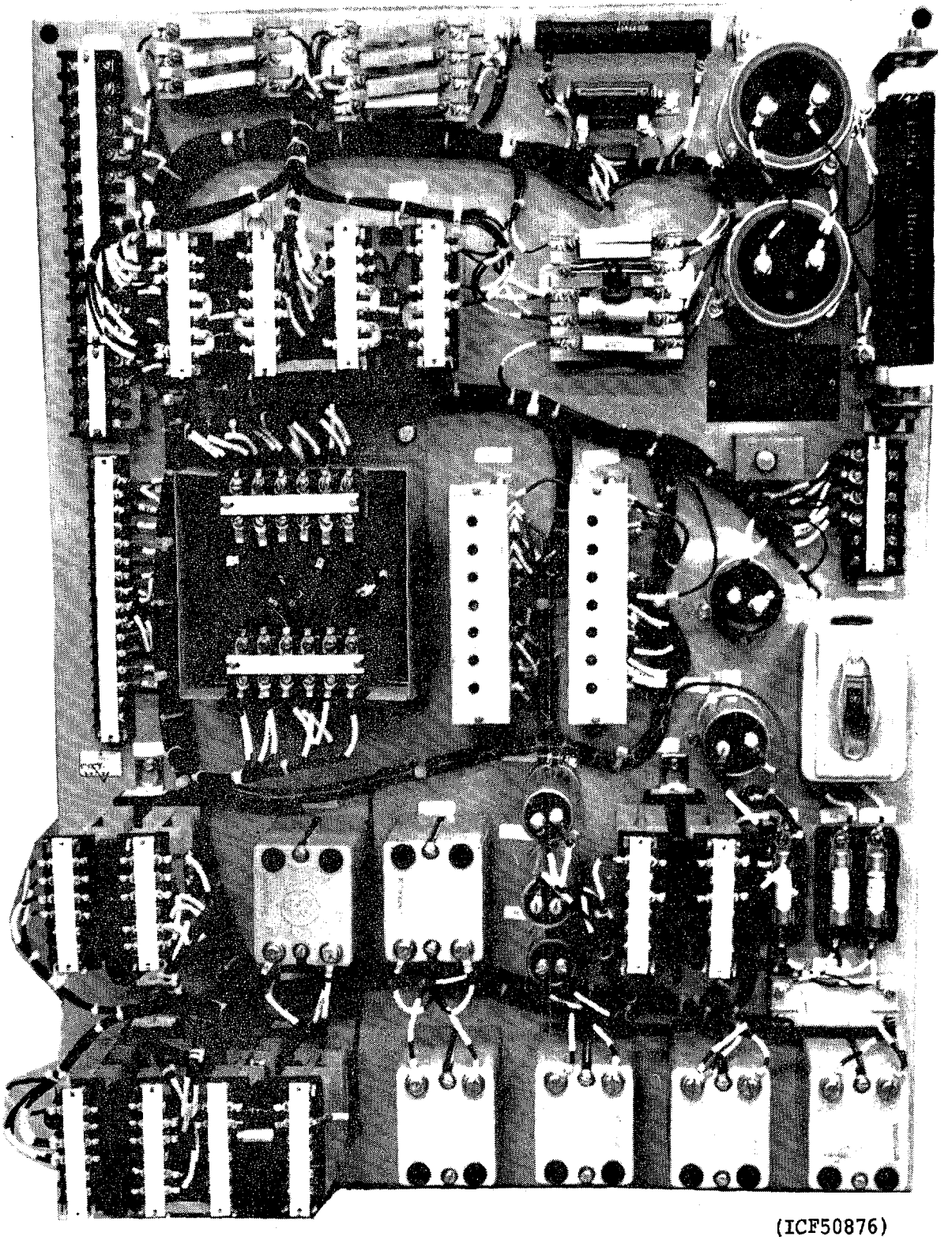
(5) Card SBMB (Optional): This card monitors the output voltages on the four buses and provides a latching alarm signal (generally taken to annunciator) if any of the voltages exceed certain limits (low/high). The alarm reset pushbuttons (four) are on the card front, along with the LED (four) alarm indicators.

A photograph of the final voltage regulator is shown in Figure 5-6.

Application Information

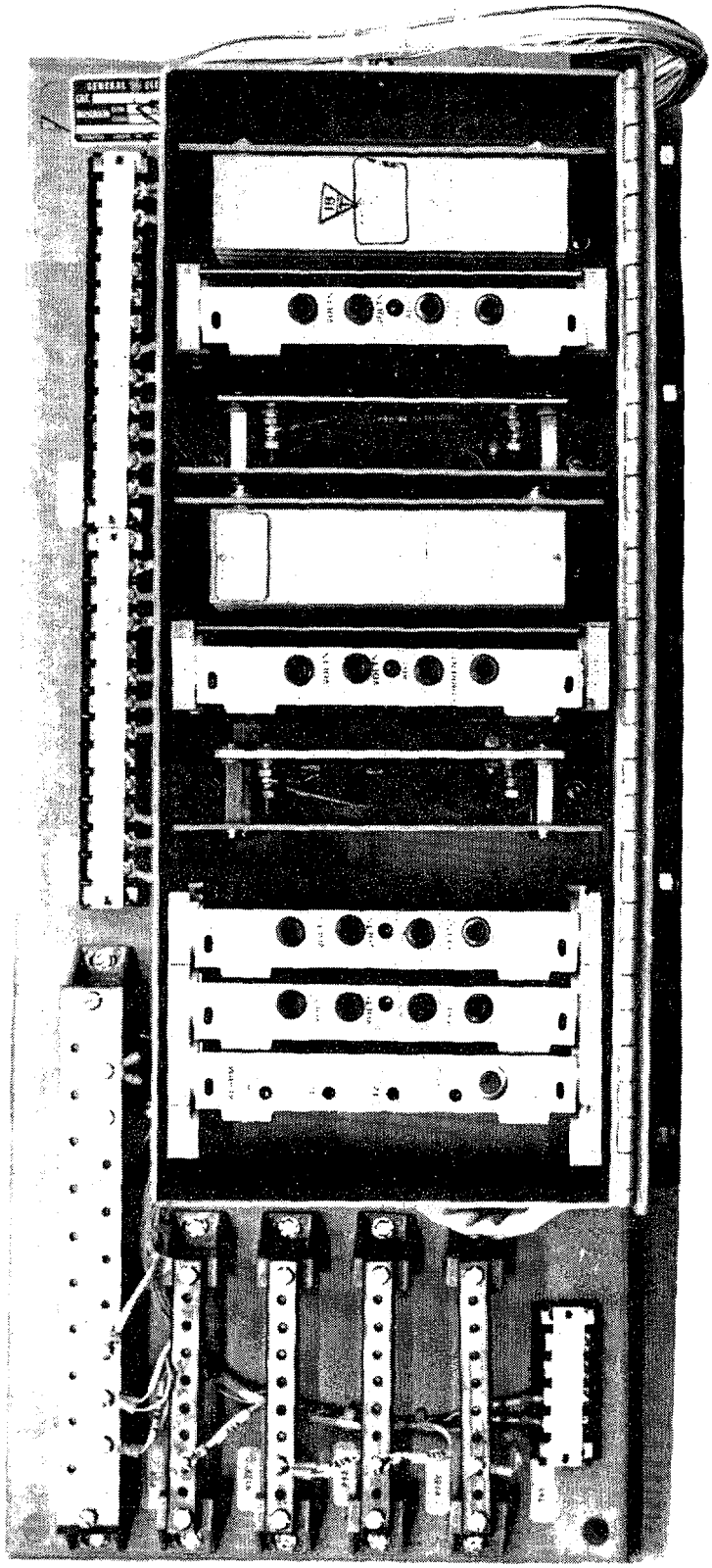
Source Selection

(1) The DC battery input characteristics are as follows:



(ICF50876)

Figure 5-5
Bulk Power Supply



(ICF-50872)

Figure 5-6
Voltage Regulator

a. Voltage range: 99 - 134 volts DC

b. The battery usually consists of 56 cells floated at 2.4 V/C.

c. No load - 2.0 V/C, discharge - 1.75 V/C.

(2) The commercial AC input characteristics are as follows:

a. Nominal volts +5% -10%; site-to-site variation of nominal is 130-105 volts RMS.

b. Frequency - 50/60 Hz; AC is sometimes grounded.

c. The commercial AC input is taken to a transformer which has five primary taps of approximately 6-volt voltage steps. The primary nominal voltages are:

Winding	volts In	Volts out (X1-X2 Winding)
H1 - H2	111	100
H1 - H3	116	100
H1 - H4	122	100
H1 - H5	128	100
H1 - H6	135	100

At a particular installation the nominal voltage level should be determined. The next higher voltage windings should be used as the input. This is done to avoid the possibility of developing over 140 volts output. Voltages in excess of 140 volts can damage relays and auxiliary equipment which are supplied by the output.

d. Output voltage - determined by input voltage and transformer winding used. Nominal peak no-load output voltage will be approximately 140 volts DC. The output regulation at an output current of 15 amperes is approximately 15 volts.

(3) The inductor alternator input characteristics are as follows:

Range of rated frequency (frame size l-7) = 12-3 kHz.

Nominal volts at 100% speed = 135 volts.

Machine-to-machine variation at 100% speed, volts = +10%.

Voltage and frequency proportional to speed.

Range of speed for each turbine = 80-100%.

Nominal operating range, generator drives = 100%.

Nominal operating range, pumper drives = 80-105%.

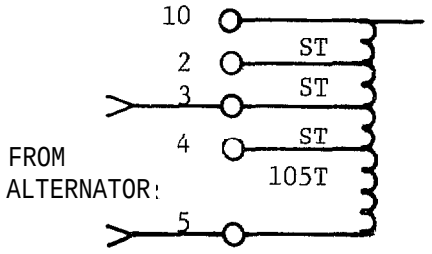
Nominal operating range, marine drives = 80-105%.

Overspeed test = 110%.

Full load (16p amperes) regulation at 0.85 pf = 20%.

The input from the inductor 'alternator is fed to an autotransformer per Figure 5-2.

The autotransformer is provided with three 5% taps. If the output voltage of the converter at TB1-4 falls below 120 volts DC or is greater than 225 volts DC, the autotransformer taps should be changed. The transformer is wired for a nominal open circuit transformation ratio of 1.09. If the output voltage is low, move wire from 3 to 4; if the output voltage is high, move wire from 3 to 2.



The current limit is set to 16 amperes (EPSA, R61, bottom).

The voltage output is set to 120 volts DC on TB1(1) to TB1(2) (EPSA, R62, top).

The crowbar voltage is set to 125 volts DC(SCBN, R3).

Bulk Supply

(1) Input voltage range is 95 to 140 volts DC.

(2) Input current is 2.1 to 7.3 amperes at 95 volts DC and 1.75 to 6.1 amperes at 140 volts DC.

(3) Outputs *are* as follows:

Terminal Points	Voltage (8%-100% Load)	No Load Voltage	Rated Current
P30	29.5 - 31	31	15 Amp.
P12X	18 - 20	34.8	6 Amp.
N12X	18 - 21	32.7	1 Amp.
P5X	10 - 13	20.8	6 Amp.
P11	11 - 11.5		30 Amp.

(4) The current signal across P30V to P28X (0.1 ohm shunt) is 0.3 volts DC to 1.5 volts DC for a zero to 15 ampere load on the P28V bus. A shunt regulator maintains a minimum to give continuous current even when the load on the P28V is zero.

(5) The supplies will not dip during a diesel start (or equivalent), defined as follows: dip to 60 volts, back to 80 volts in 65 milliseconds.

(6) Card EPSU (primary supply control)

Functions:

a. primary transistor base drive

b. primary current limit

c. control power supply

d. protection - if base and collector high, base drive removed and LED indication

Inputs:

a. collector volts (CA, CB)

b. emitter currents. EA1-3 EB1-3, SCA, SCB.

c. pulses (control) from coarse regulator card EPSV (time ratio control)

d. +51-volt zener input for control power (feeds inverter).

outputs:

a. Base drive

b. Control power supply to EPSV, 13-volt, 5 kHz.

c. LED indication for failure.

(7) Card EPSV (P28V coarse regulator).

Functions:

a. coarse voltage regulator on 28 volt supply output voltage and providing control pulses to EPSU (above).

b. control to shunt current regulator to ensure t12, -12 and t5.3 stay up. .

c. provides control power to other cards via P11 (11 volts DC) which does not disappear when inverter shuts down.

Inputs:

a. control power ACC1, ACG2 from EPSU

b. voltage feedback, AC1, AC2, corn.

c. stop-logic, "1" shutdown inverter "0" starts inverter

d. Reg. input from fine voltage regulator EPSY.

e. Signal bias operation of coarse regulator

f. Current bias operation of coarse regulator

g. Current signal across shunt P30, P28X.

outputs:

a. Pulses to EPSU (time ratio control)

b. Control supply P11, 11 volts

c. S corn - return for P11

d. SH - base drive to shunt current regulator

(8) Buffered 125VDC Source, This source is used on the input isolating cards **IC3606SIIE** for the following reasons:

a. to provide a ride-through capability; that is - battery bus dip will not interrupt input signals.

b. Form B should be used on systems with diesel starts. It has the capability of supplying 170 SIIE inputs, when power supply system is 50% loaded, with bus dips to 60 volts (recovers to 92 volts) for 70 milliseconds; initial voltage may be as low as 105 VDC. Form C has one half the ride-through capability Of form B and may be used on systems that do not have diesel starts. Form A does not have a buffered output for this function.

c. *to* limit current and voltage transients, thereby relieving **creepage** and clearance requirements on the SIIE cards.

Final Regulator.

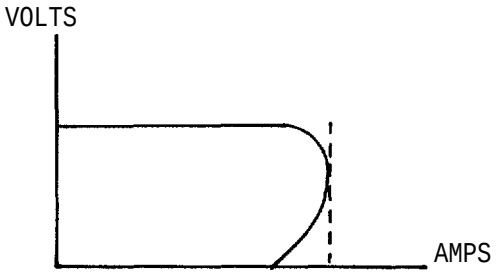
(1) The final regulator has the following output characteristics:

Volts	+28V*	+12V	-12V	+5.3V
Bus	P28V	P12V	N12V	P5V
Return	P COM	A COM	A COM	D COM
Rated Current	13.5A	7A	1.5A	6A
Current Limit	17A	8A	1.7A	7A
Ripple UP-P	250mV	10mV	10mV	100mV
Reg (LD)	300mV	30mV	30mV	150mV
Drift(Temp)	---	60mV	60mV	40mV
100% Current (Shunt-Signal	1.5V	1V	2.25V	1V

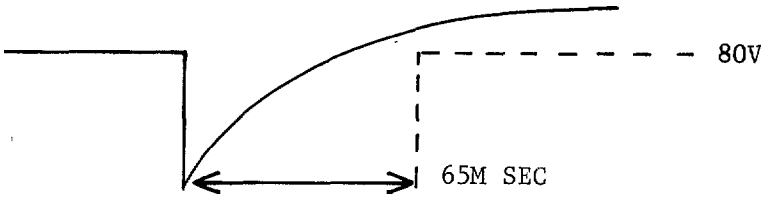
*For input of 105 **to 140 volts DC**; for input down to 95 volts, reduce 28V load to 7A or accept **P28V** drop to 26V. For lamp testing, the **P28V** supply has a momentary rating of **15A.**

(2) An SCR is used on each of the **P12V** and **P5V buses** and is called a "crowbar" circuit. Its purpose is to monitor the buses and protect the load equipment from overvoltage. The **P12V** bus crowbar operates at 15.5 to 19.5 volts; the **P5V** bus crowbar operates at 8-10.5 volts; the **N12V** bus has a diode to protect against positive-going voltages; the **P28V** has no crowbar since its load is not as voltage-sensitive as the other buses.

(3) The supplies have a current limit foldback as follows:



(4) The supplies will not dip during a diesel start (or equivalent) defined as follows:



(5) AC voltage, 350-450 Hz, 60 volts RMS is available from the bulk supply; for AC loading, reduce P28V volt-amps by an equivalent amount; non-standard.

(6) Jacks on card fronts, for both current and voltage signal.

(7) Monitors over/under voltage on all supplies, and +28-volt supply overcurrent monitoring with LED and logic annunciation.

(8) EPSY (P28 fine regulator)

Functions:

a. fine voltage regulation

b. Current limit

Inputs:

a. P11 - control supply

b. P28-28V bus, for ring bus monitor

c. P30-P28X for current limit signal

d. P30-PCOM-raw DC supply

e. Sense - remote voltage sensing

outputs:

a. Regulation - fine voltage regulation bias to EPSV

b. $\overline{CL}$ - current limit logical signal "0" = CL; to be used with monitor.

c. STOP - to shut off inverter on CL "1" = inverter OFF

(9) EPSW (P12V, P5V regulator)

Functions:

a. Regulate +5.3 volts or t12 bus

b. current limit on t5.3 volts or +12 bus

c. crowbar +5.3 volts or +12 bus

Inputs:

a. P28X - control power

b. P12X or P5X - power

c. V sensing - remote sensing

outputs:

a. Gate to crowbar SCR

b. V_m = to monitor

c. base drive to series transistor regulator

(10) EPSX (12 V regulator)

Functions:

a. Regulate - 12 volts

b. Current limit

Inputs:

a. N12X - power

b. V_m - (+12 volt supply) control power

c. V_S - remote sensing

outputs:

a. V_m - monitor voltage

b. N12V = output power

(11) SBMB (Bus Monitor)

a. monitor under/over voltage on four supplies detection at lo-15% error

b. annunciate with LED (1st fault indication)

c. signal logic; 1 = alarm (4)

a. Monitor current limit on EPSY (28V regulator), $\overline{CL}$ = 0 = current limit condition.

e. Local reset PB and remote PB

Inputs:

a. V_m for supply (monitor voltage)

b. P11; control power from EPSV

c. $\overline{CL}$ - CL from + 28 regulator

outputs:

a. $\overline{STOP}$ - "0" for normal condition; may be used to shut down inverter

b. Logic signals = alarm (4); 1 = alarm

Troubleshooting Guide

These instructions are written to aid the troubleshooter in isolating power supply troubles. This is a guide only and cannot be applied blindly without some knowledge of troubleshooting procedures and of the power supplies themselves. It is recommended that the troubleshooter, if not already familiar, briefly review the general description of the power supply system before proceeding with these instructions.

Symptom: **P5V** or **P12V** bus out of limits, but all other buses correct.

Action:

(1) It is possible that the "crowbar" protection circuit has been energized due to external voltage spiking. Verify that external instrumentation, etc. is not disturbing the bus. Remove power for several seconds (to allow SCR to reset) and reapply power.

(2) If fault still present, remove power, replace EPSW, then reapply power.

(3) If fault is still present, verify that input to final regulator module **(P12X** or **P5X)** is as defined in application section, paragraph entitled "Bulk Supply", (3); if not, the bulk supply is suspect, (i.e. the diode filter section of the bulk supply).

(4) If step (3) was correct, remove power, isolating all loading on the power supply in trouble: reapply power. If the power supply is correct now, the trouble is probably external to supply, either excessive load or overvoltage from other source.

(5) If power supply is still not correct, module **IC3601A250A (P5V)** or **IC3601A250A (P12V) is suspect.**

Symptoms: **N12V** bus is out of limits, but other buses are correct.

Action:

(1) Suspect EPSX; remove power, replace EPSX.

(2) If bus still not correct, isolate load and check **N12V** bus. If the bus returns to normal, the trouble is probably external to the power supply system.

(3) If bus still not correct, measure voltage on **N12X** (input to EPSX) and verify that it is as defined in application section, paragraph entitled "Bulk Supply", (3). If not, the bulk supply is suspect (i.e. diode filter section).

Symptom: P28v bus out of limits, but the other buses correct.

Action:

(1) EPSY is suspect; remove power and replace EPSY.

(2) If bus is still not correct, isolate the load and check the P28V bus. If the bus returns to normal, the trouble is probably external to the power supply.

(3) If bus still not correct, measure the voltage on P28x and verify that it is as defined in application section, paragraph entitled "Bulk Supply" (3); if not, the bulk supply is suspect (i.e. diode rectifier output is suspect).

Symptom: All power supply buses are low, but EPSU light emitting diode is off, and bulk supply making a noise (hum).

Action:

(1) Check that bulk supply input is 95 volts DC or greater.

(2) Check that the "STOP" signal (with respect to PCOM) labeled interconnections between bulk supply and regulator, is a logic "0". Use an oscilloscope if available. This signal should stay at zero volts. If the P28V supply is running into current limit, it will bounce between zero and approximately 2 volts DC. Also, check the voltage signal across the current jacks of the EPSY card and compare to the nominal current limit signal of 1.7 volts DC.

(3) If the trouble is still not determined, remove power, isolate the P28V load, reapply power and check for correct bus voltages. If the buses. are still low, jumper "SATOP" to "PCOM". If the bus voltages are now correct EPSY is suspect. If the bus voltages are still low, bulk supply cards and EPSU and EPSV are suspect.

(4) If the buses become correct when the P28V load is isolated, suspect P28V supply current limit trouble or primary current limit trouble. But P28V current limit was checked in step (2). Therefore, primary current limit; that is EPSU is suspect (change EPSU).

Symptom: EPSU light emitting diode is on; bulk supply will normally be shut down, i.e. will not be making a noise (hum).

Action:

(1) Remove power and check all power transistors on bulk supply heat sink HS1 and HS2.

(2) If transistors are correct, suspect EPSU and replace.

Symptom: Bulk supply not humming, EPSU LED is off, but light indicator DS1 is illuminated.

Action:

(1) Remove power and check fuses; replace any bad fuses.

(2) Remove EPSU and EPSV, then check power transistors on HS1 and HS2; replace any bad transistors.

(3) Apply power and check for battery voltage from CA to H-COM and from CB to H-COM (located on TB2). Battery voltage must be between 95-145 volts DC. If this voltage is not present, suspect bad power transistor or some other low impedance path on the front end. Remove power, isolate and remove this low impedance path.

(4) If battery voltage is correct on CA and CB, check for +51 volts DC between pins 11 and 51 in the EPSU slot.

(5) If the problem has not been identified to this point, remove power, plug in new EPSU and old EPSV cards and check system. If operation is still not correct, remove power, replace EPSV, and try again,

Black Start Inverter

A block diagram of the Black Start Inverter is shown in Figure 5-7. The primary purpose of this inverter is to provide AC power to the ignition transformers during start-up. The power source can be either commercial AC power or the station battery.

General Description. Major details of the Black Start Inverter are shown in Figures 5-a and 5-9, an elementary diagram and photograph respectively. For a detailed elementary diagram see GE Drawing IC3603A282.

Observe, in Figure 5-8, that 125 VDC appears across TB (18) and TB (19). Switch SW1 normally remains in the closed position but may be used to shut down the inverter. Relay 4w is used to cycle the inverter on and off from logic circuitry elsewhere. Resistor R30 limits the incoming current to C2 when relay 4W picks up. Relay 4WX picks up as the voltage across C2 builds up, and shorts out R30 to prevent power dissipation by the resistor.

Fuses FU1 and FU3 provide protection against a failure of C2. Fuse FU2 is a current-limiting fuse, used to limit the extent of damage in the event that transistors QB2 and QB3 fail due to a short.

Cards IC3600EPSD and IC3600EPSB are power supplies for control cards IC360EPSA and IC3600EPSC. Card IC3600EPSA, in conjunction with switching transistor QB1 and current shunt R15/16, functions as a voltage pre-regulator. That is, it regulates the voltage across C3 (or on TB(10)) to 100 VDC. It also provides a current-limiting function. Control card IC3600EPSC, in conjunction with transistors QB2 and QB3, provide voltage pulses to the primary winding of the transformer. QB1, QB2 and QB3 are not single transistors but in reality they represent banks of transistors.

Commercial AC power is connected across TB(16) and TB(17). If relay 27W is dropped out, the commercial AC power will appear across the normally-closed contact of 27VV, out TB(15), and become available to power the ignition transformers. If the inverter is energized and TB(14) is jumpered to TB(13), 27W will be picked up, and the secondary winding of the transformer will be connected to output terminal TB(15). Thus, when the inverter is shut down, the commercial power is provided as the output on TB(15). When the inverter is running, the DC voltage is inverted to AC and appears as the output on TB(15). An interlock of relay 27W is provided across TB(20) and TB(21) to indicate the operating mode of the inverter.

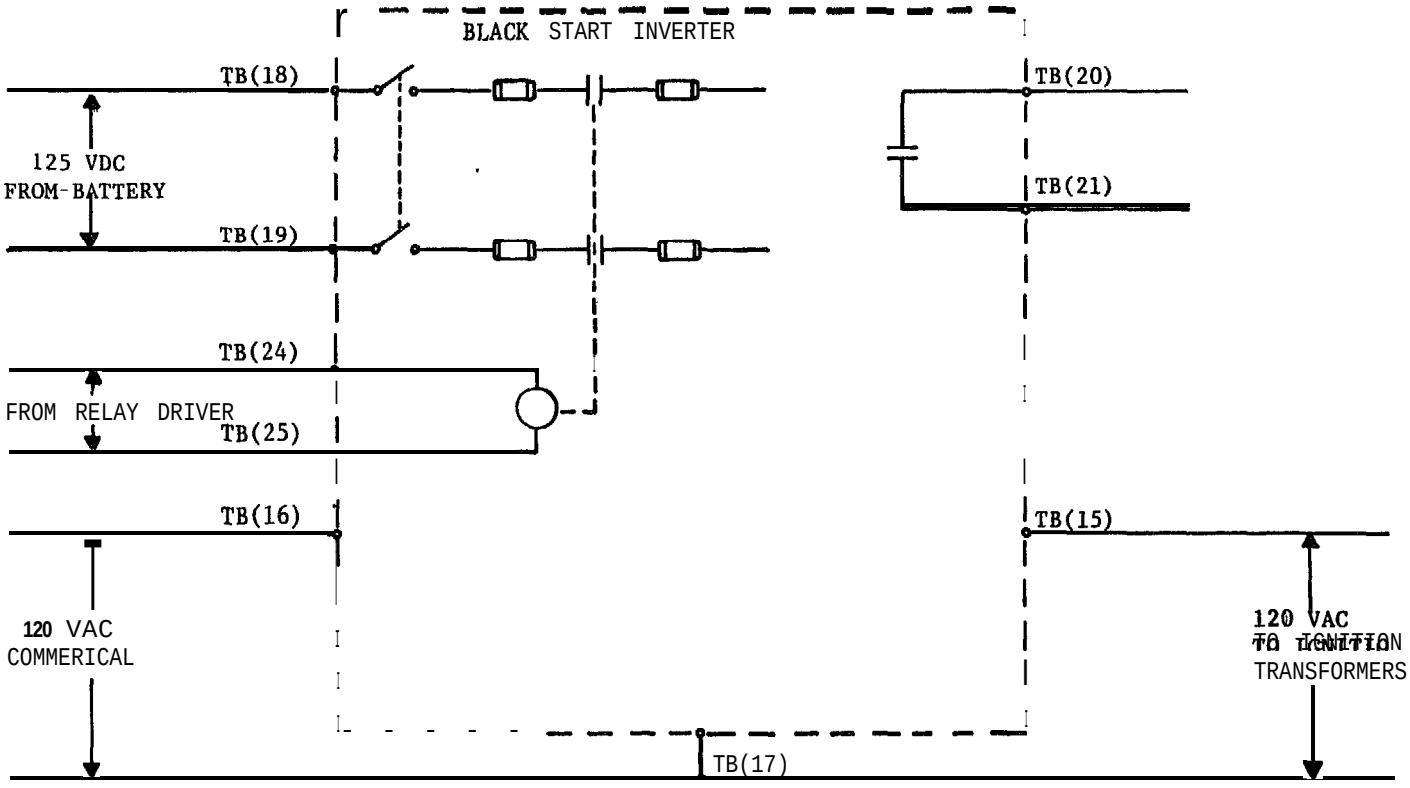


Figure 5-7
Black Start Inverter Block Diagram

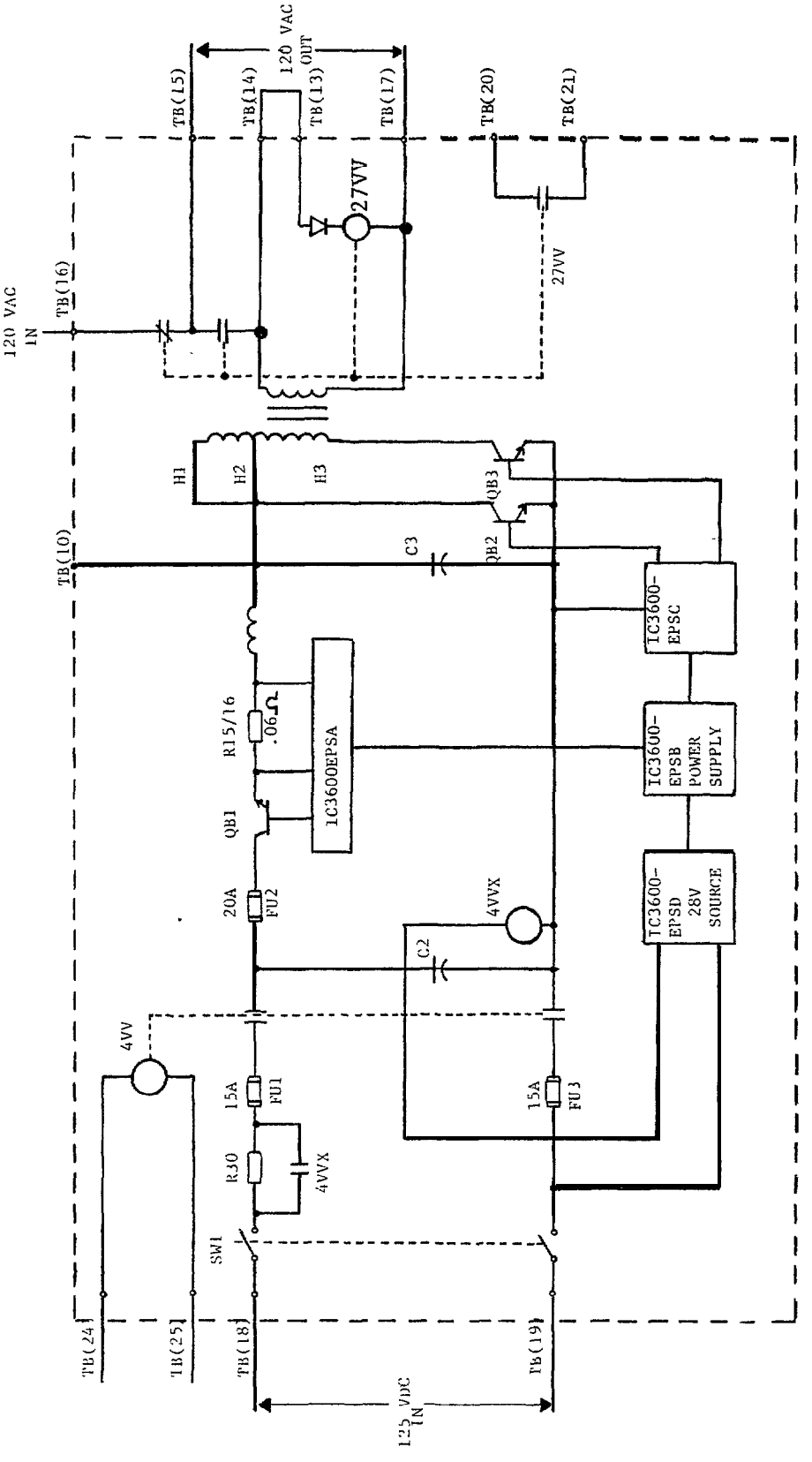
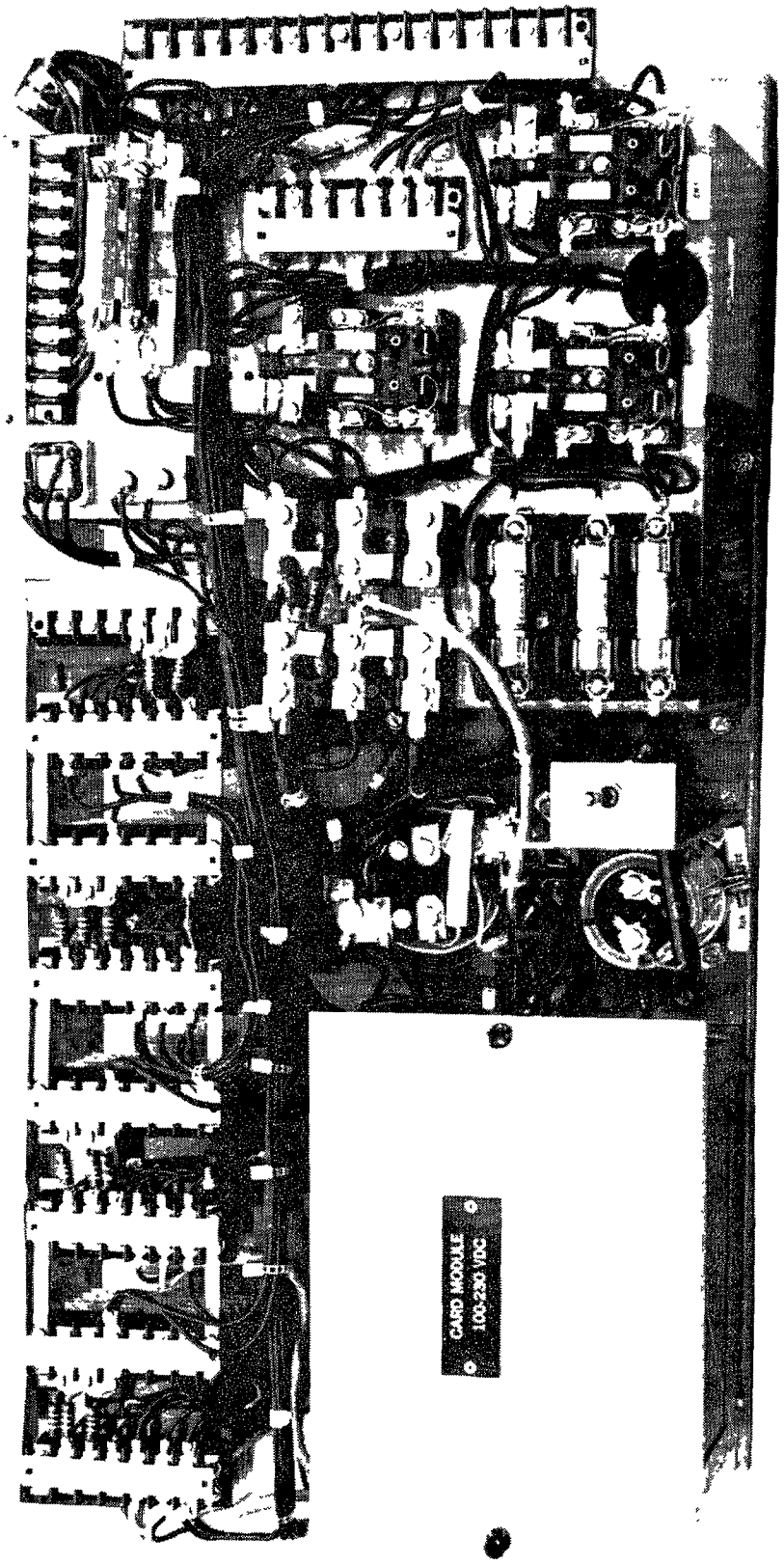


Figure 5-8
Black Start Inverter Elementary Diagram



(ICF-50881)

Figure 5-9
Black Start Inverter

Application Information.

Input Power:
105 to 250 VDC

DC current does not exceed 11 amperes on a steady state basis
DC power terminals: TB(18)+, TB(19)-.
AC terminals: TB(16) high, TB(17) low.

Signal:
Relay 4W coil voltage 28 VDC.
Terminals TB(24)+, TB(25).

Output Power:
112 VAC ±5%
AC Terminals: TB(15) high, TB(17) low.
AC Current: 3 amperes continuous
6 amperes for 3 minute&
10 amperes for 1 minute

Signal:
Relay 27W contacts
Terminals TB(20), TB(21).

Frequency: 60 Hz IC3603A282C,
50 Hz IC3603A282D,
jumper EPSC pin 18 to 20.

Troubleshooting.

Symptom: Inverter not functioning properly since relay 27W is dropped out while SW1 is closed and relay 4W is picked up.

Refer to Figure 5-8 an proceed as follows:

(1) Remove AC and DC power from the module.

(2) Check fuses FU1, FU2 and FU3.

(3) Remove three cards: EPSA, EBSB, EPSC.

(4) Apply DC power, close SW1, pick up relay 4W and check for 28±3 VDC on pins 26, 2, in the slot from which EPSB was removed. If this voltage is not present, card EPSD is probably faulty and should be replaced.

(5) Open SW1, insert card EPSB, close SW1, and check for the following voltages in slot EPSA:

a. +6 ± 1 VDC from pin 6 to pin 2.
b. -6 ± 1 VDC from pin 46 to pin 48.
c. +6 ± 1 VDC from pin 46 to pin 44.

(6) Open SW1, insert EPSA card, close SW1 and check for 100 ± 3 VDC On TB(10) to TB(19).

If this voltage is high, suspect card EPSA and the transistor bank represented by QB1. Check transistors on heat sinks HS3 and HS4 with an ohmmeter and replace EPSA.

If the voltage TB(10) to TB(19) is less than 100 ± 3 VDC, suspect card EPSA, QB1, QB2 and QB3. Isolate QB2 and QB3 by opening at transformer H2, then try again. If the measured voltage is still low, check QB1 and replace card EPSA. If this does not correct the problem, check transistors on heat sinks HS1 and HS2 (QB2 and QB3) using an ohmmeter.

(7) Open Swl, insert card EPSC, close SW1 and check that relay 27W is picked up and that the voltage between TB(15) and TB(17) is 112 VAC ± 5%. If not, suspect EPSC and the banks of resistors represented by QB2 and QB3. Check all transistors on HS1 and HS2 using an ohmmeter and replace card EPSC.

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